

MODEL DISCOVERY AGENT: LLM-ASSISTED BAYESIAN EXPERIMENT DESIGN FOR DATA-EFFICIENT DISCOVERY OF MECHANISTIC WORLD MODELS

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ABSTRACT

Predicting the answer to interventional “what if” questions — the outcome of an action never taken — requires a *mechanistic*, causal model, not a curve fit; and learning such a model requires *experiments*, because passive data leaves its mechanisms unidentified. Experiments are expensive, so the central problem is *data efficiency*. We present the Model Discovery Agent (MDA), which couples a large language model (LLM), used as a *proposer* of candidate structures, with standard Bayesian machinery — sequential Monte Carlo (SMC) for parameter and structure posteriors, simulation-based inference (SBI) for intractable likelihoods, and value-of-information (VoI) for experiment design — to discover latent mechanistic world models from few interventions. MDA operates in the M-open setting: when the truth lies outside the current hypothesis class, a predictive check flags the inadequacy and the proposer expands the hypothesis space with a new model whose parameters are then identified by designed experiments. We show that *discovery and design reinforce*: the design step identifies the mechanism the discovery step proposes, and the identified mechanism improves predictions, enabling further discoveries from the remaining unexplained residuals. On three different benchmarks — covering physics (FORCEBENCH, (Wiemann et al., 2026)), chemistry (CHEMBENCH, (Kabra et al., 2026)) and biology (NEURONBENCH, a new partially observed single-neuron electrophysiology benchmark we create) — we show that MDA sets a new SOTA in terms of data-efficient model learning and reliable interventional forecasting ability.

1 INTRODUCTION

Much of what we want from a predictive model is *interventional*: not “what will happen?” but “what *would* happen if I did *a*?” (e.g., predicting the effect of administering a drug to a patient that it has never received, launching a probe on an orbit it has never flown, or deploying a policy never enacted before). Such interventional questions cannot in general be answered by a model fit to passive observation, however flexible: two mechanisms can agree on all observed data yet disagree under intervention. Answering interventional queries requires a *mechanistic* or *causal* model of the data-generating process (Pearl, 2009; Richens & Everitt, 2024).¹ Such mechanistic models are also the foundation of true *scientific understanding* (Salmon, 1984; Krenn et al., 2022; Messeri & Crockett, 2024; Bajorath, 2025; Serre & Pavlick, 2025; Kramer et al., 2026).

Unfortunately, a latent mechanistic model is typically *unidentifiable* from observation alone: the passive data underdetermines it, and only intervening — perturbing the system and watching how it responds — breaks the degeneracy. But experiments are expensive (a lab assay, a clinical trial),

¹Note that this paper is concerned with “level 2” causality, to use the terminology of Pearl’s causal ladder (Pearl, 2009; Bareinboim et al., 2022); this can be handled with standard decision-theoretic machinery (Dawid, 2015), and does not need the more complex machinery required for “level 3” counterfactual reasoning (Dawid, 2000).

which makes the operative problem *data efficiency*: identify the mechanism, well enough to answer the queries, in as few experiments as possible. This is the classical remit of *Bayesian experimental design* — choose the intervention whose outcome is most informative (Lindley, 1956; Chaloner & Verdinelli, 1995; Rainforth et al., 2024) — but it has rarely² been combined with the *open-ended hypothesis creation* that scientific discovery demands.

To tackle these problems, we present the **Model Discovery Agent (MDA)**. This uses a large language model (LLM), which contains useful prior knowledge (Kıcıman et al., 2024), to propose candidate mechanisms, given a natural-language description of the domain and any initial observational data. We then combine this with standard Bayesian machinery: sequential Monte Carlo (SMC) for computing the posterior over parameters and structures and the model evidence, and value-of-information (VoI) maximization for choosing the next experiment. We extend the standard Bayesian machinery to the $\mathcal{M}$ -open regime, where the true mechanism may lie *outside* the proposed hypothesis class (Bernardo & Smith, 1994; Kelter, 2020). We do this by testing if the current best hypothesis fails an out-of-sample predictive check; if so, we expand the hypothesis space (using the LLM), and then design an experiment to identify the new model’s parameters (using VoI). We find that discovery and design *reinforce* each other: the experiment identifies the novel mechanism the proposal introduced, and the identified mechanism improves the model’s forecasts, enabling the detection of ever more subtle predictive errors (c.f., (Buehler, 2026)).

We validate MDA on three sets of benchmarks, covering physics (FORCEBENCH, based on (Wiemann et al., 2026)), chemistry (CHEMBENCH, based on (Kabra et al., 2026)) and biology (NEURONBENCH, a new single-neuron electrophysiology benchmark we create³). In each case, we show that MDA is substantially more data-efficient than pure LLM baselines. In summary, we make 3 contributions: we develop the MDA method; we establish new SOTA performance on two existing interactive scientific discovery benchmarks; and we create a new benchmark (NEURONBENCH), which adds features such as partial observability and stochasticity that are missing in existing benchmarks.

2 PROBLEM STATEMENT

Modeling assumptions. We consider an agent interacting with an unknown “blackbox” dynamical system, that maps an optional sequence of inputs or control signals $x_{1:T}$, for $x_t \in \mathcal{X} \subset \mathbb{R}^{d_x}$, to a sequence of noisy observations, $y_{1:T}$, for $y_t \in \mathcal{Y} \subset \mathbb{R}^{d_y}$, in response to an optional perturbation or intervention $a \in \mathcal{A}$, and an optional setting of the initial condition of the system state $\iota \in \mathcal{Z} \subset \mathbb{R}^{d_z}$. WLOG, we assume the true data generating process can be represented by a latent-state dynamical system, or state space model (SSM), as shown in Fig. 5. The latent dynamics (which may be deterministic or stochastic) are given by $z_{t+1} \sim p(z_{t+1} \mid z_t, x_t; \text{do}(a, \theta))$, where $\text{do}(a, \theta)$ represents the parameters of the system after applying intervention a .⁴ The noisy observation model is $y_t \sim p(y_t \mid z_t; \theta)$, and the initial condition is given by $z_0 \sim p(z_0 \mid \iota)$. A static input-output system is a special case with $T = 1$.

Data. We define an experiment design as $\xi = (\iota, a, x_{1:T})$, where x_t is the input at step t (if present). If $\iota = \square$, it means the initial state of the system is chosen at random from some distribution $p(z_0)$. If $a = \square$, it means we use the original unperturbed system parameters θ . The agent is presented with an initial dataset $\mathcal{D}_0 = \{(\xi_0^i, y_{0,1:T}^i) : i = 1 : N_0\}$, where each sample is drawn from the system using $y_{0,1:T}^i \sim p(\cdot \mid \xi_0^i)$. We assume the initial designs are from the default (unperturbed or “wild-type”) system, but they may use different input sequences x^i . The agent is then given a budget of B turns to interact with the system. At each step, it designs an experiment ξ_b , and then collects data $y_{b,1:T}$ from the environment, to create $\mathcal{D}_b = (\xi_b, y_{b,1:T})$. It can use this

²See Section 5 for discussion of prior work.

³Benchmark available at <https://github.com/murphyk/neuronbench>

⁴We distinguish the intervention action a , as used in the causality literature, from the action sequence $x_{1:T}$, as used in the RL and control theory literature, because they play slightly different roles: the former changes the mechanism (parameters) of the underlying system, whereas the latter corresponds to changing the set of inputs or covariates applied to a fixed system. Of course, we can always define $x_0 = a$, but we choose to keep them separate for notational clarity.

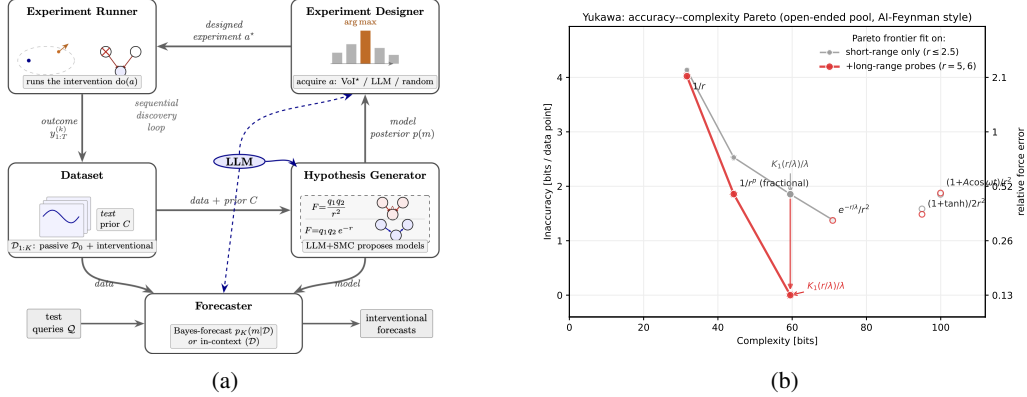


Figure 1: (a) The MDA discovery loop. See Algorithm 1 for details. (Figure based on (Elteto et al., 2026, Fig.2).) (b) Accuracy-complexity Pareto frontier for models discovered by MDA in the YUKAWA physics environment. Both axes in *bits* on a *linear* scale so the convex corner is obvious. The *y*-axis is *inaccuracy*, the absolute relative force error in bits, $|\log_2(F_{\text{pred}}/F_{\text{true}})|$. See Section 4.1 for details. (Figure based on (Udrescu & Tegmark, 2020, Fig 1)).

knowledge to update its beliefs about the underlying model, $p_b = p(m|\mathcal{D}_{0:b})$, where m specifies the SSM structure and parameters, and this belief can be used to design the next experiment.

Evaluation. After B rounds, each agent has the training set $\mathcal{D}_{\text{tr}} = \mathcal{D}_{0:B}$; the MDA agent also has its final belief state, p_B . When working with synthetically generated data, we can compare the agent’s estimated model directly with the true model using an appropriate metric. However, in the general case, we rely on measuring out-of-sample prediction error in response to a set of novel test experiments, $\xi_{\text{test}} = \{\xi^i : i = 1 : N_{\text{test}}\}$. We define the corresponding ground truth test set as $\mathcal{D}_{\text{te}} = \{(\xi^i, s(y_{1:T}^i)) : i = 1 : N_{\text{test}}\}$, where $s(y_{1:T})$ is a set of features of interest we want the model to predict. In the simplest setting we use $s(y_{1:T}) = y_{1:T}$, but for complex observations, we can use summary statistics that capture relevant high level patterns. For example, for NEURONBENCH in Section 4.3, we use $s(y_{1:T}) = \{n_k\}$, which is the number of spikes at each input strength k . We evaluate performance using the average loss, $\mathcal{L} = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \ell(\hat{s}^i, s^i)$, where $\hat{s}^i$ is the agent’s prediction for the features on test input i .

At a high level, our setup is a transductive problem similar to the ARC-AGI challenges⁵, except our domains use continuous-valued actions and observations, and are derived from real scientific problems.

3 METHODS

Overview. The MDA method is visualized in Fig. 1a; see Algorithm 1 for detailed pseudocode. At each step, the agent updates its belief state $p_b = p(m|\mathcal{D}_{0:b})$, which is a posterior distribution over models or hypotheses m . Then it chooses the next experiment by maximizing the expected value of information, $\xi_{b+1} = \arg \max_{\xi \in \Xi} \text{Vol}(\xi)$. It runs the experiment and updates its dataset by appending $\mathcal{D}_{b+1}$. After B rounds, the agent is asked to forecast the outcomes to some novel experimental conditions. We give the details below.

Sequential Bayesian inference. The belief state $p_b = p(m|\mathcal{D}_{0:b})$ is a posterior over models m , represented as a set of N_m particles. Each model m encodes the structure of the system, as in a structural causal model (SCM) (Pearl, 2009). This posterior is updated using Sequential Monte Carlo

⁵See <https://arcprize.org/arc-agi>. ARC-1 and ARC-2 are passive transduction problems, where x^i is a 2d input grid (specified by the environment, not the agent) and y^i is the resulting 2d output grid, and the goal is to learn $p(y_{\text{test}}|x_{\text{test}}, \mathcal{D}_{\text{tr}})$, where $\mathcal{D}_{\text{tr}}$ is a fixed set of 3 (x, y) pairs. ARC-3 involves dynamic interaction with a 2d grid world, where the agent actively controls the sequence $x_{1:T}$ and observes $y_{1:T}$, which is more like our setting.

(SMC), following the ModelSMC method of (Wahl et al., 2026) and SMC-S method of (Piriyakulkij et al., 2024); see Algorithm 2 for the pseudocode. We use an LLM to propose a new model given the *set* of previous hypotheses, their corresponding residual errors (derived from the data), and an initial text prompt (context) $\mathcal{C}$. We denote this proposal distribution by $p(m_b | \{m_{b-1}^n\}, \mathcal{D}_{0:b})$.

After proposing a new model (particle), we evaluate its evidence (marginal likelihood), $Z_m = p(\mathcal{D}_{0:b} | m) = \int p(\mathcal{D}_{0:b} | m, \theta) p(\theta | m) d\theta$, using the adaptive-tempered SMC method shown in Algorithm 3. (See (Naesseth et al., 2019; Chopin & Papaspiliopoulos, 2020) for more details.) Crucially, the integration over model parameters provides an automatic Occam penalty factor for complex models with many parameters (MacKay, 1991). Thus, over the course of inference, we will get a set of hypotheses that tradeoff complexity with model fit, as shown in Fig. 1b.

Likelihood functions. To compute the likelihood, $p(\mathcal{D}_{0:b} | m, \theta) = \prod_{i \in \mathcal{D}_{0:b}} p(y_{1:T}^i | \xi^i, m, \theta)$, we consider two strategies. If the latent dynamics are a deterministic function of the initial conditions, z_0 , then we can use $p(y_{1:T} | z_0, m, \theta) = \prod_{t=1}^T p(y_t | z_t, m, \theta)$ where $z_t = m_\theta^t(z_0) = m_\theta(\dots(m_\theta(z_0)))$ is z_0 pushed through the forwards model t times. If the latent dynamics are stochastic, we can use the particle filter method of Algorithm 4 to approximate $p(y_{1:T} | z_0, m, \theta) = \int \prod_t p(y_t | z_t, m, \theta) p(z_t | z_{t-1}, m, \theta) dz_{1:T}$.

For some problems (such as NEURONBENCH), individual trajectories are very noisy, so a per-time step likelihood $p(y_t | z_t)$ is not meaningful. In such cases, we convert the trajectory into a set of global summary statistics, $s_j(y_{1:T})$, and use a trajectory-level likelihood of the form $p(y_{1:T} | m, \theta) = \prod_{j=1}^J p(s_j(y_{1:T}) | m, \theta)$, as is standard in the simulation based inference literature (Deistler et al., 2025). In Section A.4 we present some initial results on learning these summary statistics $s(y)$ as well as the model itself.

Expanding and shrinking the hypothesis space. SMC can update the posterior over hypotheses (models) given observations. However, in the $\mathcal{M}$ -open case (Bernardo & Smith, 1994; Kelter, 2020), we may need to expand the hypothesis space to account for a novel mechanism. To do this, we use a *predictive check*, i.e., a held-out interventional forecast (c.f., (Kelter, 2020)). If the error is too large, MDA *expands* the hypothesis space by prompting the LLM to suggest a novel unnamed mechanism (which is endowed with broad (“uninformative”) priors). (This is analogous to the Breaker–Builder method of (Buehler, 2026), and is how MDA can create new knowledge, overcoming a limitation of pure LLM-based discovery (Zahavy, 2026).) Conversely, if the posterior has confidently identified a model that fits well, we reduce the number of hypotheses, to prevent a proliferation of near duplicates, which diminishes performance. See Section A.2 for more details on MDA’s meta-controller.

Experiment design. We choose the experiment whose outcome is most informative about which hypothesis is true: $\xi^* = \arg \max_{\xi \in \Xi} I(M; Y_\xi | \mathcal{D})$ (Lindley, 1956; Box & Hill, 1967; Chaloner & Verdinelli, 1995; Rainforth et al., 2024). This is called the Value of Information (VoI) for an experiment. For the case of deterministic latent dynamics, and Gaussian observation noise, we can derive a simple analytic expression for the VoI, shown in Eq. (5). This picks the design with highest posterior-predictive *variance* of the outcome. Since the per-structure parameter posteriors are usually fairly concentrated, this variance is dominated by cross-*model* disagreement. We can optimize the VoI for small design spaces by simply enumerating each choice and scoring it. For larger continuous spaces, we use CMA-ES (Hansen, 2016). As baselines, we also consider random designs and LLM-proposed designs (as in (Gupta et al., 2025; Wiemann et al., 2026)).

Prediction. Once we have accumulated the full dataset $\mathcal{D}_{0:B}$, and created the posterior over hypotheses, $p_B = p(m | \mathcal{D}_{0:B})$, we evaluate the model in terms of its ability to forecast the outcome of novel experiments. For simplicity, we focus on predicting the posterior mean of each scalar output, $E[Y | \xi, \mathcal{D}_{0:B}, p_B]$, which is optimal when using ℓ_2 loss. We consider 3 methods:

- **Bayes-forecast**, $E[Y | \xi, \hat{m}]$ with $\hat{m} = \arg \max_m p_B(m)$ being the MAP model.⁶

⁶In general, the Bayes-forecast can use the full Bayes model average $p(Y | \xi, \mathcal{D}_{0:B}) = \sum_m \int p(m, \theta | \mathcal{D}_{0:B}) p(Y | m, \theta, \xi) d\theta$, as proposed in (Self & Cheeseman, 1987). For example, suppose we want to predict the expected number of neuron spikes n_c at input current level c , as required in NEURONBENCH

- **LLM-forecast**, $E[Y \mid \xi, \hat{m}]$ with $\hat{m} = \text{LLM}(\mathcal{D}_{0:B})$ being an LLM-generated model, created using standard code synthesis methods. (This is the approach used in (Wiemann et al., 2026).)
- **ICL-forecast**, $E[Y \mid \xi, \mathcal{D}_{0:B}]$: this is an in-context LLM-based predictor that conditions on the collected data and directly predicts the expected output, without using any kind of explicit model (c.f., (Lee et al., 2026)).

4 EXPERIMENTAL RESULTS

In this section, we summarize some of our experimental results on various benchmarks from physics, biology and chemistry (see Table 3). We show that the MDA method reduces held-out interventional predictive error much faster (in terms of number of experiments) than the baselines. We give more details in Section C, Section D and Section E.

Common protocol. Every benchmark is run through the *same* design loop for $B \leq 8$ experiments. At each step the agent selects the next experiment with a design function f_{design} — random, LLM-proposed, or Bayesian VoI — and after each step we forecast held-out interventional outcomes with a forecaster f_{predict} (Bayes-, LLM-, or ICL-forecast; Section 3) and score them by mean-squared error against the ground truth. We report this held-out error as a function of the number of experiments; because all datasets are synthetic and the true model is known, in some domains we additionally check whether the recovered model is symbolically equivalent to the truth. The two canonical agents are “MDA” (VoI design + Bayes-forecast) and the “LLM agent” (LLM design + LLM-forecast).

4.1 FORCEBENCH: DISCOVERING FORCE LAWS

Benchmark. In this section, we give a brief description of FORCEBENCH, which is our wrapper on top of the DISCOVERPHYSICS benchmark from (Wiemann et al., 2026). (We do not change the underlying benchmark, merely the interface, to make it compatible with our other benchmarks.) FORCEBENCH requires an agent to infer an unknown but novel force law governing the behavior of two or more particles in a 2d space. The agent can control the initial location and velocity of one of the particles, as it is launched, as well as a few other environment parameters. (In practice we discretize the design space into a fixed menu of 13 different combinations, listed in Table 5.) The performance of the learned model is assessed on a test set which probes the model’s predictive performance in novel experimental settings beyond the training set. Following the paper, we report this in terms of the normalized MSE, ($\text{nMSE} = \text{MSE}/\text{test-trajectory variance}$). See Section C.1 for further details.

Modeling assumptions. The agent assumes the unknown force can be represented as a Green’s function F , and asks the LLM to propose various candidates (see Section G for details of the prompt). It then derives the acceleration using Newton’s law, and integrates this to get velocity, and then integrates this again to generate the trajectory. It assumes the likelihood $p(y_{1:T} \mid \xi, F, \theta)$ is Gaussian, as in Eq. (3), and then does posterior inference over F and experiment design following the MDA recipe.

Data efficiency experiments. In Fig. 2, we show the performance of MDA vs the baseline LLM agent aggregated over all six of the two-particle worlds (see Fig. 7 and Fig. 8 for the performance plots for all 11 worlds). For each of the 6 worlds, we sample 3 random initial conditions, and roll out 3 trajectories per IC. Both agents use the same design space, and for the LLM they either use Opus 4.7 (the best model reported in (Wiemann et al., 2026)) or the cheaper DeepSeek-v4 Pro. On the left we plot the nMSE of the forecast for up to $B = 8$ steps.⁷ On the right we plot the fraction of

discussed in Section 4.3, where c is specified as part of the experiment design ξ . The posterior mean can be approximated from the weighted set of particles using $\hat{n}_c = E[s \mid \xi = c, \mathcal{D}] = \sum_i p(m_i, \theta_i \mid \mathcal{D}) s(m_i(\xi = c, \theta_i))$, where $s(z_{1:T})$ is the number of spikes in trace $z_{1:T}$, and $z_{1:T} = m_i(\xi, \theta_i)$ is the deterministic output of running model m_i with parameters θ_i on input $\xi = c$.

⁷For the Opus LLM baseline, we also run their agent in its native “unthrottled” mode, in which it performs multiple experiments per step. Thus 16 rounds of their agent performs ~ 41 experiments on average. The nMSE of 0.013 we get using this method matches the 0.01 reported in their paper, validating our experimental pipeline.

ForceBench (DiscoverPhysics protocol, all 6 worlds): MDA lifts even a cheap base model (DeepSeek) to near-Opus accuracy in a few experiments, while the pure agent stays weak

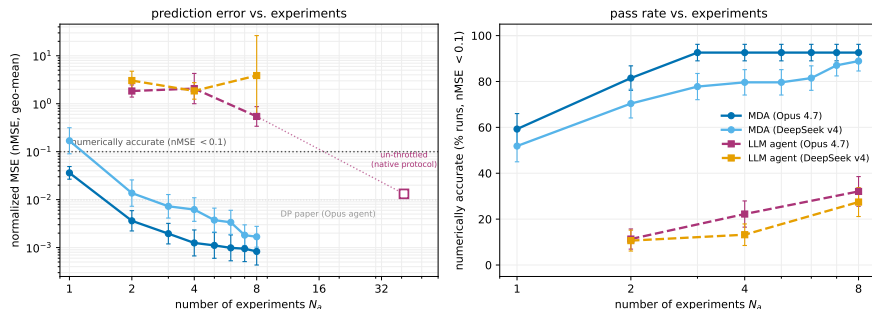


Figure 2: **Data efficiency on FORCEBENCH, aggregated over all six two-particle worlds.** Left: we plot nMSE (geometric mean over the 6×9 runs) vs number of experiments. Error bars are ± 1 standard error. The red square is the result of the “unthrottled” baseline agent, and matches the paper. Right: we plot fraction of runs where nMSE drops below the 0.1 threshold. See text for details.

runs where the prediction “passes”, following the paper’s definition of a pass as $\text{nMSE} \leq 0.1$. (We exclude the paper’s textual explanation criterion as part of the definition of “pass” because we found it to be unreliable; see Section C.7 for discussion.) From both plots we see that MDA is substantially more data efficient than the LLM baseline, and that Opus is better than Deepseek. See Table 8 for a list of the laws discovered by each agent after $B = 8$ experiments.

Example: Yukawa world. As a concrete example, we consider YUKAWA world, whose force law has the form $F = q_i q_j K_1(r/\lambda)/\lambda$, where K_1 is the modified Bessel function and $\lambda = 2$. The agent can choose the initial launch radius r_0 and speed v_0 of the target particle. The screened kernel $K_1(r/\lambda)/\lambda$ and the power laws that fit its short-range behaviour are nearly identical for $r \leq \lambda$ and diverge only at longer range (where the screening has decayed), so a probe must reach *past* the screening length to break the tie: a short-range launch leaves the candidate trajectories indistinguishable, while a long-range launch makes the true Yukawa fan out from its near-misses (visualized in trajectory space in Fig. 14, Section C). By maximizing the VoI, the agent therefore designs *long-range* probes.

The effect of the long range experiment triggers an “aha” moment for the agent. This is visualized in the Pareto curve in Fig. 1b which plots models on the accuracy–complexity frontier. With only short-range data, the true kernel sits mid-frontier with no edge over its near-misses; only after an informative long-range probe is added does the frontier shift down, making the true model drop to the convex corner — the moment where the agent truly “groks” the concept. (Note that the x -axis in Fig. 1b is a *Bayesian* description length, $-\log_2 p(m \mid \mathcal{D})$ — the posterior code-length of each candidate law. Unlike a purely *syntactic* complexity, such as the Halstead metric used in (Kasenberg et al., 2026), this is *data-dependent*: it rewards a law only to the extent the evidence supports it.)

4.2 CHEMBENCH: DISCOVERING ENZYME-KINETIC RATE LAWS

Benchmark. In this section, we briefly describe CHEMBENCH, which is our wrapper on top of ACTIVESCI BENCH-CHEM from (Kabra et al., 2026). (We don’t change the underlying benchmark, just the interface, to make it compatible with our other benchmarks.) The problem is to learn a function mapping seven controllable inputs (substrate, inhibitor, second substrate and product concentrations, enzyme loading, temperature and pH) to a reaction rate r : $r = f(C_A, C_I, C_B, C_P, \text{Enz}, T, \text{pH}; \theta)$. See Table 1 for some examples. The experimenter gets to set the 7 input variables, $\iota = x_0$, and observes the scalar response. Note that this problem is a special case of our SSM setup, since there is no temporal evolution. Following the paper, We measure performance using the held-out root-mean-squared log-error (RMSLE). If this is below $\epsilon=0.01$, we say the law is “numerically exact”. We also check for symbolic equivalence with the truth (which they call Structural Accuracy) using sympy. See Section D.1 for further details.

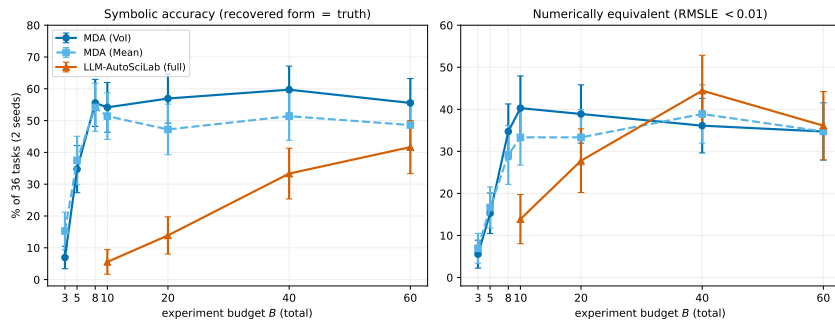


Figure 3: **Data efficiency curves for CHEMBENCH.** We plot mean performance ± 1 SE over the 36 tasks, two seeds. **Left:** symbolic accuracy. **Right:** numerical equivalence (EXACC, RMSLE < 0.01 ; the head-to-head Table 9 additionally reports the released benchmark’s own looser 0.05 threshold, for comparability with its published numbers). MDA (Vol) rises to its ceiling within ~ 8 experiments and leads on symbolic accuracy at every budget; LLM-AUTOSCI LAB only catches up by $B = 60$. On numerical accuracy, LLM-AUTOSCI LAB catches up sooner, but this is because of overfitting (see Table 1).

Modeling assumptions. The agent assumes the unknown function f can be represented as an algebraic equation, and asks the LLM to propose various candidates (see Section G for details of the prompt). It assumes a Gaussian likelihood with multiplicative noise, $p(y|\xi, f, \theta) = \mathcal{N}(y | f(\xi, \theta), \sigma_{\text{rel}}, f(\xi, \theta))$, to match the benchmark’s noise model, and its RMSLE metric. Given this model, the agent does posterior inference over f and experiment design following the MDA recipe.

Vol optimization. MDA by default uses Vol to pick the design. However, following (Kabra et al., 2026), we also create a baseline which we call MDA (MEAN), which picks the design of highest posterior-mean rate — an exploit/peak-seeking acquisition, as in Bayesian optimisation — rather than the most model-discriminating one: $\xi_{\text{mean}}^* = \arg \max_{\xi \in \Xi} \mathbb{E}_{m, \theta | \mathcal{D}}[r(\xi; m, \theta)]$. Both designs are one-step (myopic); they differ only in the objective — exploit the predicted rate versus discriminate between mechanisms. We maximise either objective over the continuous 7-D design box Ξ in one of two ways: by Monte-Carlo (draw $n_c=48$ candidate designs — log-uniform on the concentration/enzyme axes, uniform on T/pH — and take the argmax) or with CMA-ES (Hansen, 2016) over the box at the *same* 48 objective evaluations. We find the latter method is significantly better, so we use it by default.

Data efficiency experiments. In this section, we compare MDA to LLM-AUTOSCI LAB from (Kabra et al., 2026), which was the previous SOTA on this benchmark. Following their experiment protocol, we use a stratified 36-task subset (12 domains $\times$ easy/medium/hard) at two seeds. Figure 3 shows the data efficiency learning curves, which show that MDA is far more sample-efficient: it reaches its ceiling within about 8 experiments (overall SA $\approx 56\%$), whereas LLM-AUTOSCI LAB only reaches SA $\approx 42\%$ by $B = 60$.⁸ See Section D.2 for more results.

Example. Table 1 shows the laws recovered on three representative domains. MDA returns *interpretable mechanisms* — exactly the true form for substrate inhibition, and the correct inhibition/saturation structure elsewhere (although sometimes with a spurious extra factor). LLM-AUTOSCI LAB’s PySR instead returns numerically-fit but mechanistically *meaningless* expressions — nested $10^{a \log(\cdot)}$ and stretched-exponential forms — that can score a low RMSLE (even 0.001 on the hard noncompetitive domain) while being symbolically wrong: the high-exact/low-symbolic pathology discussed in (Kabra et al., 2026).

⁸Our result of 42% SA is higher than the 35.1% SA score they report in their paper, because we replaced their use of gpt-4o-mini with Opus 4.7, to be comparable to MDA. Note, however, that we stick to Qwen2.5-7B for the adaptive ensemble used by their code. (We also verified that dropping this ensemble component substantially hurt performance of their method.)

Domain	True law	MDA recovers	LLM-AUTOSciLAB recovers
substrate inhib. (easy)	$\frac{k \text{Enz } C_A}{K_m + C_A + C_A^2/K_i}$	<i>same form</i> ✓ (.007)	$\text{Enz} (a r^{C_A} + \dots) \times (.23)$
noncompetitive (hard)	$\frac{k \text{Enz } C_A}{(1 + C_I/K_i)(K_m + C_A)}$	Hill $\times$ noncomp. $\approx$ (.018)	$10^{0.87 \log(0.5 \sqrt{\text{Enz}/(C_I + \dots)})} \times (.001)$
Michaelis–Menten (easy)	$\frac{k \text{Enz } C_A}{K_m + C_A}$	+ spurious $e^{-E_a/RT} \approx$ (.017)	$10^{0.43 \log(\text{Enz } T^{0.37}/\cdot)} \times (.015)$

Table 1: **Representative recovered laws** (best config, $B=60$). Parenthesised value is held-out RMSLE; ✓ = symbolic form recovered, $\approx$ = correct structure with a spurious extra factor, $\times$ = mechanistically wrong. MDA recovers the mechanism in every case; LLM-AUTOSciLAB’s PySR fits the numbers with unphysical expressions — low RMSLE, no mechanism (the hard noncompetitive row scores RMSLE 0.001 yet is symbolically meaningless).

4.3 NEURONBENCH: DISCOVERING ION-CHANNEL MECHANISMS

Benchmark. We design a new benchmark, NEURONBENCH, by creating 6 “mystery neurons”, based on the generalized Hodgkin-Huxley (HH) model, which are a set of nonlinear ODEs for describing the spiking behavior of neurons (see Section E.2 for details of HH models). Each mystery neuron has incoming current represented by $I_{\text{Na}} + I_K + I_L + I_Z$, where Na is the sodium channel, K is the potassium channel, L is the leak channel, and Z is a novel membrane mechanism that we design, in order to prevent the LLM from simply recalling the model from memory (in most worlds an added channel, but in one a modification of the existing Na^+ channel rather than a new current). The experimental protocol allows the agent to specify the input signal (an electrical current) over time. We assume this is chosen from a set of 9 templated signal shapes. In addition, the agent can optionally apply 3 different kinds of ion channel blockers that change the underlying mechanism (see Section E.3 for details). So the total design space has $9 \times 4 = 36$ discrete options.

Modeling assumptions. The agent assumes the data can be represented by some kind of HH model m , and asks the LLM to propose various candidates (see Section G for details of the prompt). It then converts this into an ODE that defines the deterministic dynamics $p(z_t | z_{t-1}, x_t) = \delta(z_t = m(z_{t-1}, x_t; \theta))$, where the latent state is $z_t = (V(t), \text{gating variables})$ defined in Eq. (24). Finally it integrates this ODE over time to get a candidate trajectory $z_{1:T}$ from which the noiseless voltage trace $V(1:T)$ can be extracted. However, rather than evaluating the ability of the model to exactly match an observed trace (which can be very “wiggly” and hard to predict, even in the deterministic regime), we reduce the trace to a summary statistic s , and use a synthetic likelihood (Deistler et al., 2025) of the form $p(s(y_{1:T}) | \xi, m, \theta)$. For example, if $s(y)$ is the number of spikes in the trace y , we can use the Poisson likelihood $p(s(y_{1:T}) | \xi, m, \theta)$ in Eq. (32). We then do posterior inference over m and experiment design following the MDA recipe.

Data efficiency experiments. In Fig. 4, we show test error vs number of experiments for all six worlds. For each world the LLM is shown only the phenotype (the observable signature, not the mechanism) and proposes 2–5 candidate channels, which we map onto a shared channel library; the truth is sometimes *not* proposed — e.g. on Z-REBOUND the LLM omits the low-threshold inward current — a genuine $\mathcal{M}$ -open miss that we keep representable so the residual can reopen the pool. MDA then runs Poisson-evidence selection over that pool with VoI-designed experiments. We see that the Bayes-forecaster (blue) is significantly better than the in-context forecaster (purple) on every world; and within the Bayes-forecaster family, VoI and LLM-proposed experiment design perform *similarly* and both beat random design.

Stochastic extension. The worlds above use a deterministic Hodgkin–Huxley forward model, which can be represented by a deterministic ODE, so the likelihood is closed-form. In Section F.1 we make the latent dynamics *stochastic* (by adding finite-channel gating noise), turning the model into an SDE. We call this benchmark NEURONBENCHSTOCH. The corresponding likelihood $p(y | \xi, m, \theta)$ is now intractable — the one regime none of our other benchmarks reach. In Section F.1, we show that a deterministic likelihood gives poor results (the method confidently selects the wrong model), whereas a particle filter approximation to the marginal likelihood (Algo-

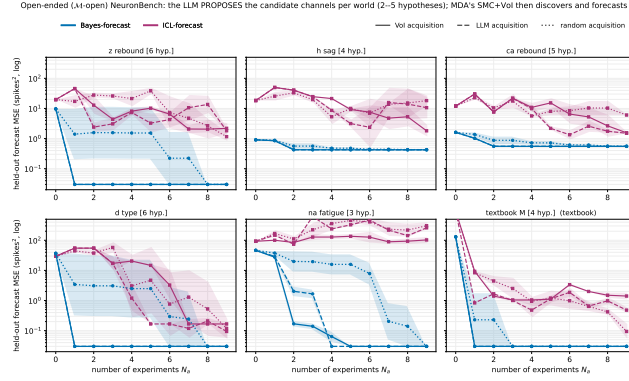


Figure 4: **Data efficiency on all six NEURONBENCH worlds** (Opus 4.7). For each world the LLM proposes the candidate channels from the phenotype (2–5 hypotheses, shown in the panel title), and then MDA designs experiments to confirm or refute these hypotheses. After each experiment, we plot the forecast accuracy on the test data. Colour = forecaster (blue Bayes-forecast, purple in-context forecast), line style = acquisition (VoI solid, LLM dashed, random dotted); held-out interventional forecast MSE (spikes², log) vs. number of experiments; shaded bands are ± 1 SE over 3 random initial conditions. The Bayes-forecaster dominates the in-context forecaster everywhere; within the Bayes family VoI and LLM design are similar and both beat random.

rithm 4) gives the correct results. Unfortunately running a particle filter inside the tempered SMC algorithm (needed to compute the evidence) — which is in turn nested inside of the SMC algorithm over models — is very slow. Fortunately we show that we can learn a suitable summary statistic function (using a 1d convolutional neural network), which speed things up by $\sim 10^4 \times$. Furthermore, because the summary is *learned*, this simulation-based likelihood adapts to whatever mechanisms the LLM proposes, rather than needing a hand-crafted feature per hypothesis.

5 RELATED WORK

- **Causal models for interventional prediction.** Predicting “what if” questions using causal models is discussed at length in (Pearl, 2009). Recently Richens & Everitt (2024) proved that an agent that can robustly predict across a full range of interventions (distribution shifts) must have implicitly learned a causal world model, and give a method to extract it from the predictor. We instead explicitly represent the causal model, so that we can leverage prior knowledge from LLMs (Kiciman et al., 2024; Ban et al., 2025), reason over our uncertainty using Bayesian methods, and provide an *interpretable model* to the user.
- **Bayesian experimental design.** Choosing the most informative experiment is the classical model-discrimination objective of (Lindley, 1956), reviewed in (Chaloner & Verdinelli, 1995; Rainforth et al., 2024); modern work scales it with amortised and gradient estimators (Foster et al., 2021). Our VoI is the Lindley objective read off a two-level SMC posterior, as derived in Eq. (5).
- **Simulation-based inference.** When the likelihood is intractable, SBI learns it or the posterior from simulations (Cranmer et al., 2020), and learned/embedding summary statistics are a whole subfield of their own (Fearnhead & Prangle, 2012; Chen et al., 2021; Radev et al., 2022).
- **LLMs for scientific discovery.** LLMs have been used to propose scientific hypotheses in many papers, including from static datasets (Romera-Paredes et al., 2024; Wahl et al., 2026; Kasenberg et al., 2026; Aygün et al., 2026; Xie & Wilson, 2026) as well as actively collected datasets from agent-designed experiments (Piriyakulkij et al., 2024; Abhyankar et al., 2026; Elteto et al., 2026; Prystawski et al., 2026; Jagadish et al., 2026). Our work is in the latter camp, differing mainly in how we handle the $\mathcal{M}$ -open regime inside SMC, the diversity of domains, and comparing to SOTA LLM baselines rather than straw men such as random designs.
- **Benchmarks for interactive scientific discovery.** Various benchmarks evaluate agents that learn scientific laws by *interactive experimentation*: we build on DISCOVERPHYSICS (Wiemann et al., 2026) and ACTIVESCIBENCH-CHEM (Kabra et al., 2026), add our own NEURONBENCH, and may target others such as NEWTONBENCH (Zheng et al., 2026) in future.

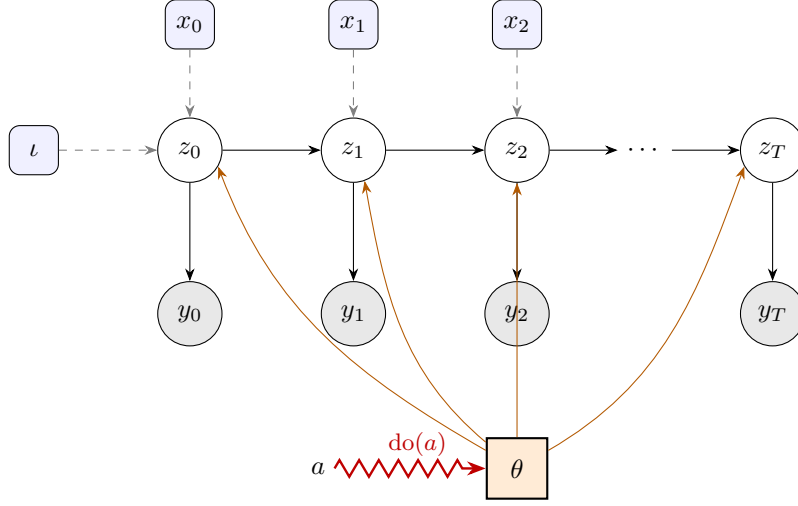


Figure 5: **The world as a controlled, intervenable state-space model** (Eq. (1)). A latent state z_t (white) evolves under the mechanism θ (orange; it parameterizes *every* transition) and emits a lossy, noisy observation y_t (grey) — in general only $y_{1:T}$ is seen. Optional exogenous inputs/covariates x_t (blue, dashed) and the initial condition ι (which sets z_0) are shifts in the *inputs* to a fixed mechanism. An intervention $\text{do}(a)$ is categorically different: the lightning bolt strikes θ itself, changing the mechanism to θ_a .

A METHOD: FURTHER DETAILS

A.1 MODELING ASSUMPTIONS

We assume the unknown dynamical system is represented by a state space model, as shown in Fig. 5. This corresponds to the following probabilistic model m :

$$z_{t+1} \sim p(z_{t+1} | z_t, x_t; \text{do}(a, \theta)), \quad y_t \sim p(y_t | z_t; \theta), \quad z_0 \sim p(z_0 | \iota). \quad (1)$$

where $\text{do}(a, \theta)$ represents the parameters of the system after applying intervention a . An experiment *design* ξ (the choice of initial conditions and control knobs the agent selects) fixes the induced intervention a ; we therefore write $\text{do}(\xi)$ for the intervened system when the design is the decision variable, as in the VoI objective below. The SSM assumption is without loss of generality, since any non-Markovian model can be converted to Markov form, as long as the latent state space is allowed to grow. If the latent dynamics are deterministic (an assumption we relax in Section F), then $z_t = m^t(z_0; \theta)$, where m^t is the forwards model iterated t times. If we additionally assume the observation noise is Gaussian, we have

$$p(y_t | z_t, m, \theta) = \mathcal{N}(y_t | m^t(z_0; \theta), \sigma^2) \quad (2)$$

Hence for conditionally independent observations, the likelihood of a parameter vector θ given model m and initial conditions z_0 is

$$p(y_{1:T} | z_0, m, \theta) = \prod_{t=1}^T \mathcal{N}(y_t | m^t(z_0; \theta), \sigma^2) \quad (3)$$

A.2 INFERENCE ALGORITHMS

This appendix gives the pseudocode for our outer active-inference loop (Algorithm 1, and the three nested inference layers of Section 3:

- Algorithm 2 computes posterior over LLM-proposed structures : $p(m | \mathcal{D}_{0:b}) \approx \sum_{i=1}^{N_m} w_i \mathbb{1}[m = m_i]$, where $w_i = p(m_i | \mathcal{D}_{0:b})$.
- Algorithm 3 uses per-structure adaptive-tempered SMC to compute the posterior over the parameters for each model: $p(\theta | m, \mathcal{D}_{0:b}) \approx \sum_{i=1}^{N_p} W_i \mathbb{1}[\theta = \theta_i]$, and the evidence $Z_m = p(\mathcal{D}_{0:b} | m)$

Algorithm 1 MDA discovery loop — the outer layer that wraps model inference (Algorithm 2) with value-of-information experiment design. It acquires *one* experiment per round (batch size 1) and re-infers $p(m \mid \mathcal{D})$ on the *growing* dataset.

```

def MDA( $B, C, \mathcal{D}_0; \tau_r, R_{\max}, \tau_p$ )  $\rightarrow p(m \mid \mathcal{D}_{0:B}), \{\mathbb{E}[Y \mid \xi_q, \mathcal{D}_{0:B}]\}_q$ 
1:  $p(m \mid \mathcal{D}_0) \leftarrow \text{MODEL-INFERENCE}(\mathcal{D}_0, C; R_m=R_{\max}, N_m=N_m^{\max})$   $\triangleright$  Alg. 2: initial pool + refinement
2: for  $b = 1 \dots B$  do
3:    $\xi^* \leftarrow \text{ARGMAX}_{\xi \in \Xi} \text{Vol}(\xi; p(m \mid \mathcal{D}_{0:b-1}))$ 
4:    $y_b \leftarrow \text{OBSERVE}(\xi^*); \mathcal{D}_{0:b} \leftarrow \mathcal{D}_{0:b-1} \cup \{(\xi^*, y_b)\}$ 
5:    $m^* \leftarrow \arg \max_m p(m \mid \mathcal{D}_{0:b-1}); r^* \leftarrow \text{RESIDUAL}(m^*, \mathcal{D}_{0:b}); p^* \leftarrow \max_m p(m \mid \mathcal{D}_{0:b-1})$   $\triangleright$  best form's fit  $r^*$  and class-posterior concentration  $p^*$ 
6:    $R_m \leftarrow R_{\max}$  if  $r^* > \tau_r$  else 0  $\triangleright \mathcal{M}$ -open: create new form when residual is too high
7:    $N_m \leftarrow N_m^{\min}$  if  $p^* > \tau_p \wedge r^* \leq \tau_r$  else  $N_m^{\max}$   $\triangleright$  ESS-adaptive pool: shrink if concentrated and well fitting
8:    $p(m \mid \mathcal{D}_{0:b}) \leftarrow \text{MODEL-INFERENCE}(\mathcal{D}_{0:b}, C; R_m, N_m)$   $\triangleright$  re-weight; explore if triggered; pool capped at  $N_m$ 
9: end for
10: return  $p(m \mid \mathcal{D}_{0:B})$  and forecasts  $\mathbb{E}[Y \mid \xi_q, \mathcal{D}_{0:B}]$  for held-out test designs  $\xi_q \sim Q$ 

```

Algorithm 2 Model inference over N_m model particles for R_m refinement rounds (so $O(N_m R_m)$ LLM calls — the dominant cost): a batch of LLM-proposed structures, or sequential refinement whose proposal conditions on the fit residuals of the *entire* particle pool (as in SMC-S (Piriyakulkij et al., 2024)), rather than a single ancestor particle (as in ModelSMC (Wahl et al., 2026)). PARAM-POSTERIOR is Algorithm 3.

```

def MODEL-INFERENCE( $\mathcal{D}, C; R_m, N_m, N_{\text{new}}$ )  $\rightarrow (p(m \mid \mathcal{D}), \hat{p}(\mathcal{D}))$ 
1: propose  $N_m$  structures  $\{m_i\}_{i=1}^{N_m}$  from the LLM given  $C$  (or enumerate the space)
   (stochastic dynamics, optional) auto-select the observation model  $o \in \{\text{PF (Alg. 4), feature synthetic lik., learned } s_\phi\}$  by cost-aware VoI (Eq. (11)); if  $o=s_\phi$ , (re)fit the learned summary on the current pool's prior-predictive simulations. PARAM-POSTERIOR's LOGLIK then uses  $o$ ; because  $s_\phi$  is only sufficient for the current pool, this refit repeats whenever the pool changes (here and at line 7), and a particle-filter spot-check flags an insufficient summary (App. A, "Sufficiency as the hypothesis pool grows").
2:  $(\cdot, \log \hat{Z}_i) \leftarrow \text{PARAM-POSTERIOR}(\mathcal{D}, m_i)$  for each  $i$   $\triangleright$  evidence  $\log \hat{p}(\mathcal{D} \mid m_i)$ 
3:  $p(m_i \mid \mathcal{D}) \leftarrow \text{softmax}_i(\log \hat{Z}_i)$   $\triangleright$  uniform structure prior
4: for  $r = 1 \dots R_m$  do  $\triangleright$  sequential refinement
5:    $\rho_j \leftarrow \text{RESIDUALS}(m_j, \mathcal{D})$  for every pooled structure  $m_j$   $\triangleright$  fit report over all prior attempts
6:    $\{m'_l\}_{l=1}^{N_{\text{new}}} \sim q(m \mid \{(m_j, \rho_j)\}_{j=1}^{\lfloor \text{pool} \rfloor}, C, \mathcal{D})$  from the LLM  $\triangleright$  propose a batch of  $N_{\text{new}} \ll N_m$  new structures jointly
7:   add  $\{m'_l\}$  to the pool; re-score (lines 2–3); evidence-prune the pool back to  $N_m$ 
8: end for
9: return  $p(m \mid \mathcal{D}) = \{(p(m_i \mid \mathcal{D}), m_i)\}, \hat{p}(\mathcal{D}) = \sum_i p(m_i) \hat{Z}_i$ 

```

- Algorithm 4 uses the bootstrap particle filter to compute the likelihood $p(\mathcal{D} \mid m, \theta)$ using $\hat{p}(y_{1:T} \mid m, \theta) = \prod_t (\frac{1}{N_z} \sum_i w_t^{(i)})$, where w_t^i is the weight of particle i . This is an unbiased estimate that plugs into the per-class tempered SMC (Algorithm 3) as its LOGLIK, which returns the model evidence $Z_m = p(\mathcal{D} \mid m)$. (In Section F.1 we discuss a way to avoid this inner PF likelihood approximation by learning a synthetic likelihood model.) If the latents are deterministic, we have $N_z=1$, and can use the likelihood in Eq. (3) without needing PF at all.

MDA meta-controller. The main MDA algorithm, shown in Algorithm 1, is a fairly standard sequential Bayesian experiment design loop. It uses an SMC subroutine to update the posterior over models $p(m \mid \mathcal{D}_{0:b})$ after obtaining $\mathcal{D}_b$ from the b 'th experiment. However, it adds two novelties, both of which turn out to be important for good performance in challenging domains (see Fig. 16).

Algorithm 3 Per-class adaptive-tempering SMC over N_p parameter particles. Returns particles, weights, and $\log Z_m$. $\text{ESS} = (\sum W_i)^2 / \sum W_i^2$; η is the target ESS fraction, R_p the number of rejuvenation moves per temperature rung, and the annealing schedule runs for at most J_p rungs — so LOGLIK is called $O(N_p R_p J_p)$ times (it is cheap, and LLM-free). ℓ_i is a *log*-likelihood; LOGLIK is the exact Gaussian for deterministic dynamics (see Eq. (3)) or, for stochastic dynamics, the auto-selected observation model — the particle filter of Algorithm 4, or the cheaper synthetic likelihood on fixed features / a learned summary s_ϕ (Eq. (8)). Parameter priors are uniform over the proposer’s declared bounds, so the random-walk Metropolis rejuvenation accepts on the tempered-likelihood ratio alone, with proposals outside the support rejected.

```

def PARAM-POSTERIOR( $\mathcal{D}, m$ )  $\rightarrow$  ( $\{\theta_i\}, \{W_i\}, \log Z_m$ )
1: sample  $\theta_i \sim p(\cdot | m)$  for  $i = 1..N_p$ ;  $W_i \leftarrow 1/N_p$ ;  $\lambda \leftarrow 0$ ;  $\log Z_m \leftarrow 0$ 
2:  $\ell_i \leftarrow \text{LOGLIK}(m, \theta_i)$   $\triangleright$  Gaussian (det.); else PF (Alg. 4) / feature- or  $s_\phi$ -synthetic lik. (stoch.)
3: while  $\lambda < 1$  do  $\triangleright \leq J_p$  annealing rungs
4:   pick  $\Delta\lambda \in (0, 1 - \lambda]$  by bisection so that  $\text{ESS}(\{W_i e^{\Delta\lambda \ell_i}\}) = \eta N_p$ 
5:    $\log Z_m \leftarrow \log Z_m + \log \sum_i W_i e^{\Delta\lambda \ell_i}$   $\triangleright$  evidence increment
6:    $W_i \leftarrow W_i e^{\Delta\lambda \ell_i} / \sum_j W_j e^{\Delta\lambda \ell_j}$ ;  $\lambda \leftarrow \lambda + \Delta\lambda$ 
7:   resample  $\{\theta_i\} \propto \{W_i\}$ ;  $W_i \leftarrow 1/N_p$ 
8:   for  $r = 1 \dots R_p$  do  $\triangleright$  random-walk Metropolis at temperature  $\lambda$ 
9:     propose  $\theta'_i \sim q(\cdot | \theta_i)$ ;  $\ell'_i \leftarrow \text{LOGLIK}(m, \theta'_i)$ ; set  $(\theta_i, \ell_i) \leftarrow (\theta'_i, \ell'_i)$  w.p.
        $\min\{1, e^{\lambda(\ell'_i - \ell_i)}\}$ 
10:   end for
11: end while

```

Algorithm 4 Bootstrap particle filter for $\log \hat{\ell} = \log \hat{p}(\mathcal{D} | m, \theta)$, the unbiased likelihood estimate that replaces the exact-likelihood line of Algorithm 3 (turning it into a pseudo-marginal sampler over $(\theta, z_{1:T})$). It samples N_z latent-state paths $z_{1:T}^{1:N_z}$ over the T observation steps; the transition is sub-stepped (for NEURONBENCH, the Euler–Maruyama discretisation of the Fox–Lu gating SDE, Eq. (33)). All the θ -dependence is in the transition (line 3); the observation noise σ is fixed, so the emission (line 4) does not depend on θ . The sum over the T observation steps is the loop (lines 2–7); the filter runs in $O(N_z T)$ time.

```

def PARTICLE-FILTER( $\mathcal{D}, m, \theta$ )  $\rightarrow \log \hat{\ell}$ 
1:  $z_0^j \leftarrow z_0$  for  $j = 1 \dots N_z$ ;  $\log \hat{\ell} \leftarrow 0$ 
2: for each observation  $t = 1 \dots T$  do
3:    $z_t^j \sim p(z_t | z_{t-1}^j, \theta)$   $\forall j$   $\triangleright$  sample the stochastic transition (Euler–Maruyama; no density needed)
4:    $w_t^j \leftarrow \mathcal{N}(y_t; g(z_t^j), \sigma)$   $\forall j$   $\triangleright$  emission on the observed coordinate  $g(z_t) = V_t$ ;  $\sigma$  fixed, no  $\theta$ 
5:    $\log \hat{\ell} \leftarrow \log \hat{\ell} + \log (\frac{1}{N_z} \sum_j w_t^j)$   $\triangleright$  incremental log marginal (running sum over  $t$ )
6:   resample  $\{z_t^j\} \propto \{w_t^j\}$ 
7: end for
8: return  $\log \hat{\ell}$ 

```

- It handles the M-open setting, in which we prompt the LLM to consider new hypotheses (beyond the current set of particles) if the residual error r^* of the current best (MAP) model, m^* , is above a threshold τ_r . (This can happen if the agent receives an informative but surprising observation.) In practice we do this by setting R_m to $R_{\max} > 0$; this enables R_m rounds in which we propose a batch of N_{new} new models, add them to the current pool, and then pick the top N_m based on their evidence. Following the SMC-S (Piriyakulkij et al., 2024), the (LLM-based) proposal kernel has the form $p_b(m_b | \{m_{b-1}^i\}, \mathcal{D}_{0:b})$, so we condition on the entire set of previous particles rather than just conditioning on a single ancestor particle, as is more commonly done. If the best residual r^* is below threshold, we set $R_m = 0$, which means we just update the weights of the current hypotheses, but do not invoke the LLM proposal to refine their structure.

- It adds an adaptive mechanism for choosing the number of hypotheses (particles) N_m : if the posterior probability p^* of the current best (MAP) model m^* is sufficiently high, and its residual r^* is sufficiently low, then we *reduce* the number of particles to $N_m^{\min}$ in Algorithm 2. This prevents a proliferation of near-duplicate hypotheses, and allows the posterior to concentrate. We re-expand the number of hypotheses if we fall outside of this convergence zone.

SMC parameters and computational cost. Table 2 shows the SMC parameters used in the experiments. The two model-level knobs N_m, R_m control the overall cost: each of the R_m refinement rounds of Alg. 2 re-fits *every* live structure (up to N_m) by a fresh N_p -particle adaptive-tempering SMC (Alg. 3).

quantity	symbol	FORCEBENCH	NEURONBENCH	CHEMBENCH	role
model particles	N_m	≤ 14	2–5	12	Alg. 2: structures / round
model rounds	R_m	8 ($=B$)	1	adaptive	Alg. 2: refinement rounds
new structures/round	N_{new}	4	1	4	Alg. 2: added per round if $R_m > 0$
parameter particles	N_p	200	200	100	Alg. 3: per-structure SMC
rejuvenation moves	R_p	3	3	3	Alg. 3: RW moves / rung
target ESS fraction	η	0.6	0.5	0.6	Alg. 3: resample trigger
max tempering rungs	J_p	80	adaptive	80	Alg. 3: schedule cap
latent particles	N_z	1	1/250	1	Alg. 4: likelihood estimation
$\mathcal{M}$ -open cap	$R_{\max}$	—	3	4	Alg. 1: max exploration rounds
residual threshold	τ_r	—	0.18	0.05	Alg. 1: $\mathcal{M}$ -open trigger
concentration thresh.	τ_p	—	—	0.9	Alg. 1: pool-shrink trigger

Table 2: **Default parameter settings for SMC** The model-level pair (N_m, R_m) dominates cost: R_m rounds $\times$ up to N_m structures $\times$ a N_p -particle inner SMC. The inner-SMC size N_p and rejuvenation count R_p match across rungs; the target ESS differs (0.6 vs. 0.5). For NEURONBENCH, we use $N_z = 1$ for deterministic model, and $N_z = 250$ for stochastic model.

A.3 ALGORITHMS FOR EXPERIMENT DESIGN

Deriving the VoI. We estimate the value of information for conducting experiment ξ , denoted $\text{VoI}(\xi)$, as follows. Model the (scalar) outcome as $Y_\xi = \mu(\xi) + \varepsilon$, with $\mu(\xi) = \mathbb{E}[y^* \mid M, \text{do}(\xi)]$ random over the posterior — a *single* deterministic forward simulation per particle, since the candidate dynamics carry no internal stochasticity (the property that made the likelihood Eq. (3) exact)— and $\varepsilon \sim \mathcal{N}(0, \sigma^2)$ independent noise. Conditioning on the mechanism class removes the epistemic spread, so $Y_\xi \mid M \sim \mathcal{N}(\mu_M(\xi), \sigma^2)$, whereas marginally $\text{Var}[Y_\xi] = \text{Var}[\mu(\xi)] + \sigma^2$. Under a Gaussian approximation both entropies are those of normals, $H[Y_\xi] = \frac{1}{2} \ln(2\pi e (\text{Var}[\mu(\xi)] + \sigma^2))$ and $H[Y_\xi \mid M] = \frac{1}{2} \ln(2\pi e \sigma^2)$, and the $2\pi e \sigma^2$ cancels in the difference:

$$I(M; Y_\xi \mid \mathcal{D}) = H[Y_\xi] - H[Y_\xi \mid M] = \frac{1}{2} \ln \left(1 + \frac{\text{Var}[\mu(\xi)]}{\sigma^2} \right), \quad (4)$$

strictly increasing in $\text{Var}_{p(M|\mathcal{D})}[\mu(\xi)]$: the *between-class* variance of the mean prediction is the value of information, monotonically. Maximising the mutual information therefore reduces to maximising this between-class disagreement, which we read off the class-evidence weights $p(m \mid \mathcal{D}) = Z_m / \sum_{m'} Z_{m'}$ (Algorithm 3 returns each $\log Z_m$) and the per-class particle means $\bar{\mu}_m(\xi) = \sum_i W_i^m \mu_{m,i}(\xi)$:

$$\xi_{\text{VoI}}^* = \arg \max_{\xi \in \Xi} \text{Var}_{p(M|\mathcal{D})}[\mathbb{E}(y^* \mid M, \text{do}(\xi))] = \arg \max_{\xi} \sum_m p(m \mid \mathcal{D}) (\bar{\mu}_m(\xi) - \bar{\mu}(\xi))^2, \quad (5)$$

with $\mu_{m,i}(\xi) = \mathbb{E}[y^* \mid m, \theta_i^m, \text{do}(\xi)]$ and $\bar{\mu}(\xi) = \sum_m p(m \mid \mathcal{D}) \bar{\mu}_m(\xi)$. Using per-class *means* $\bar{\mu}_m$ rather than noisy draws keeps VoI genuine epistemic disagreement — Lindley’s intuition (Lindley, 1956) that the best experiment is the one whose outcome current beliefs least agree on.

Contrast: the full predictive variance. An alternative acquisition keeps the within-class parameter spread instead of averaging it out. By the law of total variance the *full* two-level predictive variance

splits into the between-class term of Eq. (5) plus a within-class one:

$$\begin{aligned} \text{Var}_{p(m, \theta | \mathcal{D})} [\mathbb{E}(y^* | m, \theta, \text{do}(\xi))] &= \underbrace{\sum_m p(m | \mathcal{D}) (\bar{\mu}_m(\xi) - \bar{\mu}(\xi))^2}_{\text{between classes (mechanism disagreement)}} \\ &+ \underbrace{\sum_m p(m | \mathcal{D}) \sum_i W_i^m (\mu_{m,i}(\xi) - \bar{\mu}_m(\xi))^2}_{\text{within class (parameter uncertainty)}}, \end{aligned} \quad (6)$$

estimated empirically by the pooled particle variance $\sum_{m,i} w_{m,i} (\mu_{m,i}(\xi) - \bar{\mu}(\xi))^2$ with $w_{m,i} \propto p(m | \mathcal{D}) W_i^m$. Only the between-class term is collapsed by identifying the class. The within-class term is residual parameter uncertainty, which is negligible once the per-structure posteriors have concentrated; in this case, this full variance coincides with Eq. (5).

Optimising over a large design space. When the design space is large, we can use various gradient free optimizers to pick the design. For continuous spaces a common choice is CMA-ES (Hansen, 2016). For discrete spaces, we can use LLM-driven evolutionary search methods such as FunSearch (Romera-Paredes et al., 2024).

A.4 LEARNING THE SUMMARY STATISTICS FOR A SYNTHETIC LIKELIHOOD

Synthetic likelihoods. For simple problems we can ask how well the model predicts the entire observed trajectory, $y_{1:T}$, and measure its performance using a Gaussian likelihood, as in Eq. (3). But for more complicated signals, like neural voltage trace (or video!), it is common to use a synthetic likelihood of the form

$$p(y_{1:T} | m, \theta) \propto p(s(y_{1:T}) | m, \theta) = \prod_{j=1}^J p_j(s_j(y_{1:T}) | m, \theta) \quad (7)$$

where $s(\cdot)$ is a function that extracts J summary features from the entire observation vector $y_{1:T}$. (We omit the conditioning on ξ for brevity.) This can be used in lieu of the regular likelihood $p(y_{1:T} | m, \theta)$ in SMC. In this section, we discuss how to learn the summary function.

The collapse problem. Learning s by making the model “fit the data” is ill-posed: if s_ϕ is optimised to maximise the synthetic likelihood of the *observed* dataset, the objective is trivially maximised by a *constant* s_ϕ (which carries no information about θ or m), or by an s_ϕ that discards exactly the parameter-relevant signal. Any objective that only rewards “matching” collapses the summary. Something must force s_ϕ to *stay informative*.

The fix: learn s to be sufficient for what we infer, on a held-out simulated set. The established recipes all share one idea: use a bank of prior simulations $\{(\theta_i, m_i, y_{1:T}^{(i)})\}$ — a *held-out* set the summary must generalise across, not a single dataset to overfit — and train s_ϕ so that the latent can be *recovered* from the summary. This supervised signal both well-poses the problem and makes the constant solution impossible.

- **Semi-automatic ABC / regression** (Fearnhead & Prangle, 2012). Under quadratic loss the optimal summary is the posterior mean $s^*(y) = \mathbb{E}[\theta | y]$; one approximates it by *regressing* θ on features of y over the simulation bank, e.g. training a network $s_\phi(y) \approx \theta$ (Jiang et al., 2017). (This is called Approximate Bayesian Computation or ABC.) Note that a constant s_ϕ has *maximal* regression error, so the objective prevents collapse by construction.
- **Infomax / neural sufficient statistics** (Chen et al., 2021). Maximise the mutual information $I(s_\phi(y); \theta)$ over the simulated joint; a constant summary has *zero* mutual information, so it is the global *minimiser* of the objective, not a solution. Amortised-SBI summary networks (Radev et al., 2022) are trained jointly with a density estimator on (θ, y) pairs to the same end (see (Deistler et al., 2025) for a survey).

Both learn a low-dimensional $s_\phi(y)$ that is *predictive of* (a sufficient statistic for) the latent, and the held-out bank is what supplies the anti-collapse signal.

Example: 1d CNN for summarizing neuron voltage trace. In Section E.8, we give a concrete example of the regression approach, where we train a 1d CNN to map the neuron voltage trace $y_{1:T}$ to the label of the correct class (4 possible model types) and a corresponding model parameter (the max conductant g). The model has two output heads; the penultimate layer is a learned feature vector $s_\phi(y) \in \mathbb{R}^d$, which we use as the summary statistic.

Multivariate Gaussian likelihood. Since the components of $s_\phi(y)$ are correlated, we use a multivariate Gaussian likelihood, rather than the factored form in Eq. (7). The moments of this Gaussian are estimated by simulation — the standard synthetic-likelihood construction (Wood, 2010; Deistler et al., 2025). For each candidate (m, θ) we draw R traces $y^{(r)} \sim p(y_{1:T} \mid m, \theta)$, embed each, and set

$$p(s_\phi(y_{1:T}) \mid m, \theta) = \mathcal{N}(s_\phi(y_{1:T}) \mid \mu_{m,\theta}, \Sigma_{m,\theta}) \quad (8)$$

$$\mu_{m,\theta} = \frac{1}{R} \sum_r s_\phi(y^{(r)}) \quad (9)$$

$$\Sigma_{m,\theta} = \widehat{\text{Cov}}_r[s_\phi(y^{(r)})] + \varepsilon I, \quad (10)$$

with a small ridge ε for conditioning. (For latent variable models, we draw $(y^r, z^r) \sim p(\cdot \mid m, \theta)$, and discard z^r before fitting μ and Σ as above.)

Connection to MDA’s task-driven abstraction. MDA’s target is not full-state sufficiency but *query-sufficiency*: s_ϕ need only retain what is needed for the query distribution $\mathcal{Q}$. We can therefore train $s_\phi(y)$ to predict the *query outcomes* Y_ξ on held-out rollouts rather than to reconstruct the full state, giving a summary tuned to the downstream task. This is the observation-side analogue of learning *useful* latents: we learn not only which latent variables and mechanisms matter, but which parts of the raw observation to *attend to*.

Sufficiency as the hypothesis pool grows. A learned summary is only as informative as the simulated family it was trained on, so in the $\mathcal{M}$ -open loop sufficiency cannot be a fixed, global property — the pool the agent chooses among changes over time. Three points make it well-posed. (i) *It is task-relative and re-learned.* A statistic is sufficient for choosing among the *current* pool $\{m_1, \dots, m_K\}$ iff the likelihood ratios depend on y only through it; a classifier trained to separate the current members’ simulations has, at its optimum, exactly such a statistic (its log-odds), and adding a proposed hypothesis simply adds a class and re-fits — cheap, since the simulations are already drawn for the synthetic likelihood, and the summary dimension grows with the pool. (ii) *It can be amortised.* Training s_ϕ over the LLM’s prior-predictive mechanism *family* (rather than the current pair) generalises to in-distribution proposals without re-fitting. (iii) *It is checkable, with a model-agnostic anchor.* A particle filter needs no summary, so it is a sufficiency-safe gold standard: when a new mechanism is proposed, disagreement between the cheap synthetic likelihood and a spot-check particle filter (or a simulation-based-calibration failure) flags an insufficient summary and triggers re-learning (Section F.1 validates this synthetic-likelihood/particle-filter equivalence in the stochastic-NEURONBENCH setting). The approximation is therefore *monitored and repairable*, not assumed: a proposal whose signature lies in a data dimension s_ϕ discarded is invisible until re-learning, and the particle-filter anchor is what makes that safe. Co-training the observation abstraction with the mechanism expansion end-to-end we leave to future work.

Auto-selecting the observation model. When the latent dynamics are stochastic the likelihood is intractable, and there is a *menu* of approximations trading accuracy against compute: a bootstrap particle filter on the raw observation (assumption-free but costly, Algorithm 4); a synthetic likelihood (Eq. (7)) on a hand-crafted feature vector or on a learned summary s_ϕ (Eq. (8)) — cheap, but only as sufficient as the summary; or, when the process noise is negligible, a single deterministic rollout (Eq. (3)). No single choice is best for every discrimination problem, so we let the agent *choose* the observation model o exactly as it chooses the experiment. Writing $p_o(y \mid m, \xi)$ for the likelihood under observation model o , the agent picks

$$o^* = \arg \max_{o \in \{\text{PF}, \text{feat}, s_\phi\}} \frac{\text{MI}_o(m; y \mid \xi)}{\text{cost}(o)}, \quad \text{MI}_o(m; y \mid \xi) = H(w) - \mathbb{E}_{p_o(y \mid \xi)}[H(q_o(\cdot \mid y))], \quad (11)$$

where $q_o(m \mid y) \propto w_m p_o(y \mid m, \xi)$ is the model posterior formed under observation model o ’s likelihood (cf. Eq. (36)), and $\text{cost}(o)$ is its *compute* — the particle filter’s N_z latent trajectories vs.

a synthetic likelihood’s R simulations, so $\text{cost}(\text{PF}) \gg \text{cost}(\text{feat})$. This is Eq. (35) applied one level *up* — design over the *observation model*, not the experiment — so the agent defaults to the cheap summary and pays for the filter only where its discrimination clearly justifies the compute. We estimate MI_o by a *truth-free discrimination probe*: on the discriminating protocol, simulate single experiments from *each* candidate in turn and measure how often o ’s log-evidence gap identifies the generator, averaged over generators — never peeking at the truth.

The particle-filter spot-check. The discrimination probe averages over generators, so it scores whether o separates the pool *on average*; it cannot see a cheap summary’s *specific* blind spot. A learned s_ϕ trained on the mechanism family may confuse a single pair whose signatures collapse in its low-dimensional embedding while discriminating all the rest well — giving a deceptively high pooled score yet a confidently *wrong* answer on the world where that pair is the question. We therefore make the particle-filter anchor operational as a *spot-check*: whenever a cheap model $o \neq \text{PF}$ is selected, we re-score the collected data with *both* o and the particle filter (over the top few candidates by posterior) and compare their MAP models; on *disagreement* we fall back to the PF. The cheap summary is thus used only where it is *verifiably* sufficient — it agrees with the assumption-free filter — and the PF catches the rest. This is the safeguard that makes a fast, possibly-insufficient summary safe to deploy inside the $\mathcal{M}$ -open loop (Algorithm 2): the observation model, like the experiment, becomes something the agent *chooses* and *verifies* from data rather than a hand-set knob. Section F instantiates both — the cost-aware choice and the spot-check — on the electrophysiology benchmark, and reports how often each observation model is chosen and how often the spot-check overrides a cheap choice.

Benchmark	Domain	Model class	Observation space	Design space	Ref.
FORCEBENCH	Physics	ODE (force law)	2D trajectory (time series)	$\mathbb{R}^2 \times$ discrete (13-menu)	(Wiemann et al., 2026)
CHEMBENCH	Chemistry	algebraic rate law (sym. reg.)	single scalar (initial rate)	$\mathbb{R}^7$ (continuous)	(Kabra et al., 2026)
NEURONBENCH	Neuroscience	ODE (Hodgkin–Huxley)	1D voltage trace (time series)	input current $\times$ channel blocks	ours
NEURONBENCHSTOCH	Neuroscience	SDE (stochastic Hodgkin–Huxley)	1D voltage trace (time series)	input current $\times$ channel blocks	ours

Table 3: **Benchmarks used in this paper.** The three span three domains and several model classes — dynamical ODEs (FORCEBENCH, NEURONBENCH) and a *static* algebraic law (CHEMBENCH) — and very different observation and design spaces: from a 2D trajectory under a mixed continuous $\times$ discrete design (FORCEBENCH) to a *single scalar* rate under a continuous $\mathbb{R}^7$ design (CHEMBENCH). The same MDA engine (§3) is applied unchanged to all of them.

B EXPERIMENTAL RESULTS: FURTHER DETAILS

The datasets we use are listed in Table 3. We evaluate the forecasts for each agent α after k experiments as follows. For FORCEBENCH we first compute $\mu_{i,t,b}(\alpha) = E[Y_t^i | \xi^i, \mathcal{D}_{0:b}, \alpha]$ for each time step t of each test trajectory i , and then compute the mean squared error (MSE):

$$\text{MSE}_b(\alpha) = \frac{1}{N_{\text{test}}} \sum_{i=1}^{N_{\text{test}}} \frac{1}{T} \sum_{t=1}^T (\mu_{i,t,b}(\alpha) - \mu_{i,t,b}^*)^2 \quad (12)$$

where $\mu_{i,t,b}^*$ is the (noise free) expectation under the ground truth model. For NEURONBENCH we use $\mu_{i,j,b}(\alpha) = E[s_j(Y_{1:T}^i) | \xi^i, \mathcal{D}_{0:b}, \alpha]$ for each summary feature j , and compute the MSE by averaging over features instead of time steps. We then plot MSE vs b , for $b = 1 : B$, where $B = 8$ is the maximum number of experiments. We also plot MSE for $k = 0$, which is the performance just given $\mathcal{D}_0$, before any experiments are performed.

#	world	pairwise force magnitude $F(r, t)$	Comments
<i>Two-particle, central, radial (single fixed source at the origin)</i>			
1	GRAVITY	$k q_i q_j / r$	Simple attractive force
2	YUKAWA	$k q_i q_j K_1(r/\lambda)/\lambda, \lambda=2$	Screened (2D Helmholtz) kernel: $\sim 1/r$ at short range, exponentially suppressed beyond λ
3	COULOMB	$k q_i q_j / r^2$	Simple attractive force
4	OSCILLATOR	$k q_i q_j \cos(\omega t + \phi)/r$	Time-varying coupling that periodically reverses sign
5	FRACTIONAL	$k q_i q_j / r^{3-2\alpha}, \alpha = \frac{1}{2} (\equiv 1/r^2)$	Fractional Laplacian $-(-\nabla^2)^\alpha$ with $\alpha = \frac{1}{2}$
6	EXTRA-DIM	$k q_i q_j \Phi_{KK}(r)$	$1/r^2$ (short range) to $1/r$ (long range) transition, defined by the Kaluza–Klein image-sum kernel
<i>Non-radial or many-body (superposed background, or N mutually-interacting bodies)</i>			
7	CIRCLE	$k q_i q_j / r^{3-2\alpha}, \alpha = \frac{3}{4}$	Fractional Laplacian with ring of particles
8	ETHER	$k q_i q_j / r$	Central law + global drift, $a_i = -F \hat{r}/m_i + \alpha \hat{y}$
9	HUBBLE	$k q_i q_j / r$	Central law + position dependent Hubble flow, $a_i = -F \hat{r}/m_i + H(\mathbf{r}_i)$
10	DARK-MATTER	$k q_i q_j / r, q_j \in \{1, \underline{5}\}$	Hidden number of other particles
11	THREE-SPECIES	$k q_i q_j / r, q_j \in \{1, \underline{3}, -2\}$	3 hidden classes (one repulsive) + 5 neutral probes

Table 4: **The eleven public DiscoverPhysics worlds in a single notation.** $F(r, t)$ is the pairwise force magnitude between a receiver of charge q_i and a source of charge q_j at separation r (Green’s function of a 2D field equation; k a coupling, λ a screening length, α a fractional order, ω, ϕ a temporal modulation); $\Phi_{KK}(r)$ is the Kaluza–Klein image-sum kernel of a compactified extra dimension (an infinite tower of mirror sources that crosses over from $1/r^2$ at short range to $1/r$ at long range). Underlined quantities are *hidden* (worlds 10–11: the source charges are concealed; the task is to infer them).

C PHYSICS: FURTHER DETAILS

C.1 DETAILS ON THE BENCHMARK

FORCEBENCH (which is just a wrapper on DISCOVERPHYSICS from (Wiemann et al., 2026)) requires an agent to infer the unknown force law governing the behavior of two or more particles in a 2d space. Each particle i has an associated kinematic state: position $\mathbf{r}_i$, velocity $\mathbf{v}_i$, and a “generalized charge” $\mathbf{q}_i = (s_i, c_i)$, where s_i is the source charge, controlling how strongly particle i generates the field, and a response charge c_i controlling how strongly it feels the field generated by others. When $s_i = c_i$ for all particles, this reduces to a standard symmetric pairwise interaction. In this case, q_i might represent a charge (for electric fields) or a mass (for gravitational fields). The pairwise force takes the general form

$$\mathbf{F}_{i \leftarrow j} = F_{\text{mag}}(r_{ij}, \mathbf{q}_i, \mathbf{q}_j, t) \hat{\mathbf{r}}_{ij} \quad (13)$$

where $r_{ij} = \|\mathbf{r}_i - \mathbf{r}_j\|$ is the distance between the particles, and $\hat{\mathbf{r}}_{ij}$ is the unit separation vector from source to receiver (so $-\hat{\mathbf{r}}$ is attractive).

There are 11 different laws or worlds, shown in Table 4. We group them into 6 two-particle worlds, which follow a radial force centered on particle 1, and 5 “extra” worlds, which have slightly different semantics, as listed in the table.

From Newton’s second law, $\mathbf{F} = m\mathbf{a}$, we can derive the acceleration $\mathbf{a} = (a_x, a_y)$ of a particle as follows:

$$\mathbf{a} = -F_{\text{mag}} \hat{\mathbf{r}} / m \quad (14)$$

If there are multiple particles, we sum the forces:

$$\mathbf{a}_i = - \sum_{j \neq i} F_{ij} \hat{\mathbf{r}}_{ij} / m_i \quad (15)$$

From this, we can derive the velocity by integration, and hence generate the trajectory of each particle from its initial conditions.

The benchmark requires the agent to submit a Python function that returns the predicted trajectory. The function must satisfy the following signature:

```
def discovered_law(pos1, pos2, p1, p2, velocity2, duration, **params):
    ...
    return trajectory
```

Here `params` are free parameters of the law which can be fit to the collected data by the FORCEBENCH environment before it calls the above function. The agent can choose the initial position of particles 1 and 2, and the velocity of particle 2. (The velocity of particle 1 is fixed at $(0, 0)$.) The meaning of the control knobs p_1 and p_2 varies across the worlds: sometimes they represent masses, sometimes charges (see Table 4 for details).

The LLM baseline method from (Wiemann et al., 2026) uses an LLM to generate code which computes the acceleration function $\mathbf{a}$, from which it derives the trajectory by integration. In Table 8, we give examples of the generated code. In MDA, we instead estimate F_{mag} , and then derive $\mathbf{a}$ using Newton’s law in Eq. (14), which we pass to the integrator. (MDA also estimates its own parameters, using the posterior mean associated with the submitted model, rather than using the environment’s fitting function.) We could of course ask the LLM to generate F_{mag} instead, but this would be a different method to the one used in (Wiemann et al., 2026).

C.2 THE DESIGN SPACE

An experiment (action a) is a single probe launch in the benchmark’s own API: the probe is released from position $(r_0, 0)$ with velocity v , under two scalar coupling knobs (p_1, p_2) whose roles (source charge, probe inertia, ...) are part of what must be discovered.

The design space Ξ for TWOPARTICLEWORLDS is the fixed menu of 13 such launches in Table 5. This space was chosen by an LLM to cover the relevant dimensions. The design space for MULTIPARTICLEWORLDS is shown in Table 6. VoI (and the LLM/random acquisition baselines) selects one action per round. Note that these are discrete spaces, to make the VoI maximization problem simple.

action a	r_0	v (launch)	p_1	p_2	purpose
1–8	$\{1.5, 2, 3, 4, 5, 6, 8, 10\}$	$[0, 0]$ (radial drop)	1	1	radial profile (short→long r_0)
9–10	$\{2, 4\}$	$[0, 0.4]$ (tangential)	1	1	orbit shape (angular momentum)
11	4	$[0, 0]$	2	1	identify the role of p_1
12	3	$[0, 0]$	1	2	identify the role of p_2
13	4	$[0, 0]$	2	2	vary both knobs

Table 5: **The design space Ξ for the 6 TWOPARTICLEWORLDS** — a fixed menu of 13 probe-launch experiments. Each is a probe released from $(r_0, 0)$ with velocity v under coupling knobs (p_1, p_2) ; VoI selects one per round. The seed launch $\mathcal{D}_0$ is action 3 (the passive radial drop at $r_0=3$). Rows 1–8 sweep the radial force profile — the long $r_0 \geq 5$ drops probe where any screening has decayed, decisive for Yukawa/extra-dim (VoI selects $r_0=5, 6$; Figs. 14, 1b); rows 9–10 add angular momentum; rows 11–13 vary the two coupling knobs to identify their roles (source charge vs. probe inertia, $a=F/p_2$). The held-out interventional test set uses more extreme knob settings ($p_1 \in \{3, 4, 5\}$, $p_2 \in \{3, 5\}$).

C.3 INTERACTIVE APP

To make the task concrete, we built *PhysicsPlayground*, a self-contained web app that lets a reader play a simplified version of the game that the agent must solve. Figure 6 shows a screenshot. The top row is a *transduction puzzle* in the style of ARC-AGI but for a physical law: two *training* experiments (a launch radius r_0 and the resulting orbit $r(t)$, the raw trajectory the discovery algorithm fits) and two test forecasts — a launch at a new radius, and a launch under a *perturbed source* ($\text{do}(\text{mass} \times 2)$), the interventional “what if” the method targets. At the bottom of the screen is the playground, where the user can launch their own orbits, read the animated trajectories, and work out the force law. Finally they submit their forecast for each test launch in the top right, and they can then choose to reveal the truth to self-score.

world	system	each experiment sets	#	discovers
ether	central + drift $\vec{a}=(0, \alpha)$	5 orbiters, $r \in [3, 8]$, $v=2.8$	6	F, α
Hubble	central + $H\vec{r}$	5 orbiters, $r \leq 8$	6	F, H
circle	11-body ring	ring R , launch v , R -scaled t	6	exponent p
dark-matter	+ K hidden masses	continuous (x, y, v_x, v_y)	∞	# & loc. of masses
three-species	30 bg., hidden couplings	probe <i>direction</i>	4	couplings $\rightarrow$ species

Table 6: **Design spaces for the 5 MULTIPARTICLEWORLDS.** Unlike the fixed 13-launch menu of the two-particle worlds (Table 5), these are heterogeneous. Ether and Hubble use a fixed set of 5-probe orbiter launches on top of a central force plus a background term (a uniform drift α , or a Hubble expansion H); circle sweeps the radius of an 11-body self-gravitating ring with per-radius measurement times (a *wide*-ring sweep breaks the scale degeneracy that otherwise hides the force exponent); dark-matter designs a *continuous* tracer launch (x, y, v_x, v_y) by VoI to localise unseen point masses; three-species probes the 30-particle background from a few directions to recover each particle’s hidden coupling, then clusters the couplings into species (count k chosen by BIC). “#” is the number of candidate experiments (∞ = a continuous design box).

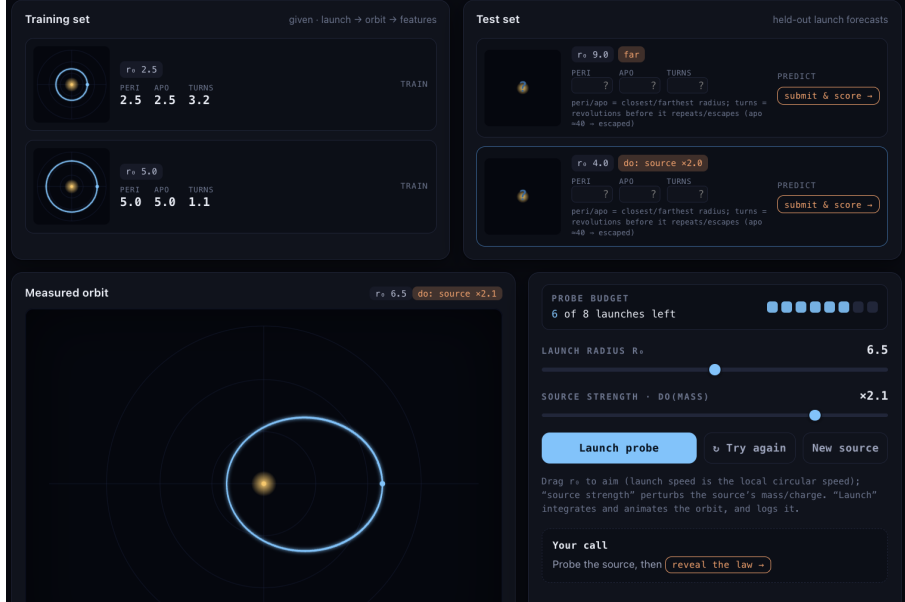


Figure 6: **App for FORCEBENCH.** The goal is to identify a hidden central-force law from a few probe orbits, then predict held-out launches — including one under a perturbed source. Training orbits (top left), held-out interventional test forecasts (top right), and the reader’s own budgeted experiment bench with an animated measured orbit (bottom). Available at <https://claude.ai/code/artifact/565fe6cc-a355-4c19-bf7e-b44e766cf87e>.

C.4 PARAMETERS AND THEIR PRIORS

On FORCEBENCH the candidate structures are *open ended*: the LLM proposes a force magnitude $F(r, q_i, q_j, t; \theta)$, where each model has its own free parameters and bounds. Each free parameter takes a uniform prior over the proposer-declared bounds; see Table 7. The joint prior factorises as

$$p(m, \theta) = p(m) \prod_{k=1}^{C_m} p_k(\theta_k) \quad (16)$$

where we use a *structure* prior of the form

$$p(m) \propto e^{-\lambda C_m} \quad (17)$$

where C_m is the number of free parameters of structure m , and $\lambda = 2.5$ is the Occam penalty per free parameter. Thus the posterior over structures is

$$p(m \mid \mathcal{D}) \propto Z_m e^{-\lambda C_m} \quad (18)$$

$$Z_m = \int p(\mathcal{D} \mid \theta, m) p(\theta \mid m) d\theta \quad (19)$$

where Z_m is the SMC marginal likelihood. The explicit $e^{-\lambda C_m}$ term is added as an additional regularizer, since on near-deterministic data ($\sigma=0.03$), a more flexible form can win Z_m by fitting the observation *noise*. (Note that penalizing the length of the *representation* of the function F — computed either by string length or Halstead complexity (Halstead, 1977) — did not work as well, since that ignores the flexibility of the underlying “elementary” functions that are used.)

We currently fix the observation noise to $\sigma = 0.03$. However, this could invite overfitting, since a more flexible model can win marginal evidence by driving residuals below that noise floor. The principled remedy is to be Bayesian about σ — put an inverse-gamma prior on the residual variance and *marginalize* it, giving a Student-t marginal likelihood $-(a + M/2) \log(b + \frac{1}{2} \text{SSE}(\theta))$ over all M residuals, whose scale is tied to the known noise floor via b . This removes the arbitrary fixed- σ dependence and, being scale-invariant in the residuals, no longer rewards fitting below a floor.

Empirically, however, σ -marginalization *alone* does not cure the overfitting, and can worsen model selection: with the hundreds of residuals these many-body worlds provide, the marginal likelihood over-rewards *any* reduction in SSE by a factor $\sim M/2$ (a Lindley-paradox-like effect), so a flexible form that absorbs the observation noise wins, and the Occam factor alone does not compensate. Fortunately the explicit prior on models, $p(m)$, to encourage simplicity suffices. This combination is robust and, on Hubble, converts a near-miss (the pool’s spurious time-modulated $1/r$, a small ε fitting the noise) into the clean $1/r$ result reported above.

Quantity	Symbol	Prior / value	Role
<i>Inferred: a candidate’s free coefficients θ (prior = Uniform over the declared bounds):</i>			
coupling strength	k	Uniform(0.01, 5)	force magnitude
screening length	λ	Uniform(0.5, 40)	Yukawa / range cutoff
radial exponent	p	Uniform(0.5, 2.5)	power-law falloff $1/r^p$
oscillation frequency	ω	Uniform(0.1, 6)	time-varying force
oscillation phase	ϕ	Uniform(0, 2π)	time-varying force
<i>Structure prior (Bayesian Occam over the parameter count):</i>			
number of free params	C_m	$p(m) \propto e^{-2.5 C_m}$	penalise flexibility
<i>Fixed (not inferred):</i>			
charge / inertia roles	q_i, q_j, m	$q_i=1$; p_1, p_2 set charge, inertia	driving force
integrator step	Δt	0.005 (symplectic)	forward model
measurement times	t	$\{0.5, 1, 1.5, 2, 3, 4\}$	readout grid
seed launch	$\mathcal{D}_0$	one passive drop at $r_0=3$	warm-start data
position noise	σ	0.03 (fixed Gaussian)	likelihood

Table 7: **Parameters and priors for the FORCEBENCH force-law rung** (the physics analogue of the ephys parameter conventions, App. E). The candidate laws are LLM-proposed, so θ is not a fixed list; each free parameter takes a uniform prior over the proposer’s declared bounds — the recurring parameters (those of the six ground-truth laws) and representative ranges are shown, and a candidate has $C_m=1\text{--}3$ free parameters. Only these coefficients are inferred; everything else is fixed. The structure prior $p(m) \propto e^{-2.5 C_m}$ is the Bayesian–Occam penalty of §4.1; the likelihood is a fixed- σ Gaussian on the probe positions at the measurement times, and each structure’s marginal evidence is read off an adaptive-tempering SMC ($N_p=200$ particles, target ESS 0.6, $R_p=3$ random-walk moves at half the particle SD per tempering rung).

C.5 LAWS DISCOVERED FOR TWOPARTICLEWORLDS

Table 8 shows the laws discovered by MDA and the pure LLM agent after $B = 8$ experiments on TWOPARTICLEWORLDS. Looking at the details of the discovered laws, we see that sometimes the result looks different from the truth but is mathematically equal. For example, on FRACTIONAL the truth is $F = k q_i q_j / r^{3-2\alpha}$ with $\alpha = 0.5$, and MDA proposes the simpler but equivalent expression $F = k q_i q_j / r^2$.

method	best submitted law	nMSE	%pass _{<0.1}	%≡
gravity — True law $F=k q_i q_j / r$, $k=0.16$				
MDA	$F=k * q_i * q_j / r$ $k=0.159$	4.4×10^{-5}	100	100
LLM	$a_x = -C * p_1 * x / (p_2 * r^2)$ $C=0.158$	3.8×10^{-1}	22	11
Yukawa — True law $F=k q_i q_j K_1(r/\lambda)/\lambda$, $k=0.16$, $\lambda=2$				
MDA	$F=k * q_i * q_j * k_1 (r / \text{lam}) / \text{lam}$ $k=0.152$, $\text{lam}=2.064$	8.3×10^{-4}	100	89
LLM	$a_x = F * dx / r$ $n=4.06$, $a=1.0$, $b=1.0$	3.3×10^0	33	11
Coulomb — True law $F=k q_i q_j / r^2$, $k=1$				
MDA	$F=k * q_i * q_j / r ** 2$ $k=0.986$	5.5×10^{-2}	56	67
LLM	$f = -p_1 * p_2 / (r^2 * r)$ $\text{eps}=0.002$	2.8×10^{-1}	22	89
oscillator — True law $F=k q_i q_j \cos(\omega t + \phi) / r$, $k=0.80$, $\omega=\pi/2$, $\phi=0$				
MDA	$F=q_i * q_j * k * \cos(w * t + \text{phi}) * \exp(-r / \text{lam}) / r$ $k=1.096$, $\text{lam}=4.548$, $w=1.569$, $\text{phi}=-0.035$	1.3×10^{-2}	100	0
LLM	$a_x = F * dx / r$ $G=0.01$, $n=5.0$, $a=1.0$, $b=1.0$	9.7×10^{-1}	33	0
fractional — True law $F=k q_i q_j / r^{3-2\alpha} (\equiv 1/r^2)$, $k=0.16$, $\alpha=\frac{1}{2}$				
MDA	$F=k * q_i * q_j / r ** 2$ $k=0.159$	2.5×10^{-6}	100	100
LLM	$a = -k * p_1 / (p_2 * r^2)$ $k=0.035$	1.1×10^{-1}	56	56
extra-dim — True law $F=k q_i q_j \Phi_{KK}(r)$, $R=0.5$				
MDA	$F=k * q_i * q_j / r ** 2 + \text{sigma} * q_i * q_j$ $k=0.254$, $\text{sigma}=0.023$	9.4×10^{-3}	100	89
LLM	$a_{\text{mag}} = k * p_1 / (p_2 * r)$ $k=0.163$	6.4×10^{-1}	22	22
grand mean MDA		9.2×10^{-4}	93	74
grand mean LLM		5.4×10^{-1}	31	31

Table 8: **Laws discovered for the six FORCEBENCH worlds by MDA and the pure LLM agent** (Opus 4.7, $B = 8$ experiments; regenerated from the same runs as Fig. 2). For each world we show the true force law and each method’s best (lowest-error) submitted law with its fitted parameters (MDA submits a force magnitude F ; the LLM writes an acceleration line, of which we show a_x or the radial a , whichever is simpler). *nMSE* is the DiscoverPhysics normalized MSE (MSE/test-trajectory variance), geometric mean over the 9 runs. *%pass_{<0.1}* is the fraction of runs with *nMSE* below the paper’s 0.1 threshold (dropping their explanation score); *%≡* is the fraction *exactly* form-equivalent to the true law (form-MSE < 10^{-3} at unit charges, isolating the form from the charge-role).

We score each run two ways. The *exact-form* rate (*%≡*) is the fraction of runs whose submitted law is functionally equivalent to the truth after removing parameters and constants; we judge this by the *form-MSE* at unit charges ($p_1=p_2=1$), which isolates the radial/temporal form from the charge-role handling, and call a run exact when this form-MSE is below 10^{-3} . The *numeric* rate (*%pass_{<0.1}*) instead uses the DiscoverPhysics benchmark’s own criterion: the *normalized* MSE, $\text{nMSE} = \text{MSE}/\text{Var}$ where *Var* is the variance of the held-out test trajectories (accounting for the different total particle travel across worlds), counting a run as passing when $\text{nMSE} < 0.1$. We drop the benchmark’s second gate — an LLM-judged explanation score ≥ 0.9 — because we found it unreliable (Section C.7).

Under these metrics MDA recovers the exact form in 74% of runs and passes numerically in 93%, versus 31% and 31% for the LLM agent *budget-matched* to one experiment per round (the same $B = 8$ budget MDA uses). This gap is one of *data efficiency*, not capability. The DiscoverPhysics benchmark lets an agent submit a *batch* of experiments each round, so its nominal 16-round budget collects far more than 16 experiments; run un-throttled in its native batched protocol, our LLM

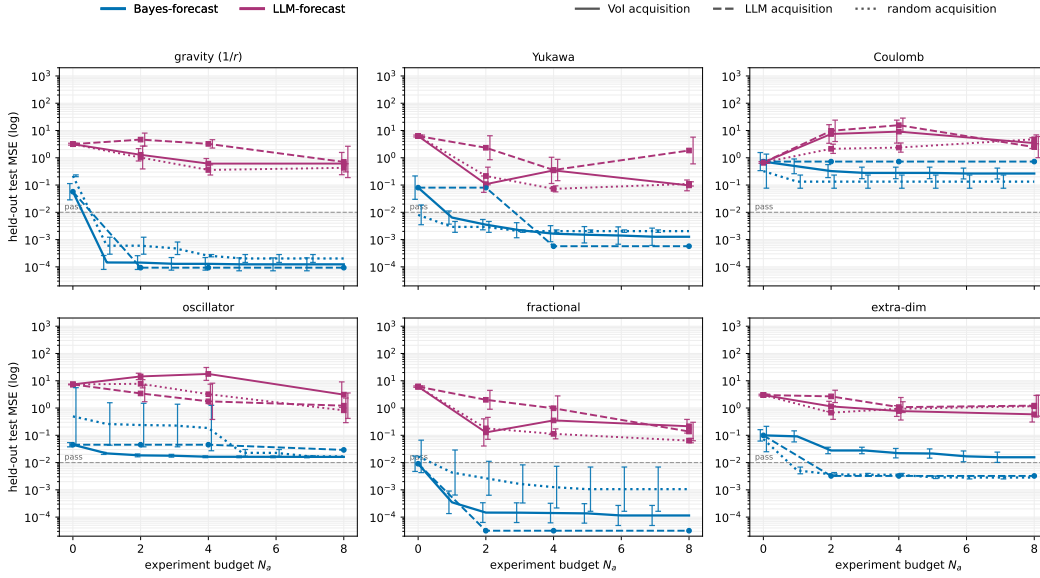


Figure 7: **Per-world data efficiency on TWOPARTICLEWORLDS** (Opus 4.7; the compact aggregate is Fig. 2 in the main text). One panel per world. Colour = forecaster (blue Bayes-forecast, purple LLM-forecast); line style = acquisition (VoI solid, LLM dashed, random dotted). The initial value at $N_a = 0$ is the result based on D_0 before any experiments. Uncertainty is ± 1 SE in $\log_{10}$ over 9 runs (3 random initial conditions $\times$ 3 draws per IC): a shaded band on the continuous best-so-far Bayes-forecast/VoI traces, and error bars on the $N_a \in \{0, 2, 4, 8\}$ budget points of both forecasters. Within the Bayes-forecaster family the VoI and LLM design strategies perform similarly and both generally beat random.

agent uses ~ 41 experiments and reaches nMSE 0.013, essentially reproducing the paper’s strongest agent (Opus, nMSE 0.01; (Wiemann et al., 2026)). MDA reaches that same accuracy with only 8 one-per-round experiments — a $\sim 5\times$ data-efficiency advantage (Fig. 2, right).

Is the advantage the curated design menu? MDA chooses experiments from the fixed 13-launch menu of Table 5, hand-built to contain informative probes, whereas the pure agent chooses initial conditions freely. One might worry that this curated design space — rather than the Bayesian inference — is what drives MDA’s lead. To test this we ran an Opus agent *given the same menu*: each round it picks a menu experiment, and after 8 experiments it submits its own best-fit force law (no SMC). It reaches only 22% numeric pass and 17% exact-form — *no better* than the free-choice Opus agent (31%), and far below MDA’s 93%. So the menu is not the source of MDA’s advantage: handed the identical design space, an LLM’s own propose-and-fit inference remains far weaker than MDA’s SMC-evidence selection and VoI design. (Conversely, the base-model sweep of Section C.8 shows the gap does narrow with a much stronger agent, Table 5 — so the advantage is the inference, and its size depends on how good the free-form agent’s own inference is.)

C.6 DATA EFFICIENCY CURVES

In Fig. 7 we plot the data efficiency curves for each of the six TWOPARTICLEWORLDS, from which the aggregated results in Fig. 2 are obtained. In Fig. 8 we plot similar curves for each of the five MULTIPARTICLEWORLDS. We see that MDA beats LLM agent by a large margin.

The above figures, and Fig. 2 in the main text, follows the DiscoverPhysics protocol and reports only the *numeric* metric. Figure 9 adds the *symbolic* (exact-form) view we use as a secondary, more stringent check: the fraction of runs whose submitted force law is the ground-truth form *exactly* (form-MSE $< 10^{-3}$ on held-out unit-charge cases, isolating the functional form from the charge-role). MDA recovers the exact form for $\sim 70\%$ of runs within a few experiments, roughly twice the Opus agent’s rate; note this exact-form test is stricter than, and can diverge from, numeric accuracy (a strong agent may write a law it cannot accurately integrate, and a predictive law need not be the

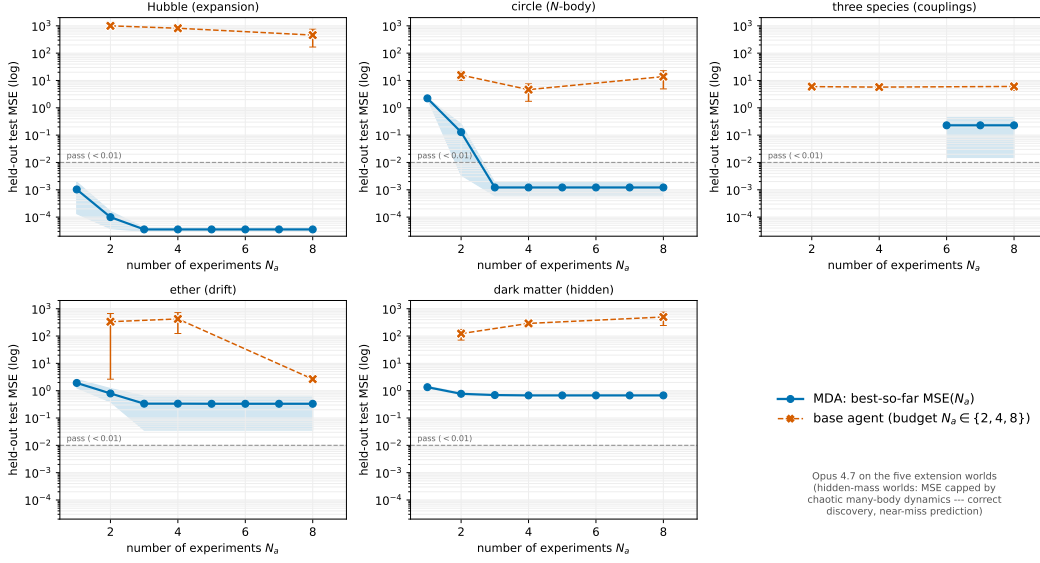


Figure 8: **Data efficiency on MULTIPARTICLEWORLDS** using Opus 4.7. We plot held-out forecast MSE vs. experiments for MDA (solid, mean $\pm$ SE over seeds) against the pure agent (dashed). MDA is orders of magnitude better at every budget. Several worlds clear the pass line (error of 0.01 or less) within three experiments. Ether and dark matter plateau above the line, but this is the intrinsic ceiling of their scoring (due to near-singular free-fall and a chaotic many-body system), not a discovery failure (since MDA recovers the correct drift and the hidden monopole).

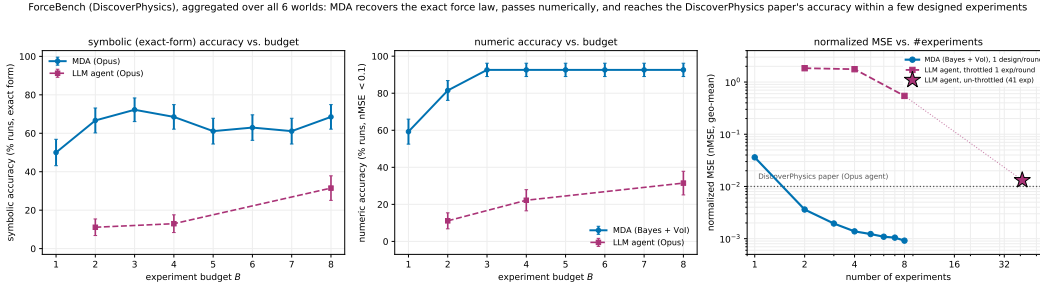


Figure 9: **Numeric and symbolic data efficiency on FORCEBENCH** (Opus 4.7; the numeric-only, multi-base-model version is Fig. 2). (left) Symbolic (exact-form) accuracy vs. budget: MDA (blue) vs. the pure LLM agent (purple). (middle) Numeric accuracy (nMSE < 0.1): MDA reaches $\sim 93\%$; the agent $\sim 31\%$. (right) nMSE vs. number of experiments, with the un-throttled Opus agent (star) reproducing the DiscoverPhysics paper’s ~ 0.01 at its native ~ 41 -experiment budget. Error bars are ± 1 SE over the 6×9 runs.

canonical form) — see Section C.8 for the Fable comparison, where the gap between the two metrics is largest.

C.7 EXPLANATION METRIC

A low held-out MSE does not *certify* a correct model: a law can be “right for the wrong reasons,” fitting observed orbits without capturing the mechanism. (Wiemann et al., 2026) proposed to fix this by asking each agent to return a text explanation to accompany its predicted law; this is then evaluated using an LLM judge. However, we have found this metric to be unreliable. For example Fig. 10 shows that the explanation score is essentially flat in the number of experiments, and often moves *non-monotonically* (more data making it worse), because the discovered functional form converges within the first couple of experiments and the residual movement is run-to-run variation in how the LLM *phrases* the same law, filtered through an 11-level judge.

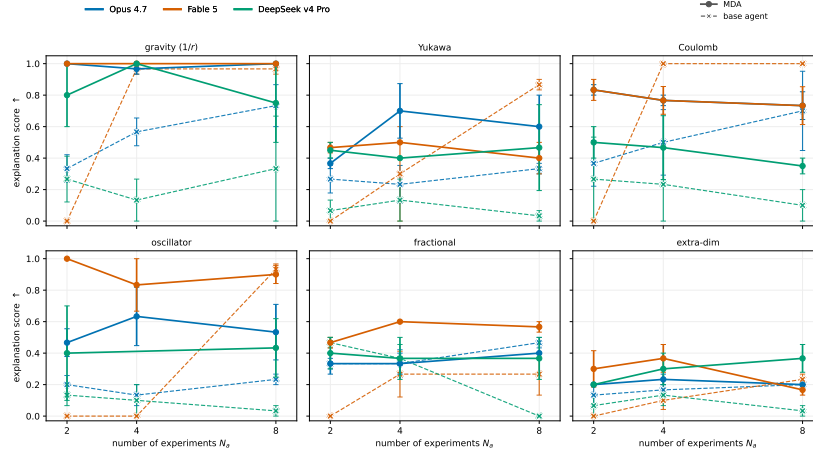


Figure 10: **The LLM explanation score vs. number of experiments (six worlds, Opus 4.7).** Flat and unreliable: sometimes monotonically *decreasing* with more data (Coulomb $0.83 \rightarrow 0.77 \rightarrow 0.73$), sometimes non-monotonic (Yukawa $0.37 \rightarrow 0.70 \rightarrow 0.60$) — either way more data can lower the score, a weak instrument (contrast the interventional forecast, which improves monotonically, Fig. 2).

An alternative is to test the ability of the model to predict under different kinds of novel distribution shifts, which is equivalent to testing its robustness to interventions on the mechanism. As proved in (Richens & Everitt, 2024), an agent that can perform such out-of-distribution predictions reliably must have learned a causally correct model of the world. In fact FORCEBENCH already evaluates models performance in this way: it measures MSE on test sets that combine one long-horizon probe and two single-knob interventions. Focusing on predictive performance on interventional test sets is not only more robust, but it also more general, since it does not require comparing to some (usually unknown) “true model”.

C.8 ROBUSTNESS TO THE BASE MODEL

The headline comparison uses Opus 4.7 as the shared base model, where the pure LLM agent is weak (Table 8). This raises a fair question: is MDA’s advantage an artifact of a *particular* (weak) agent, and would it vanish with a stronger base model? In this section, we consider three base models spanning a wide capability range: Opus 4.7, Fable 5, and DeepSeek v4.

Figure 11 shows the MSE results across the 6 worlds. We see that Fable is able to catch up with MDA’s data efficiency in 5 out of 6 of the worlds. However, we note that we beat Fable at small number of experiments using a much cheaper model (DeepSeek), provided we augment it with MDA (which has negligible cost). Further experimentation with Fable on MULTIPARTICLEWORLDS and other scientific domains is left to future work (since running Fable is expensive).

In Fig. 12 we aggregate results across worlds, but also show symbolic accuracy, not just MSE. Two patterns emerge. First, **MDA is essentially model-agnostic**: its numeric pass rate is 89–94% and its exact-form rate 74–83% regardless of the base model — the Bayesian machinery (SMC evidence, VoI design) does the heavy lifting, and a stronger proposer helps only at the margin. Second, **the pure agent is highly base-model-dependent**: its numeric pass rate swings from 26% (DeepSeek) and 31% (Opus) up to 81% for Fable 5, a much stronger recent model. So the striking gap against the Opus agent narrows sharply against Fable.

Even against the strongest agent we tested, **MDA’s extra inference on the same proposals never hurts and sharply helps**: MDA attains the higher numeric pass rate (94% vs. 81%; Fig. 12, middle) and reaches $\text{nMSE} \approx 10^{-3}$ in ~ 2 designed experiments, an accuracy the Fable agent reaches only at $B=8$ (right). On the joint metric that credits a run only when it is *both* exact-form and numerically correct, the two methods tie (74% each). The one axis on which the Fable agent leads is *pure* exact-form recovery (93% vs. 78%), and that lead is partly illusory: a free-form agent writes its own integrator, so it can name the exact law without being able to compute with it. On `coulomb`, for

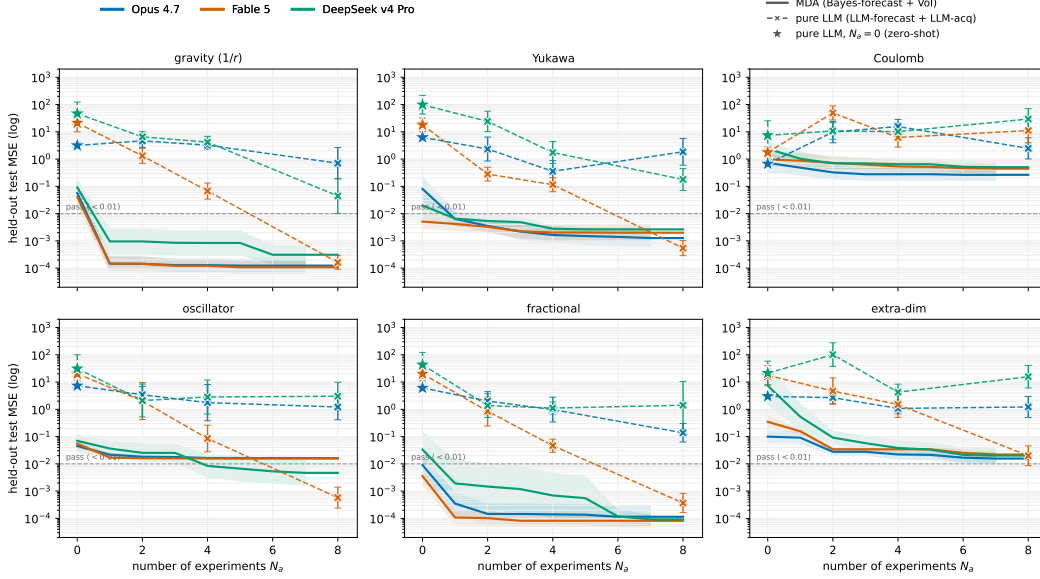


Figure 11: **Effect of changing the proposer LLM on data efficiency:** held-out test MSE vs. experiment budget B , per world, for three proposer LLMs (Opus 4.7, Fable 5, DeepSeek v4 Pro; colour). **Solid:** MDA (Bayes-forecast + Vol acquisition), the best-so-far MSE(B) trajectory with a ± 1 SE band. **Dashed:** the matched pure-LLM agent (LLM-forecast + LLM-acq) at $B \in \{0, 2, 4, 8\}$, with ± 1 SE error bars and anchored at its $B=0$ zero-shot law ($\star$). Uncertainty is over 9 runs (3 seeds $\times$ 3 draws), geometric mean ± 1 SE in $\log_{10}$. Grey: the pass threshold (0.01). Across proposers, MDA drives the held-out error to the identifiability floor within a few experiments, whereas LLM agent behavior plateaus well above it on most worlds, except in the case of Fable.

instance, the Fable agent scores 100% exact-form but only 11% numeric, whereas MDA — using the same proposals with a vetted forward model — scores 67% numeric.

The residual exact-form gap is *not* because MDA cannot propose the exact law: it uses the same LLM proposer (Fable), and a truth-equivalent form is present in its $N_m \approx 14$ -candidate pool in the large majority of runs. It is a model-*selection* effect: the Bayesian evidence sometimes outvotes the truth in favour of a slightly more flexible form that fits the observation noise marginally better (e.g. a spurious weak time-modulation on top of the correct radial law). We counter this with a **parsimony submission rule**: at submission, among the pool forms that share the winning force-profile shape $F(r)$, we return the fewest-parameter member — evicting the over-elaborated near-duplicates the $\mathcal{M}$ -open exploration introduces, the same ESS-eviction idea used in CHEMBENCH (Section D). This recovers the exact form on the time-modulation misses, lifting MDA’s exact-form rate 74% $\rightarrow$ 78% and its joint metric to parity with the agent (74%), at negligible numeric cost (94% $\rightarrow$ 93%; green vs. blue in Fig. 12, left). The remaining gap (dominated by *oscillator*, where MDA recovers a predictive but non-canonical form) reflects structure identifiability, not inference quality. We read the overall pattern as the honest boundary of the result: MDA’s contribution is robust, model-agnostic accuracy under a tight budget — an inference layer that dominates the predictive metric and at least matches a strong free-form agent on structure, not an unbounded lead over any conceivable agent.

C.9 EXAMPLE: COULOMB WORLD

In this section, we visualize behavior of MDA when applied to COULOMB world, as shown in Fig. 13. On the left, we show what happens when a probe is launched near the source. At unit charge ($p_1=1$) the true law $k q_i q_j / r^2$ and the charge-blind overfit k/r^2 trace the *same* orbit (grey) — fit on unit-charge data, they are identical there, so *no* probe placement, at any radius or launch, can tell them apart. Turning the source charge to $p_1=4$ — a $\text{do}(a)$ on the mechanism — scales the true law’s force fourfold (green) while the overfit is unmoved (orange): the orbits split, and that split is what the observations measure.

ForceBench with the Fable-S base model on both arms: MDA's Bayesian inference on Fable's proposals dominates numeric accuracy and nMSE convergence (mid, right); parsimony narrows the exact-form gap (left, green), where the strong unaided agent is otherwise ahead

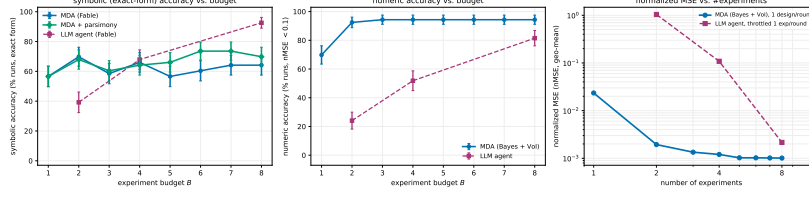


Figure 12: **Data-efficiency curves with the strongest base model (Table 5) on both arms**, aggregated over all six FORCEBENCH worlds. (left) Exact-form accuracy vs. budget: the parsimony submission (green) lifts MDA (blue) toward the strong unaided agent (purple), which leads on *pure* structure recovery. (middle) Numeric accuracy (fraction of runs with normalized MSE < 0.1): MDA reaches $\sim 94\%$ within ~ 3 designed experiments and dominates the agent everywhere. (right) Normalized MSE vs. number of experiments: MDA converges to $\sim 10^{-3}$ in ~ 2 experiments, an accuracy the throttled agent reaches only by $B=8$. So even against a strong proposer, MDA's Bayesian inference on the *same* proposals never hurts and sharply helps on the predictive metric. Error bars are ± 1 SE over the 9 runs (3 seeds $\times 3$ LLM draws) per world.

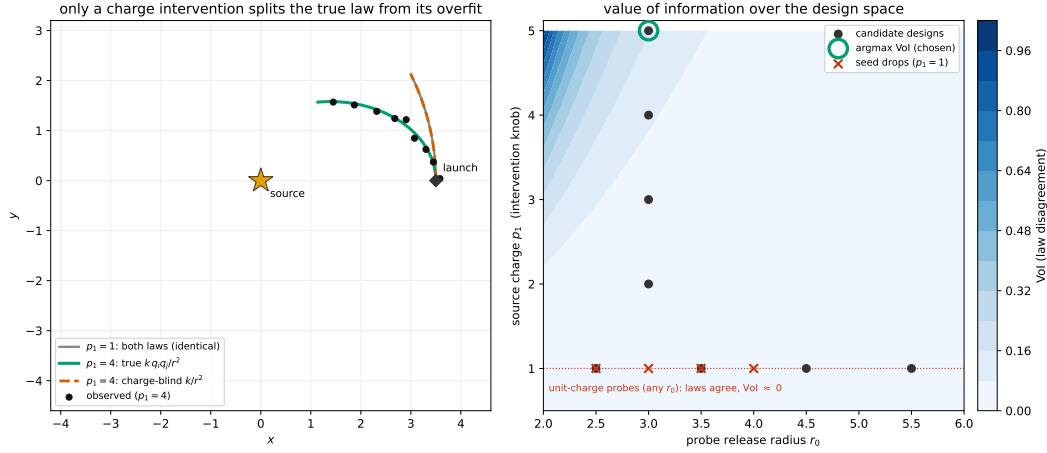


Figure 13: **Visualising COULOMB world and its design space.**

On the right, we plot the VoI over a 2d slice of the design space, namely the release radius r_0 (an initial condition) $\times$ source charge p_1 (an intervention knob). The red $\times$ are the *seed drops*: the unit-charge probes the agent has already collected (the initial, un-designed observations both laws are fit to). VoI is ≈ 0 all along the unit-charge axis, and rises only with the charge, so MDA's VoI-driven design step reaches for a *charge intervention* (green ring), not a farther probe. Thus we see that changing a causal (mechanism) knob, not just the initial location, is needed to distinguish a correct law from a curve-fit.

C.10 EXAMPLE: YUKAWA WORLD

Whereas COULOMB's discriminating design is a charge *intervention* (Fig. 13), YUKAWA's is a spatial *extrapolation*. The screened kernel $K_1(r/\lambda)/\lambda$ and the power laws fit to short-range data are nearly identical for $r \leq \lambda$ and diverge only beyond it, so a probe confined within λ cannot tell them apart while one reaching past λ can (Fig. 14). This is why the VoI design reaches for the long-range $r_0=5, 6$ drops (Table 5), and why the true kernel only reaches the convex corner of the Pareto frontier (Fig. 1b) once such a probe is added.

C.11 EXAMPLE: DISCOVERING HIDDEN PARTICLES

In this section we give a simplified example of the DARK-MATTER world, where the challenge is to identify both the number and location of hidden particles.

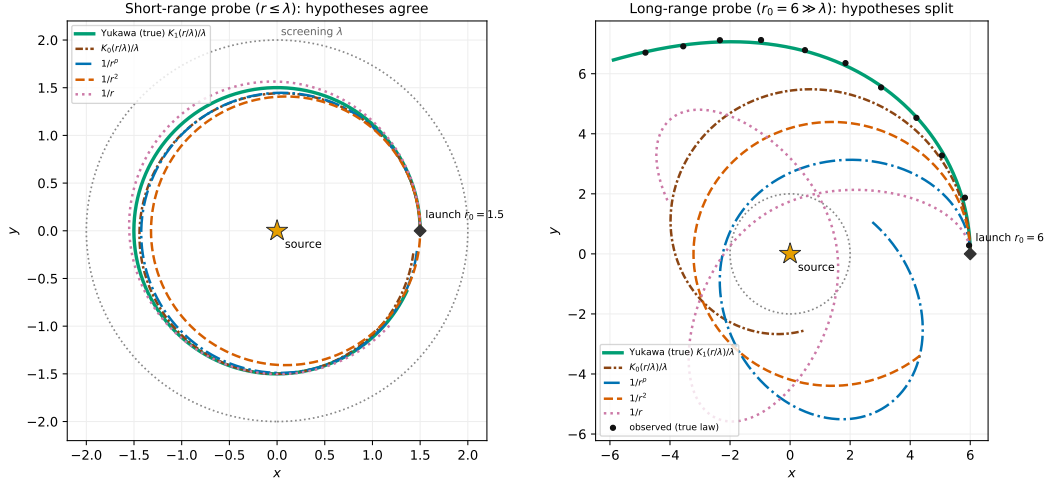


Figure 14: **Probe orbits under the candidate force laws for YUKAWA world: a short-range vs. a long-range design.** The screened Yukawa kernel $K_1(r/\lambda)/\lambda$ and the power laws fit to the short-range seed data nearly coincide for $r \leq \lambda$ and diverge only beyond it. (*left*) Launched within the screening length ($r_0=1.5$), every candidate law traces almost the same orbit — they cannot be told apart. (*right*) Launched well beyond it ($r_0=6$, matching the long-range probes of Fig. 1b): the true screened kernel (green, with the observed data) has decayed, so it holds a wide slow arc, whereas the un-decayed power-law near-misses are far too strong at this range and plunge inward — the hypotheses fan out.

The model class. Neutral test *probes* move in a *known* static 2D Poisson field: a source of coupling q at position s pulls a probe at x with force $q/(2\pi\|x-s\|)$ toward s , and the field superposes over sources. One *visible* source of known coupling sits at the origin; the world also contains K *hidden* sources whose positions and couplings are concealed. A probe released from rest therefore falls not toward the visible source but toward the *total* mass, so it appears to accelerate toward empty space — the dark-matter tell (Fig. 15, middle, arrows). The structure m is the count K ; its parameters are the $3K$ hidden coordinates and couplings. The design knob $a \in \mathcal{I}$ is the probe launch configuration (x, y) , a point in the plane, encoded as a $\text{do}(\cdot)$; we record the probe under position noise $\sigma = 0.03$. Because the sources are point masses, not a density field, the forward model is small- N and the whole rung runs on CPU.

The task. The true world has one visible source ($q = 2$ at the origin) and one hidden mass ($q = 4$ at $(3.5, 2)$). The method is handed three *seed* probes released far from the hidden mass, so they feel it only as a weak far-field deflection: enough to reveal that *some* unseen mass exists, but too little to say *where*. It must (i) decide how many hidden masses there are, (ii) localize the one that exists, and (iii) choose where to place the next probe. We fit $K \in \{0, 1, 2\}$ by adaptive-tempering SMC over the hidden coordinates (Algorithm 3, $N_p = 1000$) and combine by marginal evidence (§A.2); $K=0$ is a genuine zero-parameter model (visible field only), so comparing it to $K=1$ is Bayesian model selection, and comparing $K=1$ to $K=2$ is Bayesian Occam — a second mass must *earn* its three parameters against the prior volume they cost.

Model selection and localization. The evidence is decisive (Fig. 15, left): $p(K=1 \mid D) \approx 0.97$, with $K=0$ excluded outright (its visible-only field cannot bend the probes toward empty space) and $K=2$ rejected by Occam (fitting a second, redundant mass buys a negligible likelihood gain for a three-parameter prior-volume penalty). The recovered mass sits at $(3.53, 1.97) \pm (0.13, 0.09)$ with coupling 3.99 ± 0.07 — correct in count, position, and strength. That the count is inferred, not assumed, is the point: this is the latent-*existence* question (§2) — *is there* an unobserved cause, and how many — answered by evidence.

Where to look, in a 2D design space. Which placement best localizes the mass? We score each candidate by the query-relevant VoI (Eq. (5)) for a downstream query — a test probe released near the hidden mass, whose outcome depends on the hidden-mass position. The VoI landscape (Fig. 15,

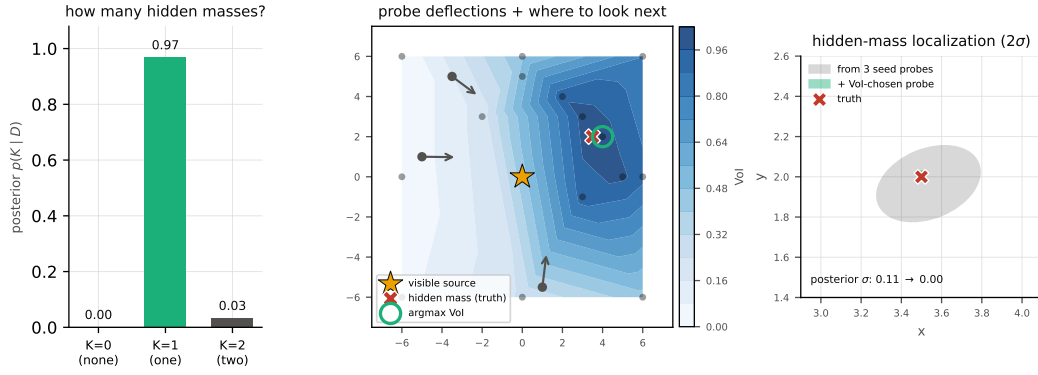


Figure 15: **The hidden-mass rung.** **Left:** trans-dimensional model selection $p(K | D)$ from the seed probes — the deflections demand a hidden mass ($K=0$ excluded), and Bayesian Occam rejects the surplus second mass ($K=2$). **Middle:** the scene. The visible source (star) sits at the origin, but the seed probes (released from the dots) deflect toward empty space (arrows) — toward the hidden mass (red cross). The blue field is the query-relevant VoI over candidate next-probe placements; it peaks on the hidden mass, and the argmax (green ring) sits essentially on it. **Right:** the $K=1$ posterior over the hidden-mass position (2σ ellipses): the three seed probes localize it only loosely, and the single VoI-chosen probe collapses the uncertainty onto the truth.

middle) peaks sharply on the hidden mass: a probe placed there measures it directly, while probes on the far side (visible-dominated) are nearly useless. Running the argmax probe drives $p(K=1)$ to 1.0 and collapses the position posterior from $\sigma \approx 0.11$ to ≈ 0.01 (Fig. 15, right) — the localization the seed probes could not reach.

Method	Easy		Medium		Hard		Overall	
	SA	Ex	SA	Ex	SA	Ex	SA	Ex
MDA (VoI)	66.7	83.3	45.8	79.2	54.2	79.2	55.6	80.6
MDA (Mean)	54.2	79.2	58.3	87.5	33.3	66.7	48.6	77.8
LLM-AUTOSCI LAB (us)	41.7	58.3	41.7	75.0	41.7	75.0	41.7	69.4
LLM-AUTOSCI LAB (reported)	55.6	88.9	22.2	37.0	42.9	52.4	35.1	50.9

Table 9: **Performance on CHEMBENCH at $B=60$** , 36-task stratified subset (12 domains $\times$ 3 tiers). SA = symbolic accuracy (%); Ex = EXACC at the 0.05 threshold (%; for comparability with the reported row); **bold** = best among the top three rows. The bottom row is LLM-AUTOSCI LAB’s published numbers from (Kabra et al., 2026, Table 3), based on a different, unreleased 36-task subset, and using gpt-4o-mini instead of our use of Opus 4.7.

D CHEMISTRY: ADDITIONAL DETAILS

D.1 BENCHMARK

The problem is to learn a static algebraic function of seven controllable inputs (substrate, inhibitor, second substrate and product concentrations, enzyme loading, temperature and pH) to a reaction rate y :

$$y = f(C_A, C_I, C_B, C_P, \text{Enz}, T, \text{pH}; \theta), \quad (20)$$

The unknown is *which kinetic mechanism is active*. There are 9 canonical single mechanisms (Michaelis–Menten, competitive / uncompetitive / noncompetitive / product inhibition, substrate inhibition, Hill cooperativity, Arrhenius temperature dependence, ping-pong bisubstrate), and 48 compound mechanisms, created from combinations of these elementary mechanisms (e.g., ping-pong $\times$ Arrhenius, MM $\times$ competitive $\times$ Arrhenius and Hill $\times$ Arrhenius), yielding a total of 57 rules or worlds. The dataset is divided into easy, medium, and hard tiers, based on the mechanisms used and their corresponding parameters (some of which make the response hard to detect).

Performance of a submitted law is evaluated using the procedure described in (Kabra et al., 2026, App.C). First we compute the held-out root-mean-squared log-error $\text{RMSLE} = \left[\frac{1}{N} \sum_{i=1}^N (\log(1+\hat{y}_i) - \log(1+y_i))^2 \right]^{1/2}$ over $N=1000$ test points. Then we compute whether the law is numerically equivalent to the true law using $\text{EXACC} = \mathbb{1}[\text{RMSLE} < \epsilon]$, a quantity they call the “exact accuracy”. The released benchmark code uses $\epsilon=0.05$ for chemistry (and 0.01 for physics). Our data-efficiency curves (Fig. 3) use the stricter App. C threshold $\epsilon=0.01$, whereas the head-to-head Table 9 reports $\epsilon=0.05$ to match LLM-AUTOSCI LAB’s published numbers. They also compute symbolic equivalence to the true law using sympy.

D.2 FURTHER RESULTS

Table 9 reports symbolic accuracy (SA) and exact accuracy (EXACC, at their 0.05 threshold for comparability), per difficulty tier and overall, at a max budget of $B=60$ experiments. We see that MDA beats their method overall (SA 56 vs. 42 and EXACC 81 vs. 69), as well as on every tier. Crucially, MDA wins on the hard tier, where LLM-AUTOSCI LAB’s low-error solutions are numerically accurate but mechanistically meaningless (e.g. recovering $10^{0.87 \log(0.5 \sqrt{\text{Enz}/\dots})}$ at $\text{RMSLE}=0.001$, which passes even the strict 0.01 threshold, yet the symbolic-equivalence check marks it as wrong).

D.3 ABLATIONS

In Fig. 16, we show the effects of ablating various parts of the MDA method (36-task subset, $B = 60$ experiment budget).

- We start by using vanilla SMC, as in the ModelSMC paper (Wahl et al., 2026), combined with Monte Carlo (random sampling) based optimization of the VoI.
- Next we add $\mathcal{M}$ -open exploration. This lifts overall symbolic accuracy 36 $\rightarrow$ 50% — driven by the easy tier, where the residual-directed re-prompt corrects the single-mechanism form

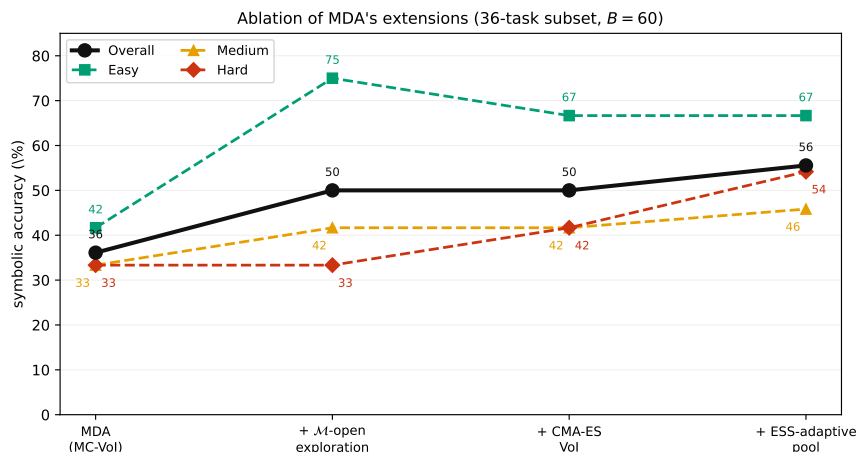


Figure 16: **Ablation of MDA’s extensions** beyond ModelSMC: symbolic accuracy per tier and overall, added incrementally on the 36-task subset at $B=60$. Each extension helps a distinct tier. $\mathcal{M}$ -open exploration lifts the *easy* tier (42→75%, single-mechanism correction; also the only route to any compound recovery) but leaves the hard tier flat; CMA-ES VoI and then the ESS-adaptive pool lift the *hard* tier (33→42→54% — sharper designs, then evicting the over-elaborated near-duplicates exploration introduces). The full config is the two-seed result of Table 9; intermediate configs are seed 0.

(easy SA 42→75%). This is the *only* route to any compound recovery (0→11%, since the fixed library of 9 primitives cannot express compound mechanisms). Interestingly, this does *not* help the hard tier, where the mechanism’s signal is too weak under the extreme parameters.

- Next we add CMA-ES optimization of VoI. This sharpens the design, raising exact numerical accuracy 36→42% and hard-tier SA 33→42%.
- Finally we add the ESS-adaptive pool. This lifts overall SA 50→56% and hard-tier SA 42→54% by evicting the over-elaborated near-duplicates the exploration introduces. Together they turn the ModelSMC base (36% SA) into a method that wins every tier of the fair comparison.

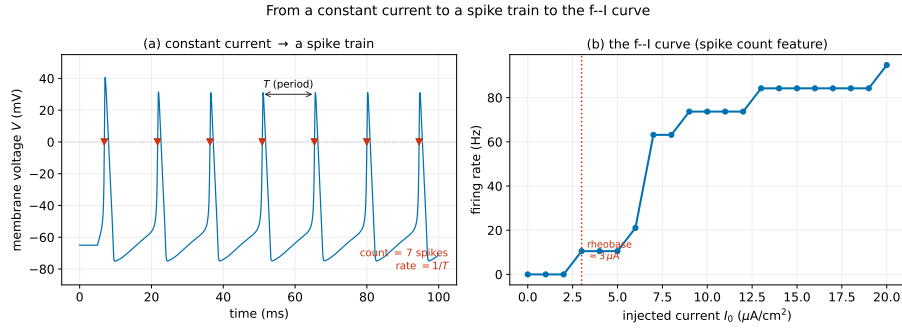


Figure 17: **The f-I curve, and why we count spikes.** (a) A constant supra-threshold current makes the model fire a periodic *spike train*; the readout is simply the *spike count* (red markers) — or, per unit time, the firing rate $1/T$. (b) Sweeping the injected current traces the *f-I curve* (firing rate vs. current): flat and zero below the *rheobase* (the smallest current that fires, red), then rising. This matches the intuitive “how many spikes” readout.

E NEURONBENCH

This appendix covers the deterministic form of the benchmark: background on neuron electrophysiology and Hodgkin–Huxley models, the problem specification, the (tractable, synthetic-feature) solution methods, the benchmark results and baselines, and case studies. The stochastic form — where the likelihood becomes intractable — is deferred to Section F.

E.1 PRIMER ON NEURON ELECTROPHYSIOLOGY

In this section, we give a brief introduction to neuron electrophysiology.

We can view a neuron as a device that turns an injected current into a voltage trace. At rest the membrane voltage V sits near -65 mV. A small (*sub-threshold*) injected current depolarises V a little and it relaxes back — a passive, RC-like response. A large enough (*supra-threshold*) current triggers an *action potential* or *spike*: voltage-gated Na^+ channels open regeneratively, V shoots to $\sim +40$ mV in under a millisecond, then K^+ channels open and pull it back down. Spikes are the neuron’s output; their *count* (or rate) as a function of the injected-current amplitude is the *f-I curve* (frequency–current), the standard input–output summary of a cell: see Fig. 17.

Crucially, the behavior of the neuron depends on its inputs, as illustrated in Fig. 18. Here we show the voltage over time, under 3 different experimental conditions: the standard model, stimulated with a $10 \mu\text{A}$ step signal, which generates repeating spikes (blue); the same model stimulated with a $2 \mu\text{A}$ step signal, which fails to trigger a response (dotted black); and the model modified by applying TTX blocker and then stimulated with a $10 \mu\text{A}$ step signal, which also fails to trigger a response (red line). This illustrates why experiment design is critical in this domain.

E.2 PRIMER ON HODGKIN-HUXLEY MODELS

In this section, we give a brief primer on generalized Hodgkin-Huxley models. The model is named after Alan Hodgkin and Andrew Huxley who invented it in 1952 to explain the ionic mechanisms underlying the initiation and propagation of action potentials in the squid giant axon. Since then, the model has been generalized and is widely used to mechanistically explain the spiking behavior of many kinds of neurons. Hodgkin and Huxley received the 1963 Nobel Prize in Physiology or Medicine for this work.

The model they came up with can be represented as an electric circuit, as shown in Fig. 19. This example contains Na, K, M and L ion channels, but the generalized model can contain different combinations of the 6 channels listed in Table 11, each of which have their own parameters and dynamics. We can write the generalized model as a set of nonlinear ODEs, which follow from

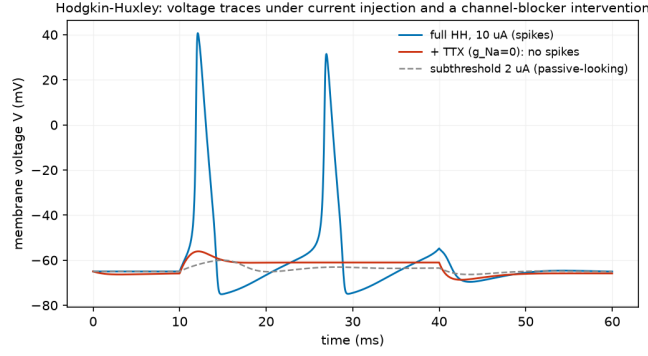


Figure 18: **Example spike traces from a single neuron under different conditions.** Membrane voltage under current injection: a supra-threshold step ($10 \mu\text{A}$) elicits overshooting action potentials (blue); the sodium blocker TTX ($g_{\text{Na}}=0$, a do on the mechanism) abolishes them (red); a sub-threshold current gives a passive response (grey).

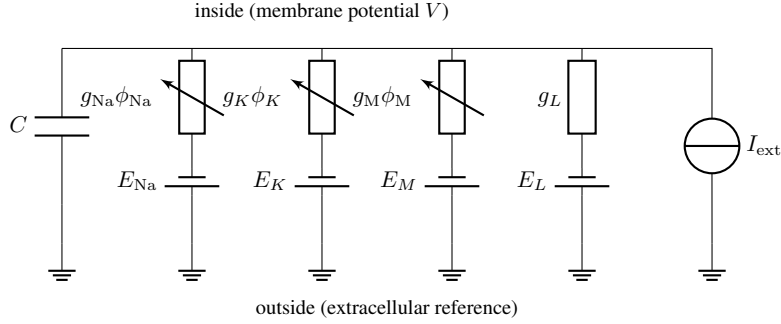


Figure 19: **The Hodgkin-Huxley equivalent circuit (Eq. (21)).** The membrane is a capacitor C ; each ion channel is a branch with a variable conductance $g_c \phi_c$ (opening/closing gates ϕ_c) in series with a battery E_c (the reversal potential). The injected current I_{ext} charges the capacitor and flows through the open channels; a *blocker* deletes a branch ($g_c \rightarrow 0$). Which branches are present is the *structure*; the conductances g_c are the *parameters*.

Kirchoff's current law:

$$C \frac{dV(t)}{dt} = I_{\text{ext}}(t) - \sum_{c \in \mathcal{C}} I_c(t) \quad (21)$$

$$I_c(t) = g_c \phi_c(t) (V(t) - E_c) \quad (22)$$

$$\phi_c(t) = m_c^{p_c}(t) n_c^{q_c}(t) h_c^{r_c}(t) \quad (23)$$

$$\frac{dx_c(t)}{dt} = \frac{T_{x,c}^\infty(V(t)) - x_c(t)}{\tau_{x,c}(V(t))}, \quad x \in \{m, n, h\} \quad (24)$$

Here C is the capacitance, $V(t)$ is the voltage, I_c is the current for channel c , $\mathcal{C}$ is the set of channels associated with this neuron, and $\phi_c(t)$ is the fraction of the channel that is open. Thus the current in the channel is given by $I_c = g_c \phi_c (V - E_c)$: (maximal conductance) $\times$ (fraction open) $\times$ (driving force). The fraction open ϕ_c (which changes over time) is based on a product of gating terms — denoted by m_c , n_c and h_c — each raised to an integer power (p_c , q_c , r_c ; how many independent gates the channel has): see Table 11 for the list. Each such gating term x_c relaxes towards a voltage-dependent target $T_{x,c}^\infty(V)$ with its own time constant $\tau_{x,c}(V)$ (fast for activation, slow for inactivation), given by

$$T_{x,c}^\infty(V) = \frac{\alpha_x(V)}{\alpha_x(V) + \beta_x(V)}, \quad \tau_{x,c}(V) = \frac{1}{\alpha_x(V) + \beta_x(V)} \quad (25)$$

gate	channel	role	target $T_{x,c}^\infty(V)$	speed
m_{Na}	Na^+	activation	rises with depolarisation	fast
h_{Na}	Na^+	inactivation	<i>falls</i> with depolarisation	slow
n_K	K^+	activation	rises with depolarisation	slow

Table 10: **The three classic Hodgkin–Huxley gates.** *Activation* gates (m_{Na}, n_K) open as the cell depolarises; the *inactivation* gate (h_{Na}) closes. Each relaxes to a voltage-dependent target $T_{x,c}^\infty(V)$ with its own time constant $\tau_{x,c}(V)$; the fast/slow separation between m_{Na} and $\{h_{Na}, n_K\}$ is what generates and terminates the spike. Other channels (Table 11) carry their own gates m_c, n_c, h_c with the same form but different half-voltages and kinetics.

Channel	carries	current	role in the response	blocker (a do)
Na^+	sodium	$I_{Na} = g_{Na} m_{Na}^3 h_{Na} (V - E_{Na})$	regenerative spike <i>upstroke</i>	tetrodotoxin (TTX)
K^+ (delayed rectifier)	potassium	$I_K = g_K n_K^4 (V - E_K)$	<i>repolarises</i> the spike	TEA
Ca^{2+} (high-threshold)	calcium	$I_{Ca} = g_{Ca} m_{Ca}^2 h_{Ca} (V - E_{Ca})$	<i>alternative</i> , slower spike up-stroke	cadmium (Cd)
M-type K^+	potassium	$I_M = g_M m_M (V - E_K)$	slow; <i>spike-frequency adaptation</i>	XE991
A-type K^+ (transient)	potassium	$I_A = g_A m_A^p h_A (V - E_K)$	transient outward; <i>delays</i> firing onset	4-AP
leak	mixed	$I_L = g_L (V - E_L)$	sets the <i>resting potential</i> ; passive	—

Table 11: **The voltage-gated ion channels — the building blocks.** A *blocker* is a drug that removes one channel by setting its conductance $g_c=0$; these are the mechanism-level interventions $do(a)$ available on this rung (e.g. TTX abolishes Na^+ -based spikes but not Ca^{2+} -based ones). The parenthetical drug names in this table refer to these blockers, and are what the design loop gets to apply.

where expressions for α_x and β_x can be found at https://en.wikipedia.org/wiki/Hodgkin-Huxley_model. As example, the classic spiker is the following three-channel model

$$C\dot{V} = I_{ext} - g_{Na} m_{Na}^3 h_{Na} (V - E_{Na}) - g_K n_K^4 (V - E_K) - g_L (V - E_L) \quad (26)$$

Here the Na^+ channel carries an activation gate m_{Na} (cubed) and an inactivation gate h_{Na} , and the K^+ channel a single activation gate n_K (to the fourth); Table 10 summarises the three. The names m, n, h are historical: what actually distinguishes a gate is its target curve $T_{x,c}^\infty(V)$ (whether it *opens* or *closes* as V rises) and its time constant $\tau_{x,c}(V)$. It is the *separation of timescales* — fast m_{Na} activation admitting Na^+ for the upstroke, before the slower h_{Na} inactivation shuts it off and the slower n_K activation repolarises — that makes the spike a transient, regenerative event.

It is worth noting that HH is only one point on a spectrum of models at different levels of abstraction. There are more detailed stochastic models that capture individual cellular responses at a more granular level. There are also simplified models, such as the two-variable FitzHugh–Nagumo model, and the leaky integrate-and-fire model. Finally, if we set the membrane time constant to zero and binarise the output, we get the McCulloch–Pitts unit (McCulloch & Pitts, 1943), $y = \phi(\sum_i w_i x_i - b)$, which is the basis of artificial neural networks. So there is no single “true model”. Instead, scientists seek the *coarsest valid causal abstraction* that is sufficient for the things they want to understand or predict (Beckers & Halpern, 2019; Rubenstein et al., 2017).

E.3 OUR BENCHMARK

We design a benchmark, NEURONBENCH, by creating 6 “mystery neurons”, each composed of a plain $Na+K+leak$ spiker plus one extra membrane mechanism, chosen from the list in Table 12: five are *novel* mechanisms and the sixth is a recallable textbook M-current control. Each of the five novel mechanisms is deliberately tuned to be *silent under every textbook probe*, i.e., the plain and novel neurons fire *identically* to standard current steps and channel blockers. This requires the agent to propose novel experimental protocols that it has not already memorized.

The task The agent is told that it will be presented with some voltage trace data from a neuron of unknown type, and is asked to propose various candidate mechanisms (the exact prompts are shown in Section G.3). It is also given the menu of stimulation protocols (Table 13) and channel blockers, and a fixed experiment *budget*. From a handful of designed experiments it must (i) *propose its own* candidate mechanisms m and return a posterior $p(m \mid \mathcal{D})$ over them, and (ii) forecast the cell’s response to held-out interventions it never ran. The truth is never revealed to the agent; it is used only for scoring. (A solver may of course restrict its hypothesis space — e.g. MDA fits a pool of conductance archetypes — but the benchmark neither supplies nor assumes an enumerated candidate set.)

Evaluation. The task is *counterfactual trajectory forecasting*: on a disjoint set of held-out protocols the agent never ran, it must predict the cell’s response — a spike count and a voltage trace per protocol. Because the hypothesis space is open we score *behaviour*, not model labels, on two levels. (i) The **spike-forecast MSE** (the headline metric): the mean-squared error of the predicted test-window spike counts. (ii) The **feature-forecast MSE** (a secondary, finer metric for model-based deep-dives): the standardised MSE, over the per-trace summary feature vector $s(y)$ of Eq. (30) between the agent’s predicted trace and the truth.

Note that the feature-forecast requires predicting a full trace, which is then converted to summary features. Generating a trace is hard to do for a pure LLM based forecaster, but is easy for a model-based one. We reduce the prediction to a set of features in order to make the comparison to ground truth more meaningful (see Section E.4).

Specification of the novel channels. Each novel channel has roughly the same gated form as the textbook ones:

$$I_Z = g_Z m_Z^p h_Z^q (V - E_Z), \quad T_{m,Z}^\infty(V) = \sigma\left(\frac{V - V_{1/2}^m}{k_m}\right), \quad T_{h,Z}^\infty(V) = \sigma\left(-\frac{V - V_{1/2}^h}{k_h}\right), \quad (27)$$

with $\sigma(u) = 1/(1 + e^{-u})$ the logistic (Boltzmann) sigmoid, half-voltages $V_{1/2}^m, V_{1/2}^h$, slopes $k_m, k_h > 0$, and fixed time constants τ_m, τ_h : activation rises with V and inactivation falls, while a *negative* activation slope $k_m < 0$ instead makes the channel hyperpolarisation-activated (as for I_h), and an inactivation half-voltage $V_{1/2}^h$ below rest makes it *de-inactivated by hyperpolarisation* (available only after a hyperpolarising pre-pulse). This Boltzmann form is generic across the *novel* channels but is *not* the textbook parameterisation shown in Eq. (25), which are monotonic curves of the same qualitative shape but not identical logistic sigmoids.

Design space. The agent gets to control the external current $I_{\text{ext}}(t) = x_t$ injected at each step. In our benchmark, we assume the current is chosen from one of the 9 sequence options in Table 13. In principle the agent can also apply a single channel blocker (tetrodotoxin (TTX) zeroing g_{Na} , TEA zeroing g_{K} , cadmium (Cd) zeroing g_{Ca} , or none), giving a nominal 9×4 actions. But the novel mechanisms are by construction *silent under blockers* as well as under textbook steps — a blocker deletes a channel branch equally in the plain and novel cells, so it cannot separate them — so the blockers are non-discriminating for these worlds. We therefore report all experiments over just the 9 current-clamp protocols (blockers remain available in the released benchmark and the interactive app, but are unused in the runs, for simplicity).

Interactive app. Figure 20 shows a screenshot for a web app we built that lets users try this benchmark for themselves. The app is available at <https://github.com/murphyk/neuronbench>.

E.4 THE LIKELIHOOD: SUMMARY FEATURES, NOT THE RAW TRACE

Because a spike is a ~ 1 ms all-or-none event, a sub-millisecond timing mismatch between model and data produces a ~ 100 mV pointwise error even for an essentially correct model. Hence a likelihood that factorizes over time steps

$$p(y_{1:T} \mid \xi, m, \theta) = \prod_{t=1}^T p(y_t \mid z_t, \sigma^2) \quad (28)$$

Mechanism	(g_Z, E_Z)	activation [†]	inact. [†]	behavioural signature (revealing protocol)
Z-REBOUND (I_Z)	(4, +120)	(−57, 5, 4, 2)	(−88, 4, 130)	spike-count collapse after a hyperpolarising conditioning pulse
H-SAG (I_h)	(5, −30)	(−95, −5, 140, 1)	—	voltage sag + post-inhibitory rebound on a hyperpolarising step
NA-FATIGUE	—	slow inactivation added to h_{Na}		use-dependent spike-count run-down over paired long pulses
CA-REBOUND (I_{CaT})	(3.2, +120)	(−54, 6, 2, 2)	(−87, 4, 22)	low-threshold rebound <i>burst</i> on release from hyperpolarisation
D-TYPE (I_D)	(9, −77)	(−30, 10, 3, 1)	(−80, 5, 200)	delayed / suppressed firing after a hyperpolarising pre-pulse
TEXTBOOK-M (I_M)	(2.5, −77)	(−35, 10, 60, 1)	—	spike-frequency adaptation on a long step (recallable by name)

Table 12: **The six worlds of NEURONBENCH.** Each current is added to a Na+K+leak spiker via Eq. (27); the row label is the Fig. 4 panel name. [†]the activation/inactivation columns are the tuples $(V_{1/2}, k, \tau, \text{power})$ and $(V_{1/2}, k, \tau)$ of Eq. (27). Conductances g_Z in mS/cm²; reversals E_Z , half-voltages and slopes in mV; time constants in ms; p, q are gate powers ($q=1$ when an inactivation gate is present, else 0). All are tuned to be *indistinguishable from the plain spiker under textbook steps and blockers* and separable only by the matched non-textbook protocol in the last column (a hyperpolarising conditioning pre-pulse for the de-inactivating currents). NA-FATIGUE adds no channel: it slows the inactivation of the existing Na⁺ gate h_{Na} . The I_M control is a standard non-inactivating K⁺ current the LLM *can* name and probe.

#	protocol	segments (Δt ms, I μ A)	probes
1	brief step	(40, 12)	fast onset
2	long step	(300, 10)	spike-frequency adaptation
3	strong step	(120, 18)	high-rate firing
4	weak step	(120, 5)	near-threshold f-I
5	hyperpol. conditioning + test	(250, −30), (150, 12)	de-inactivation / depol. block
6	hyperpol. pre-pulse + weak test	(250, −30), (120, 0), (60, 6)	rebound at low drive
7	paired long pulses	(300, 12), (60, 0), (300, 12)	use-dependence / slow inactivation
8	depol. conditioning + test	(250, 15), (150, 12)	depolarising history
9	brief hyperpol. conditioning + test	(40, −30), (150, 12)	fast de-inactivation

Table 13: **The nine-protocol menu of external currents that can be applied** over which VoI is enumerated on NEURONBENCH. Each protocol is a sequence of (duration, amplitude) current segments; a leading hyperpolarising segment is a conditioning pre-pulse. Rows 1–4 are standard current-clamp steps; rows 5–9 are the non-textbook protocols that expose the hidden mechanisms of Table 12 — exactly one is decisive for each, so only a designed (VoI-chosen) experiment identifies the mechanism.

is dominated by nuisance spike-timing noise and is useless for spiking data. We instead use a *feature* (synthetic / simulation-based) likelihood, the standard choice in this field. That is, we replace the above with Eq. (7), which we repeat here:

$$p(y_{1:T} | \xi, m, \theta) \propto \prod_{j=1}^J p(s_j(y_{1:T}) | \xi, m, \theta) \quad (29)$$

where s_j is the j th feature (a scalar) derived from the entire trajectory $y_{1:T}$.

Deterministic synthetic likelihood. The above synthetic likelihood does *not* factorise over time. However, because we assume the latent dynamics are deterministic and the initial state is known, it is tractable to compute: we solve the ODE for $z_{1:T}$, read off the predicted features $s_j(m, \theta) = s_j(z_{1:T})$, and evaluate the kernel in closed form. By contrast, if the dynamics are stochastic, the likelihood $p(s_j(y) | m, \theta) = \int p(s_j(y) | z_{1:T}) p(z_{1:T} | m, \theta) dz_{1:T}$ has no closed form, so we have to marginalize out over the latent paths, as we discuss in Section F.1.



Figure 20: **NEURONBENCH**. Screenshot of our app, which lets users interact with the same environment we give our agents (except the agents see numerical data, not images.) The top left is the training set, $\mathcal{D}_{tr}$, the top right is the test set, $\mathcal{D}_{te}$, and the bottom row is the interactive environment. The agent can choose a sequence of input currents $x_{1:T}$ by specifying the magnitude and duration of a step pulse (shown in orange). The agent can also choose from a finite set of interventions, corresponding to blocking different ion channels (shown as white boxes). The resulting output current $y_{1:T}$ is shown in the green trace. App is available at <https://claude.ai/code/artifact/2848d02d-cdcl-4c1c-99fe-c0034e9714fb>.

The summary statistics. The summary statistic we use are spike counts in the test and conditioning windows, their use-dependent run-down, within-pulse adaptation, and two sub-threshold voltage summaries, as illustrated in Fig. 21. These are computed as follows:

$$s(y) = \left(\underbrace{n_{\text{test}}}_{\text{test spikes}}, \underbrace{n_{\text{pre}}}_{\text{pre-pulse spikes}}, \underbrace{n_{\text{pre}} - n_{\text{test}}}_{\text{run-down}}, \underbrace{n_{\text{test}}^{\text{early}} - n_{\text{test}}^{\text{late}}}_{\text{adaptation}}, \underbrace{\min_t V(t)}_{V_{\min}}, \underbrace{\bar{V}_{\text{end}}}_{\text{steady state}} \right), \quad (30)$$

where $n_{\text{test}}, n_{\text{pre}}$ count upward zero-crossings of V in the test window (after any conditioning pre-pulse) and before it, $n_{\text{test}}^{\text{early}} - n_{\text{test}}^{\text{late}}$ splits the test window in half, and $\bar{V}_{\text{end}}$ is the mean voltage over the steady-state tail of the trace (the final few percent, after the stimulus ends). This is chosen so that both the rate-signature worlds (NA-FATIGUE, TEXTBOOK-M) and the sub-threshold/burst worlds (H-SAG, CA-REBOUND) leave a signal.

Gaussian likelihood. For real-valued summaries we use a Gaussian kernel

$$p(y_{1:T} | m, \theta) = \prod_j \mathcal{N}(s_j(y_{1:T}) | s_j(z_{1:T}), \sigma_j) \quad (31)$$

where $z_{1:T} = \text{unroll}(m, \theta, z_0)$ is the trajectory deterministically generated by solving the ODE defined by m and θ from the initial condition z_0 .

Poisson likelihood. In some cases, the features are just the spike counts $\{n_k\}$ at different input currents a_k . Since this is a set of non-negative integers, the natural observation model is a product of Poissons:

$$p(y_{1:T} | m, \theta) \propto \prod_k \text{Poi}(n_k(y_{1:T}); \lambda_k(m, \theta)), \quad \lambda_k(m, \theta) = n_k(z_{1:T}(m, \theta)). \quad (32)$$

Here the rate λ_k of the k -th count is obtained by running the candidate forward — solving its ODEs under protocol a_k to get the voltage trace $z_{1:T} = V_\theta(\cdot; a_k)$ — and then applying the *same* spike-count feature map $n_k(\cdot)$ used on the data, i.e. counting upward threshold crossings of that simulated trace. That scalar predicted count *is* the Poisson mean; the observed recording supplies the Poisson “data” $n_k(y_{1:T})$. So the deterministic model sets the mean spike count per protocol and the Poisson supplies the trial-to-trial spike-count dispersion (in practice we average a few repeats per protocol).

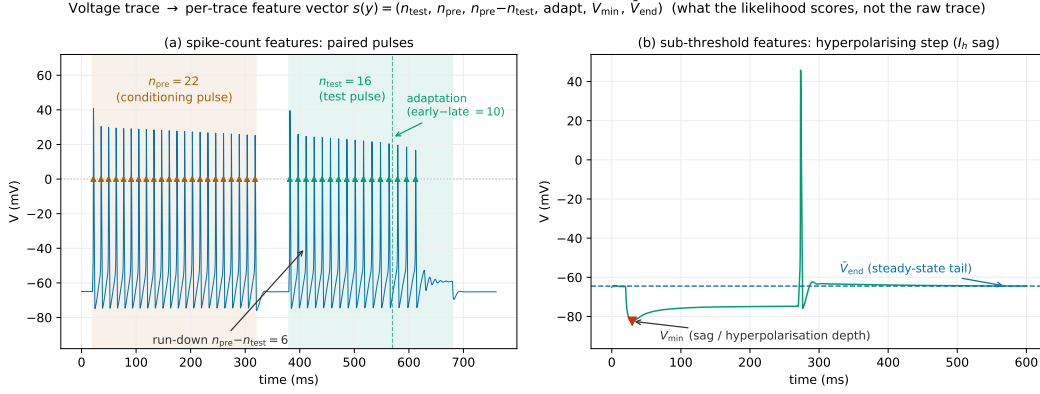


Figure 21: **From a raw voltage trace to the per-trace feature vector $s(y)$ of Eq. (30)**, computed on real NEURONBENCH traces. (a) On a paired-pulse protocol the spike-count features are the test- and pre-pulse counts ($n_{\text{test}}, n_{\text{pre}}$; upward 0 mV crossings, triangles), their use-dependent *run-down* $n_{\text{pre}} - n_{\text{test}}$ (here the slow-NANA-FATIGUE cell fires less on the second pulse), and the within-pulse *adaptation* (early-half minus late-half of the test window, dashed divider). (b) On a hyperpolarising step the sub-threshold features are the voltage minimum V_{min} (the I_h sag / hyperpolarisation depth) and the steady-state tail $\bar{V}_{\text{end}}$. These six numbers — not the raw trace — are what the synthetic likelihood and the feature-forecast metric score.

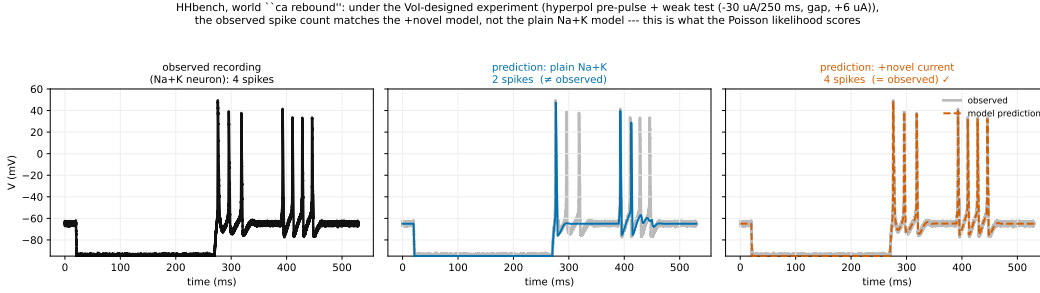


Figure 22: **Visualizing the predictions of two different models on CA-REBOUND.** We show the *observed* recording (left) and the spike response *predicted* by each candidate structure (plain Na+K neuron in middle, augmented model with novel channel on right) under the VoI-designed experiment. The plain Na+K neuron fires 2 spikes and misses the rebound (middle, $\neq$ observed); the neuron with the extra current fires the 4-spike rebound burst that matches the data (right, $=$ observed). The Poisson likelihood scores exactly this: the spike count each *deterministic* model predicts is the Poisson rate (Eq. (32)), so the observed count identifies the mechanism.

Example: CA-REBOUND. In this section, we visualize the predictive distribution $p(y_{1:T}|m)$ for two different hypotheses m — the plain Na+K+L neuron, and the correct Na+K+L+Z neuron — and show the resulting summary statistics. These results are for the CA-REBOUND world, and are obtained under a VoI-designed experiment ξ (here a hyperpolarising pre-pulse that de-inactivates the hidden Ca2+ current, then a weak test). Fig. 22 visualizes a trace $y_{1:T}$ and its summary statistic (4 spikes), followed by a prediction $E[y_{1:T}|m, \xi]$ and the summary statistics we derive from each prediction (2 spikes and 4 spikes). We see that despite the trajectory being reduced (in this case) to a single integer, $s(y_{1:T})$, the synthetic likelihood can discriminate the correct model from the incorrect one, if the design ξ is chosen properly.

E.5 PARAMETERS AND PRIORS.

Table 14 lists what is inferred and under what prior. The only free parameters are the maximal conductances g_c of the *present* channels; each is given an independent log-normal prior centred on its literature nominal value, with a log-space SD of 0.7 — a 1σ band of roughly $[\text{nominal}/2, \text{nominal} \times 2]$, deliberately broad since the target is a real cell of unknown size. The membrane capacitance, reversal potentials, and the voltage-dependent gating kinetics $T_{x,c}^\infty(V), \tau_{x,c}(V)$ are held fixed (a channel is

Quantity	Symbol	Prior / value	Role
<i>Inferred (only for channels present in the structure m):</i>			
Na ⁺ conductance	g_{Na}	LogNormal(log 120, 0.7 ²)	spike upstroke
K ⁺ (delayed rect.)	g_{K}	LogNormal(log 36, 0.7 ²)	repolarisation
Ca ²⁺ (high-thr.)	g_{Ca}	LogNormal(log 12, 0.7 ²)	alt. spike carrier
A-type K ⁺ (transient)	g_{A}	LogNormal(log 45, 0.7 ²)	onset delay
M-current (slow K ⁺)	g_{M}	LogNormal(log 1.0, 0.7 ²)	spike-freq. adaptation
leak	g_{L}	LogNormal(log 0.3, 0.7 ²)	resting potential
<i>Fixed (not inferred):</i>			
capacitance	C	1.0 $\mu\text{F}/\text{cm}^2$	membrane
reversal potentials	$E_{\text{Na/K/L/Ca}}$	+50/ − 77/ − 54.4/ + 120 mV	driving forces
gating kinetics	$T_{x,c}^{\infty}(V), \tau_{x,c}(V)$	Hodgkin–Huxley forms	channel identity
<i>Observation model (feature-kernel tolerances σ_j):</i>			
sub-threshold count	σ	0.3 spikes (tight; enforces rheobase)	likelihood
supra-threshold count	σ	1.2 spikes	likelihood
input resistance	σ	0.06 mV/pA	likelihood

Table 14: **Parameters and priors for NEURONBENCH.** Only the maximal conductances of the channels present in a candidate structure are inferred (so the minimal Na+K+leak model has three free g_c , the Na+K+M+leak winner four); everything else is fixed. Conductances g_c are in mS/cm². The log-normal priors are centred on literature nominals with a broad 0.7 log-SD; the SMC uses $N_p=70$ particles, target ESS 0.5, $K_p=3$ random-walk-Metropolis moves per tempering rung, and a 0.12 log-space proposal SD.

its kinetics; only its density g_c is free), and the feature-kernel tolerances σ_j are the fixed observation model.

E.6 RESULTS ON THE BENCHMARK

Figure 4 shows the results of MDA and the LLM baseline on NEURONBENCH. For these results, we used the Poisson likelihood on the spike count, as in Eq. (32). We see that MDA is substantially more data efficient.

E.7 MODEL SELECTION DEEP DIVE

In this section, we give a worked example of model selection, to better understand how MDA works. We focus on the H-SAG world. The LLM is shown only the *phenotype* (a depolarising sag during hyperpolarisation, then a rebound) and *proposes* the candidate channels: HCN/ I_h , T-type Ca²⁺, Kir, A-type K⁺, and persistent Na⁺, which map onto four distinct dynamical hypotheses $\{I_h, \text{T-type}, \text{D-type K}, \text{plain}\}$. This is genuinely $\mathcal{M}$ -open: the phenotype is consistent with several real channels (all can rebound), so proposing them is correct, not a mistake, and only an experiment can decide. A textbook depolarising input step leaves all four candidates *identical* (posterior unmoved from the uniform prior), whereas the VoI-designed hyperpolarising probe concentrates the posterior on I_h in one shot. Figure 23(a) shows why: under that probe only I_h produces the depolarising sag; the others stay pinned at the hyperpolarised floor. After discovering the right model structure, SMC continues to refine the posterior over the parameters, as shown in Fig. 24.

E.8 LEARNING SUMMARY STATISTICS USING A 1D CNN

In this section we give a concrete example of the approach discussed in Section A.4 for learning a set of summary statistics. In NEURONBENCH we already know the “right” hand-crafted answer, namely the features shown in Eq. (30), but this section shows how a learning based approach can give comparable performance; we will use this in earnest for the stochastic benchmark in Section F.

Extracting summary features using a neural network. We replace $s(y_{1:T})$ with a learned encoder $s_{\phi}(y_{1:T})$ which we train as follows: (1) simulate a bank of $(m, \theta) \rightarrow y_{1:T}$ raw voltage traces

HHbench (h sag), M-open discovery: the LLM PROPOSES the candidate channels (top); a Gaussian synthetic likelihood on the feature vector identifies I_h (bottom) (a) under the Vol-designed hyperpolarising experiment, only I_h shows the depolarising sag

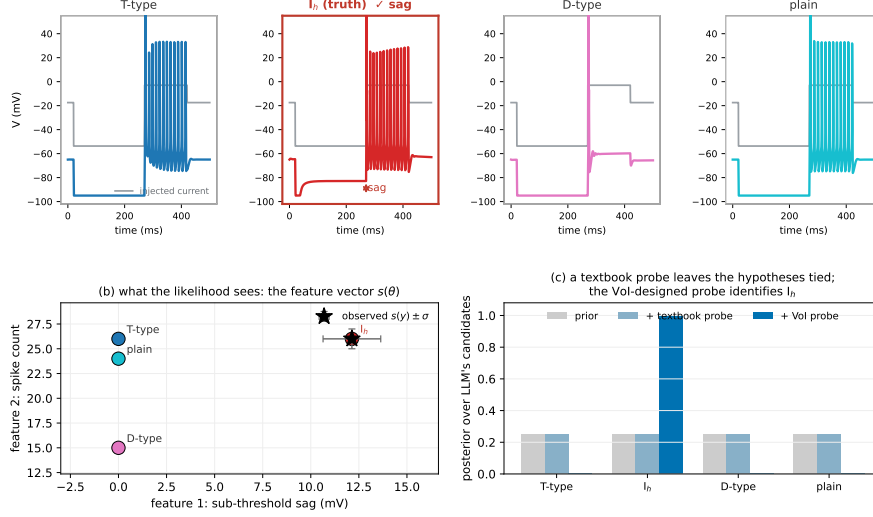


Figure 23: **M-open structure discovery on NEURONBENCH (H-SAG world).** The LLM is given only the phenotype and *proposes* the candidate channel mechanisms; they are mapped onto four dynamical hypotheses (the truth I_h is not revealed). (a) Under the Vol-designed experiment — a hyperpolarising conditioning step (grey, injected current) then a depolarising test — only I_h (red) shows the slow depolarising *sag* during hyperpolarisation; the T-type, D-type, and plain candidates stay pinned at the hyperpolarised floor. (b) *What the likelihood actually sees*: the feature vector $s(\theta) = (\text{sag}, \text{spike count})$ for each candidate (coloured points) and the observed features $s(y) \pm \sigma$ (black star). The Gaussian synthetic likelihood scores only these two numbers — not the raw trace — and the observed point coincides with I_h , the only candidate with a large sag. (c) The posterior over the LLM’s own candidate set: uniform at the prior, *unmoved* by a textbook probe (which cannot distinguish them), and collapsed onto I_h after the single Vol-designed probe. Together with Fig. 24 this exercises the whole loop — LLM proposal, M-open selection, then parameter refinement — on NEURONBENCH.

under the design protocol menu; (2) train a small 1-D CNN encoder $s_\phi(y_{1:T}) \in \mathbb{R}^d$ with two supervised heads — *classify* the channel structure m (four candidates, as in Fig. 23) and *regress* its conductance g — on that bank; the supervised targets are what forbid collapse.⁹ The two heads are used only to *train* the encoder — the softmax over structures is a discriminative posterior $p(m | y)$, not a likelihood, and both heads are discarded at inference. What we keep is the penultimate-layer embedding $s_\phi(y_{1:T}) \in \mathbb{R}^d$ ($d=8$), which plays exactly the role of the hand-crafted feature vector of Eq. (30).

The likelihood on the learned summary. We then use the multivariate Gaussian likelihood in Eq. (8) to define $p(s|m, \theta)$. This is the multivariate generalisation of the per-feature Gaussian kernel of Eq. (31): the deterministic hand-feature case reads the mean off a *single* rollout with hand-set tolerances σ_j , whereas the learned (and stochastic) case estimates *both* the mean and the covariance from the R simulations already drawn for the likelihood. Equation (8) plugs into the model-SMC unchanged, model selection being driven by the log-density gap $\log \mathcal{N}(s_\phi(y) | \mu_a, \Sigma_a) - \log \mathcal{N}(s_\phi(y) | \mu_b, \Sigma_b)$ between candidates.

Training cost, data, and re-fitting as the pool changes. The encoder is deliberately tiny — three 1-D convolutional blocks ($1 \rightarrow 16 \rightarrow 32 \rightarrow 32$ channels) into an 8-dimensional embedding with a classification and a regression head — so both the data and the compute are negligible. The training bank is 350 simulated traces per candidate structure (1400 total for the four

⁹The released open-world auto-select integration (Section F) uses this same 1-D CNN, but *pre-trained once* over the channel-archetype family and then *frozen*. Because the convolutional encoder ends in global average pooling it is protocol-length agnostic, so one encoder serves every world, protocol, and candidate pool without the per-pool re-fit — the amortised, family-trained summary of “Sufficiency as the hypothesis pool grows” (Section A). Only the Gaussian moments of Eq. (8) are re-estimated per candidate at scoring time; the particle-filter spot-check remains the out-of-family backstop.

HHbench (h sag): a Gaussian synthetic likelihood on trace features $s(\theta)$ refines the hidden channel's conductance; Vol-designed experiments do so fastest

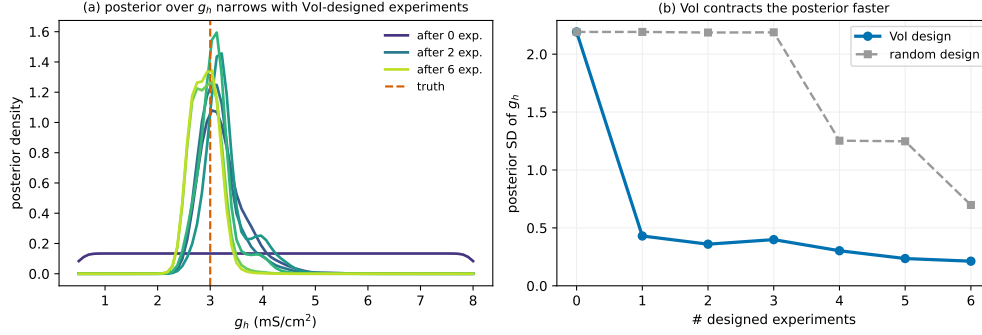


Figure 24: **Parameter refinement on NEURONBENCH** (H-SAG world: a hidden hyperpolarisation-activated I_h current, the second half of the discover-then-refine loop begun in Fig. 23). Once the mechanism is identified, its maximal conductance g_h is inferred from the *Gaussian synthetic likelihood* on trace features $s(\theta)$ (the gating kinetics are held fixed, per Table 14); because the model is deterministic this likelihood needs no simulation-based estimation. (a) The posterior over g_h contracts from the uniform prior onto the truth (dashed) as Vol-designed experiments accumulate. (b) Vol — which selects the hyperpolarising probes that make the sag, and hence $s(\theta)$, depend on g_h — contracts the posterior standard deviation $\sim 8\times$ in a single experiment, whereas random design, spending most probes on uninformative depolarising steps, lags several-fold. The menu here is augmented with a battery of hyperpolarising steps at different depths.

$\{I_h, \text{T-type, D-type, plain}\}$ hypotheses of Fig. 23), split 80/20 into 1120 train / 280 test; each trace is a single forward solve under the protocol menu, downsampled to ~ 750 samples. Training runs for 40 epochs (batch 64, Adam at 2×10^{-3}) and completes in about a minute on a laptop CPU/MPS — no GPU cluster. The simulations dominate the wall-clock, and they are *the same rollouts already drawn for the synthetic likelihood*, so the marginal cost of learning s_ϕ over hand-specifying it is essentially free.

This also answers what happens when the outer model-SMC pool changes. We do *not* maintain one global summary: sufficiency is task-relative to the *current* candidate set, so when the LLM proposes a new mechanism we add a class and re-fit — a $\sim$ one-minute job on simulations already in hand — and the summary dimension simply grows with the pool. Re-fitting per round is cheap enough to do routinely, and can be avoided altogether by amortising: training s_ϕ once over the LLM’s prior-predictive mechanism *family* (rather than the current pair) generalises to in-distribution proposals without any re-fit. Either way the particle-filter anchor (Section F.1) catches an insufficient summary should a proposal fall outside what s_ϕ was trained to attend to. See Section A.4 (“Sufficiency as the hypothesis pool grows”) for the full argument.

Results. In Fig. 25(a), we show that, on held-out traces, the learned summary identifies the hidden channel with 100% accuracy — *above* the hand-crafted [sag, spike-count] baseline (92%) — and recovers the conductance to 0.56 mS/cm² mean absolute error. Here the two hand features are the sub-threshold “sag” — the slow depolarising recovery from the trough during a hyperpolarising step, the signature of the hyperpolarisation-activated inward current I_h (HCN) — and the spike count; together they discriminate the four candidate channels.

Figure 25(b) asks *which windows* the learned encoder actually relies on, via an *occlusion* analysis: we mask each time window of the input and measure the resulting drop in the I_h logit. (We use occlusion rather than a raw input-gradient saliency because the latter is nearly uniform for this network — a known pathology of vanilla saliency — and so localises nothing.) The importance concentrates on the *spike* windows — the rebound burst after the hyperpolarising release and the depolarising spike train (mean importance 0.54) — and is markedly lower across the sub-threshold sag phase (0.20) and the quiescent stretches (0.25). So the network rediscovers that the discriminative signal lives in the *firing pattern*, the $\{n_k\}$ spike-count features.

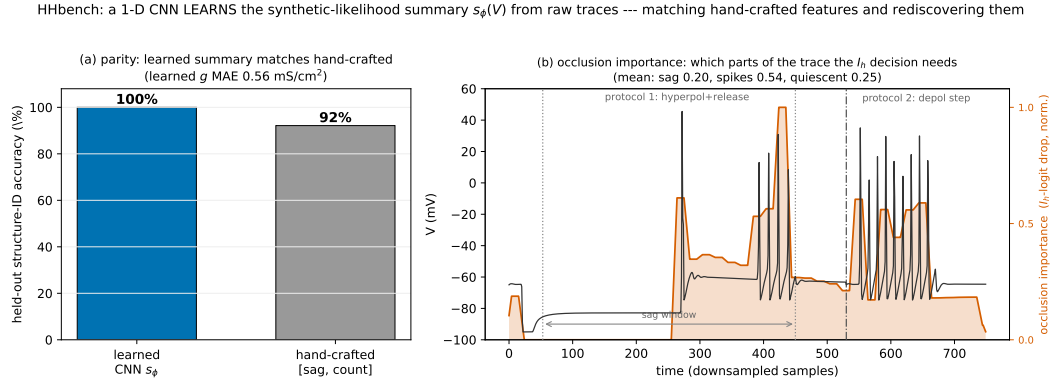


Figure 25: Learning the NEURONBENCH synthetic-likelihood summary from raw traces (proof-of-concept; runs on a laptop). A 1-D CNN encoder $s_\phi(V_{1:T})$ is trained on simulated $(m, g) \rightarrow V$ traces to classify the channel structure and regress its conductance. (a) On held-out traces the learned summary identifies the structure at 100% — above the hand-crafted [sag, spike-count] baseline (92%) — and recovers g to 0.56 mS/cm² MAE. (b) *Occlusion* importance for the I_h class (orange, right axis: normalised drop in the I_h logit when each time window of the input is masked) over a noise-free I_h trace (black) that concatenates two protocols (separated by the dash-dot line): a hyperpolarising step+release (the I_h sag window marked by the grey dotted bars) and a depolarising step. The decision rests on the *spike* windows (the rebound burst and the depolarising spike train; mean importance 0.54), not the sub-threshold sag (0.20) or the quiescent stretches (0.25): the network rediscovers that the $\{n_k\}$ spike-count features carries the discriminative signal. (I_h is the hyperpolarisation-activated inward (HCN) current; the “sag” is its slow depolarising recovery during a hyperpolarising step. A raw input-gradient saliency is near-uniform here and thus omitted.)

F NEURONBENCHSTOCH

F.1 STOCHASTIC LATENT DYNAMICS: BACKGROUND

The worlds above use a *deterministic* Hodgkin–Huxley forward model, so the likelihood is available in closed form (Section E.4). Real neurons are stochastic: with a finite number of ion channels, gating fluctuates (channel noise), the latent dynamics become an SDE. This is the regime real experiments occupy, and the one setting our other benchmarks (deterministic ODEs + observation noise) do not exercise.

To create a stochastic neuron, we add finite- N channel noise via the Fox–Lu diffusion approximation (Fox & Lu, 1994), with the channel count N tuning the intrinsic noise from near-deterministic ($N \rightarrow \infty$) to strongly stochastic.

In more detail, each gate x_c is really an ensemble of N two-state ion channels, each switching open $\leftrightarrow$ closed as a continuous-time Markov chain with the voltage-dependent rates $\alpha_x(V), \beta_x(V)$ of Eq. (25); the deterministic HH gating ODE is the $N \rightarrow \infty$ mean-field limit of the open fraction. The Fox–Lu diffusion approximation (Fox & Lu, 1994) keeps finite N by replacing that mean field with a Langevin (stochastic differential) equation — the deterministic drift plus a Gaussian channel-noise term whose variance scales as $1/N$:

$$dx_c = [\alpha_x(V)(1-x_c) - \beta_x(V)x_c] dt + \sqrt{\frac{\alpha_x(V)(1-x_c) + \beta_x(V)x_c}{N}} dW_t, \quad x \in \{m, n, h\}, \quad (33)$$

with dW_t an independent Wiener increment per gate. The diffusion coefficient is the sum of the two transition fluxes divided by N (the system-size / Ω -expansion correction to the channel master equation), so more channels means smaller fluctuations and $N \rightarrow \infty$ recovers the deterministic gate. We integrate Eq. (33) by Euler–Maruyama and substitute the noisy gates into the membrane equation (21), making N a single knob from near-deterministic to strongly stochastic. Fox–Lu is the standard cheap channel-noise model; see Goldwyn & Shea-Brown (2011) for how it compares to exact Markov-chain channel simulation.

F.2 THE BENCHMARK

We convert the deterministic six-world NEURONBENCH into a stochastic form by substituting the Fox–Lu SDE (Eq. (33)) for the deterministic gate evolution in Eq. (21). We use the same design space. This defines NEURONBENCHSTOCH.

F.3 LIKELIHOODS

The marginal likelihood for model m is given by

$$Z_m = p(y_{1:T} \mid m, \theta) = \int p(y_{1:T} \mid z_{0:T}, m, \theta) p(z_{0:T} \mid m, \theta) dz_{0:T}, \quad (34)$$

where $y_{1:T}$ is the observed voltage trace and $z_{0:T}$ the latent gating path. This requires marginalising over the stochastic latent path $z_{0:T}$ — a high-dimensional path integral with no closed form, because the Fox–Lu transition density $p(z_t \mid z_{t-1})$ is itself intractable. Below we discuss how to approximate this integral using a bootstrap particle filter (Algorithm 4), as well as various other faster approximations.

Particle filtering. The agent fits a stochastic state-space model (1) where the latent state $z_t = (V_t, \{x_c(t)\})$ (voltage and gates) evolves by the discretised Fox–Lu transition $p(z_t \mid z_{t-1}, \xi)$ of Eqs. (21) and (33), and the voltage is observed with Gaussian noise, $y_t \sim \mathcal{N}(V_t, \sigma^2)$. Candidate models m differ in *structure* (which channels are present) and in the conductances θ ; the channel count N (the noise scale) is a known part of the model here. The one-step transition density $p(z_t \mid z_{t-1}, \xi)$ has no closed form — it is a nonlinear diffusion over the interval — but the bootstrap particle filter never needs it. It only *samples* the transition (one Euler–Maruyama step, i.e. a Gaussian draw on the gates, Eq. (33)) as its proposal, and only *evaluates* the tractable observation density $\mathcal{N}(y_t \mid V_t, \sigma^2)$ to reweight the particles, from which the marginal likelihood Z_m can be estimated. So the intractable-likelihood regime needs only a *simulator* of the latents plus an evaluable observation model, exactly what a mechanistic ODE/SDE provides — no transition density is ever computed.

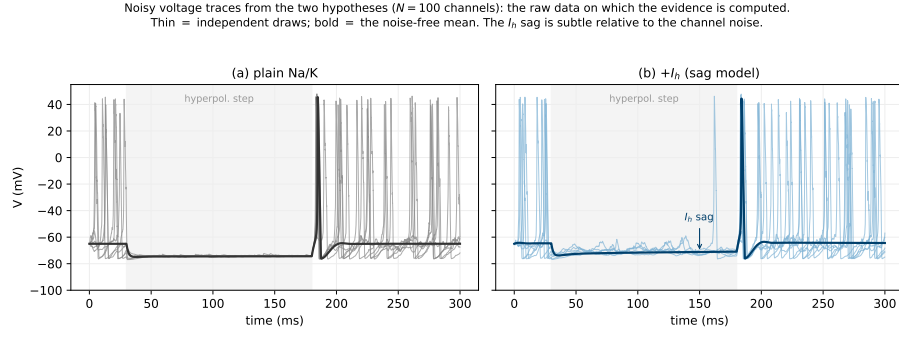


Figure 26: **Stochastic-latent NEURONBENCH: the raw data.** Noisy voltage traces from the two competing hypotheses under a moderate hyperpolarising-step protocol, at $N=100$ channels (thin: independent draws; bold: the noise-free mean). (a) A plain Na/K cell. (b) The same cell plus a hyperpolarisation-activated I_h current (the H-SAG model of Table 12), whose only signature is a small depolarising *sag* during the step (arrow). Because that sag is comparable in size to the channel noise, the two hypotheses overlap and cannot be told apart by eye — this is the data on which the evidence of Fig. 27 is computed.

Stochastic-latent HHbench: finite- N channel noise makes $p(y | m, \theta)$ intractable, so MDA estimates it by simulation (particle filter), with a learned-summary synthetic likelihood as a cheap surrogate

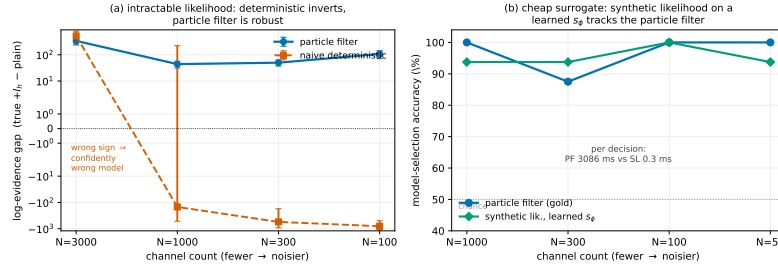


Figure 27: **Stochastic-latent NEURONBENCH: estimating the intractable likelihood by simulation.** Both panels score the I_h -vs-plain decision on the data of Fig. 26, sweeping the channel count N (fewer = noisier). (a) The log-evidence gap $\log Z_1 - \log Z_2$ vs. N . A likelihood that ignores the process noise (a single deterministic rollout + Gaussian observation, orange) degrades and, below $N \approx 1000$, *inverts* — a negative gap means it confidently selects the *wrong* mechanism. A bootstrap particle filter (blue), which estimates $p(y | m, \theta)$ by propagating the latent gating SDE, stays robustly positive. (b) A synthetic likelihood on a learned summary s_ϕ (green) reproduces the particle filter’s model-selection accuracy across the noise sweep at $\sim 10^4 \times$ less compute (~ 0.3 ms vs. ~ 3 s per decision). Bars/points are ± 1 SE over independent noise realisations; see Section A.4 for s_ϕ .

Why the deterministic likelihood breaks. To illustrate why we cannot just use a deterministic ODE model (and hence a deterministic likelihood, as we did in Eq. (3)), we consider a simple example where we need to distinguish just two hypotheses: a plain Na/K cell vs the novel H-SAG model I_h defined in Table 12. The noisy voltage traces from the two hypotheses are shown in Fig. 26 — the I_h sag is subtle relative to the channel noise, so they overlap. In Fig. 27(a) we plot the log-evidence gap, $\log Z_1 - \log Z_2$, vs noise level N , where Z_1 is the evidence for the I_h hypothesis and Z_2 for the alternative Na/K hypothesis. We see that a likelihood that treats the intrinsic channel noise as zero degrades as the noise grows and, at $N=1000$ channels, *inverts*: it confidently selects the wrong mechanism. By contrast, the particle filter algorithm in Algorithm 4, which propagates the latent gating SDE with N_z particles and weights each by the observation, stays robustly correct at every noise level.

A learned-summary synthetic likelihood as a cheap surrogate. The particle filter is accurate but costly (~ 4 s per candidate model per experiment, $N_z=600$). A *synthetic likelihood* on a learned summary statistic, as in Eq. (8), reproduces the particle filter’s decision $\sim 10^4 \times$ less compute, as shown in Fig. 27(b).

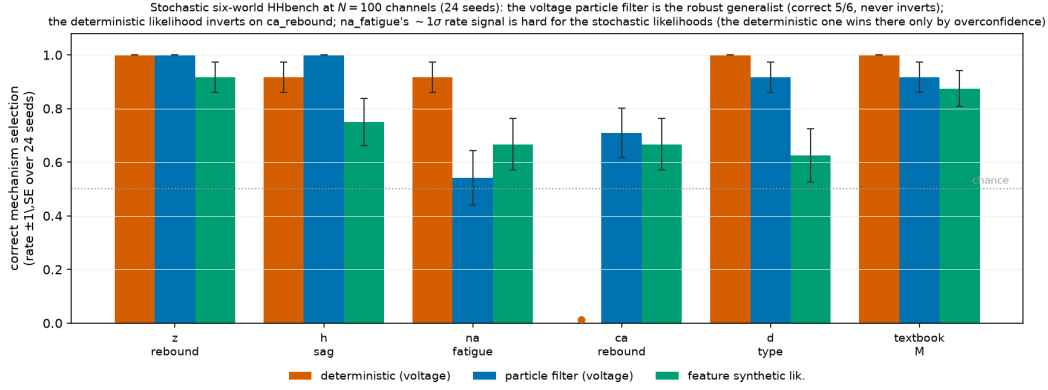


Figure 28: **Three observation models on the stochastic six-world battery (fixed hypothesis space)** ($N=100$ channels; correct-selection rate ± 1 SE over 24 seeds; a dot marks an exact zero). The voltage *particle filter* (blue) is the robust generalist — correct on five of six worlds and never inverting (its one weak world is the SNR-limited NA-FATIGUE, at chance). The *deterministic* likelihood (orange) *inverts* on CA-REBOUND (wrong on every one of the 24 seeds). The *feature synthetic likelihood* (green) stays above chance on every world and *edges* the particle filter on the rate world NA-FATIGUE, but is weaker on the worlds whose discriminating signal is a sub-threshold or timing *shape* (H-SAG, D-TYPE) that the summary vector compresses. On NA-FATIGUE only the deterministic likelihood is clearly correct; however the deterministic likelihood catastrophically fails (inverts) on CA-REBOUND.

Which observation model? In this section we consider a simplified version of the 6 stochastic worlds where we only have two hypotheses (truth and a distractor). We use $N=100$ latent channels, so the dynamics are fairly stochastic. We consider the *deterministic* likelihood (one noiseless rollout + Gaussian voltage noise), the bootstrap *particle filter* on the raw voltage, and a *synthetic likelihood on the feature vector* $s(y)$ of Eq. (30), estimated by simulation, as in Eq. (8). The results are shown in Fig. 28. We see that the voltage particle filter is the robust generalist: correct on five of six worlds and, crucially, it *never inverts* (its one weak world is the SNR-limited NA-FATIGUE, where it sits at chance). The deterministic likelihood does more than degrade — on CA-REBOUND it *confidently inverts* (wrong on every seed), reproducing the single-world failure of Fig. 27 in a fresh world: treating each voltage sample as independent Gaussian evidence accumulates spike-timing jitter into a large, wrong gap. The feature likelihood is complementary: it stays above chance on every world and *edges* the particle filter on the spike-rate world NA-FATIGUE (where the raw-voltage PF is at chance), but is weaker on the worlds whose discriminating signal is a sub-threshold or timing *shape* (H-SAG, D-TYPE) that the summary vector compresses. There is thus no single best observation model: shape signatures want the voltage filter, spike-rate signatures want the feature likelihood, and neither escapes the overconfidence that sinks the deterministic one on CA-REBOUND — which is what motivates holding *both* and auto-selecting (Fig. 29, below).

Auto-selecting the observation model. Because no single observation model wins on every world (Fig. 28), the agent should hold *both* the feature synthetic likelihood and the voltage particle filter and pick per world — and it can do so *without* knowing the truth. Before committing to an experiment it runs a cheap probe: on the world’s discriminating protocol it simulates single experiments from *each* candidate in turn and measures how often each observation model’s log-evidence gap identifies the generator, averaged over which candidate generated the data. This is pure *discrimination power* — which likelihood best tells the hypotheses apart, never peeking at the truth.

Formally this is the cost-aware observation-model selection of Eq. (11) (App. A), applied here over $o \in \{\text{PF}, \text{feat}\}$: the agent estimates each MI_o by the truth-free discrimination probe above and, charging the PF its extra compute ($\text{cost}(\text{PF})/\text{cost}(\text{feat}) \approx N_z/R \gg 1$), defaults to the cheap feature likelihood and pays for the filter only where its discrimination clearly justifies it. Ignoring cost it reduces to picking the more discriminating arm (Fig. 29).

Figure 29 shows the result at $N=100$: the probe correctly routes the burst world CA-REBOUND to the particle filter (feature likelihood $0.49 \rightarrow$ PF 1.00) and the rate world NA-FATIGUE to the feature likelihood (0.82, where the voltage PF is below chance at 0.43), so the auto-selected arm attains the

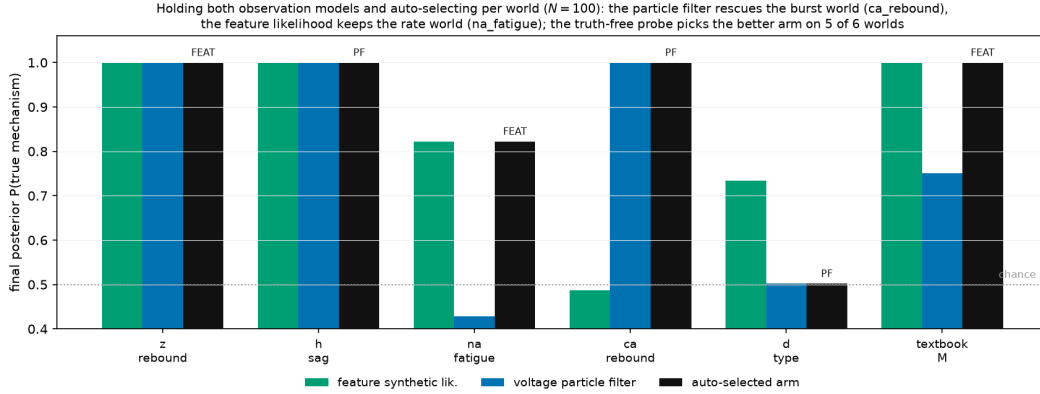


Figure 29: **An auto-selected observation model** ($N=100$, six worlds, fixed hypothesis space; final posterior of the true mechanism under repeat-aware VoI). The feature synthetic likelihood (green) and voltage particle filter (blue) are complementary — the PF rescues CA-REBOUND’s burst while the feature likelihood keeps NA-FATIGUE’s spike-rate signature (where the PF is below chance). The agent auto-selects (black) by a *truth-free* discrimination probe — which likelihood best separates the candidates on the discriminator, averaged over each candidate generating the data (the letter marks the chosen model, PF or FEAT). It attains the better arm on five of six worlds; on D-TYPE the single-shot probe favours the PF, which then underperforms the feature likelihood under the budgeted design.

better of the two on *five of six* worlds. The exception is D-TYPE: the probe favours the PF (which separates it well in a single shot, 0.92), but under the feature-MI-driven design the PF underperforms at budget (0.50 vs. the feature likelihood’s 0.73) — the single-shot probe does not perfectly predict the full-loop outcome. Even so, the observation model, like the experiment, becomes something the agent *chooses* from data rather than a hand-set knob.

The learned summary and the spot-check in the open-world loop. The two-hypothesis study above pits the particle filter against a *fixed*-feature synthetic likelihood. The released open-world benchmark adds a third, still cheaper option — a synthetic likelihood on a *learned* summary s_ϕ (a frozen 1d CNN, Section A.4) — so the cost-aware selection of Eq. (11) now ranges over $o \in \{\text{PF}, \text{feat}, s_\phi\}$, guarded by the particle-filter spot-check of Section A. Running the full $\mathcal{M}$ -open battery (six worlds, three seeds, the LLM proposing its own candidate channels at $N=100$), the cost-aware probe selects a *cheap* model on every run, and the spot-check overrides it to the PF wherever the cheap posterior disagrees with the filter. Of the 18 runs the final observation model was the learned s_ϕ on 2, the fixed feature likelihood on 5, and the particle filter on 11 — all 11 reached via a spot-check *disagreement*. So the frozen s_ϕ is used only where it is *verifiably* sufficient (its posterior matches the filter’s, e.g. H-SAG and D-TYPE), while the PF anchor catches the confusable cases. The sharpest is CA-REBOUND: the cheap summaries confidently prefer a slow- Na^+ run-down, but the filter — and the T-type Ca current the LLM proposed — win, cutting its feature-forecast error from 4.5 (feature-only) to 0.85, with mean mechanism recovery 0.94 across the battery (19 LLM calls, \$0.55 total). Figure 30 visualises one such disagreement on D-TYPE.

F.4 DATA EFFICIENCY CURVES

The data efficiency curves from applying MDA and baseline LLM to NEURONBENCHSTOCH are shown in Fig. 31. Note only do we see that MDA is more sample efficient, but also that the ICL forecasting approach is basically hopeless in this noisy setting (Bayes ~ 0.15 vs. ICL ~ 46 spikes²). We also see that random designs are similar to (and arguably a tiny bit better than) VoI designs.

F.5 WIRING THE REPEAT COUNT INTO THE VOI DESIGN SPACE

When the data is noisy, it is useful to be able to repeat experiments, to average the noise down. We therefore enlarge the design to $\xi = (\text{protocol}, r)$, where r is a repeat count costing r units of budget: averaging r repeated trials shrinks the spike-count noise $\sim 1/\sqrt{r}$, so re-running the informative protocol is itself a design lever the agent can pull.

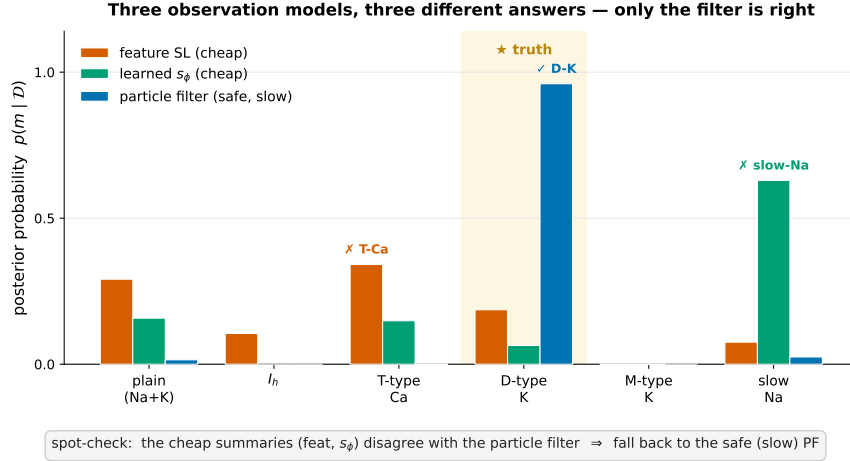


Figure 30: **The particle-filter spot-check** (D-TYPE, fixed archetype pool, $N=100$). The two cheap observation models disagree with each other *and* with the particle filter: the fixed-feature synthetic likelihood picks T-type Ca and the learned s_ϕ picks slow- Na^+ , while only the assumption-free particle filter recovers the true D-type K current ($\star$). Because a selected cheap model’s MAP disagrees with a PF spot-check on the collected data, the auto-select of Eq. (11) falls back to the safe — but $\sim 7\times$ slower — filter. The truth-free discrimination probe alone cannot catch this: it averages over generators, so a summary that confuses one pair while separating the rest still scores well.

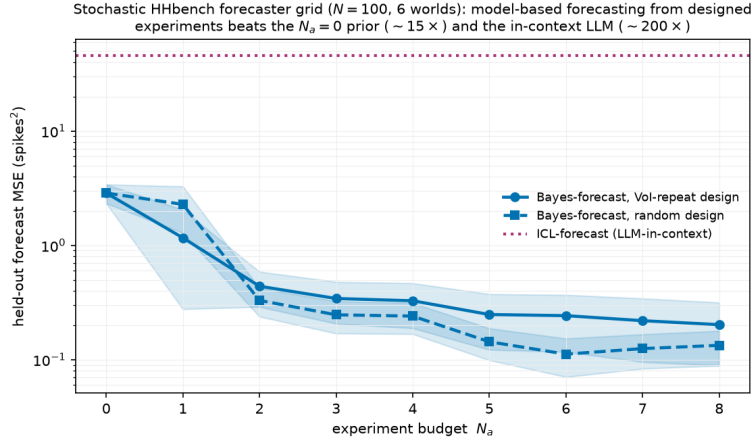


Figure 31: **Stochastic-NEURONBENCH forecaster grid** ($N=100$, six worlds, fixed hypothesis space, held-out interventional-forecast MSE vs. the experiment budget N_a ; log scale). The model-based Bayes-forecast (blue; Vol-repeat solid, random design dashed) built on the design-loop posterior falls $\sim 15\text{--}20\times$ from the $N_a=0$ prior and sits more than two orders of magnitude below the in-context LLM forecaster (purple dotted). As on the physics and deterministic-neuron rungs, the forecaster axis (model-based vs. LLM) matters more than the acquisition axis (VolI vs. random). Bands are ± 1 SE over worlds $\times$ seeds.

Expanded Vol. The expanded Vol equation becomes

$$\xi^* = \arg \max_{\xi} \frac{\text{MI}(m; \bar{s}_r(y) \mid \xi)}{\text{cost}(\xi)}, \quad (35)$$

where $\bar{s}_r(y) = \frac{1}{r} \sum_{t=1}^r s(y^{(t)})$ is the feature vector averaged over the r trials, and $\text{cost}(\xi) = r$ is the number of repeats. Here the per-trace summary $s(y)$ is the six stochastic-battery features in Eq. (30).

Concretely, let $w_m = p(m \mid \mathcal{D})$ be the current posterior over the candidate mechanisms. The r -averaged features have the (simulation-estimated) Gaussian synthetic likelihood $p(\bar{s}_r \mid m, \xi) = \mathcal{N}(\bar{s}_r \mid \mu_{m,\xi}, \frac{1}{r} \Sigma_{m,\xi})$, with $\mu_{m,\xi}, \Sigma_{m,\xi}$ read off R simulated traces per candidate, where Σ is

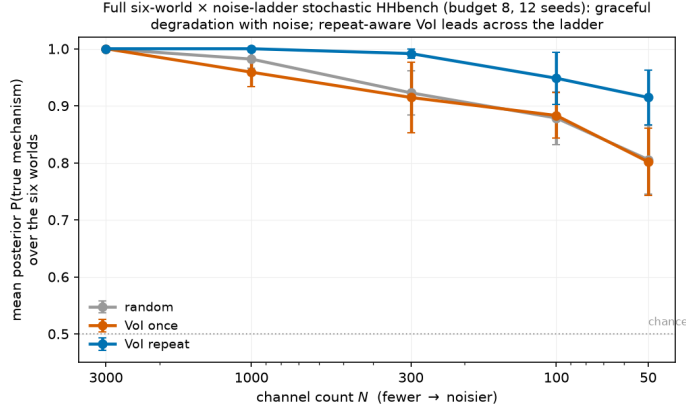


Figure 32: **Full six-world $\times$ noise-ladder stochastic NEURONBENCH.** Mean posterior probability of the true mechanism over the six worlds vs. the channel count N (log axis; near-deterministic at left, noisiest at right), for the three acquisition policies (budget 8, 12 seeds). All degrade gracefully with noise; repeat-aware VoI (blue) leads across the whole ladder and its lead over each-once/random *widens* as N falls — re-running the discriminator is the decisive lever exactly where the per-experiment signal is weakest. Bars are ± 1 SE over the six worlds. These aggregates are a mild *lower* bound: every world, including CA-REBOUND, is scored under the single feature likelihood, whereas the auto-selected agent of Fig. 29 would route CA-REBOUND to the voltage particle filter.

diagonal when using the summary vector in Eq. (7). (The $\frac{1}{r}$ factor being the variance reduction from averaging.) The information gain is the expected drop in the entropy of the model posterior,

$$\text{MI}(m; \bar{s}_r | \xi) = H(w) - \mathbb{E}_{p(\bar{s}_r | \xi)}[H(q(\cdot | \bar{s}_r))], \quad q(m | \bar{s}_r) = \frac{w_m p(\bar{s}_r | m, \xi)}{\sum_{m'} w_{m'} p(\bar{s}_r | m', \xi)}, \quad (36)$$

with prior entropy $H(w) = -\sum_m w_m \log w_m$ and posterior-predictive feature mixture $p(\bar{s}_r | \xi) = \sum_m w_m \mathcal{N}(\bar{s}_r | \mu_{m, \xi}, \frac{1}{r} \Sigma_{m, \xi})$. We evaluate Eq. (36) by Monte Carlo over that mixture: draw J samples $m^{(j)} \sim w$, $\bar{s}_r^{(j)} \sim \mathcal{N}(\mu_{m^{(j)}}, \frac{1}{r} \Sigma_{m^{(j)}})$, form each model posterior $q(\cdot | \bar{s}_r^{(j)})$, and average the per-sample gain $H(w) - H(q(\cdot | \bar{s}_r^{(j)}))$ over the J draws ($J=200$ here). Because averaging shrinks the noise, VoI can now *spend budget re-running the discriminator* to beat the channel noise, rather than being forced onto uninformative decoy protocols. This is the VoI objective of Eq. (5) with the repeat count promoted to a first-class part of the design.

Results. As a proof of concept, we run the whole six-world battery (using a fixed set of hypotheses) across the full channel-count ladder — from near-deterministic ($N=3000$) to strongly stochastic ($N=50$) — using the synthetic factored Gaussian likelihood with the fixed summary features from Eq. (30). In Fig. 32 we plot the mean posterior on the truth over the six worlds. All acquisition policies *degrade gracefully* with noise, from certainty at $N=3000$ to 0.8–0.9 at $N=50$. Repeat-aware VoI leads at every rung, and — the key point — *its margin widens as the noise grows*: from a tie at $N \geq 1000$ to 0.91 vs. 0.80 over each-once/random at $N=50$. Spending budget on repeats is exactly the lever that matters most when the per-experiment signal is weakest.

G LLM PROMPTS

This appendix reproduces the prompts used by the three domains of this paper (physics, chemistry, biology). Throughout, the LLM runs at temperature 0.2–0.4 with JSON-mode responses and every call is cached for reproducibility; the base model is stated per experiment (Opus 4.7 unless noted, with the base-model robustness sweep of Section C.8 using Fable 5 and DeepSeek v4, and the CHEMBENCH head-to-head using a matched proposer). MDA uses the LLM only to *propose* structures and, in the baseline arms, to *acquire* experiments and *forecast*; inference, VoI design, and fitting are exact Bayesian computations.

G.1 FORCEBENCH (FORCE LAWS, §4.1)

The physics rung invokes the LLM in four distinct roles, which fall into two groups: the *proposer*, which is the only LLM call inside MDA’s discovery loop, and the *verbaliser/judge*, which are the benchmark’s own explanation-scoring machinery and play no part in discovery, inference, or forecasting.

G.1.1 MDA PROPOSER (THE ONLY LLM CALL IN THE DISCOVERY LOOP)

The proposer sees the world context, the probe data collected so far, and a language specification that steers it toward field-equation Green’s functions rather than curve-fits; it returns candidate force laws as JSON (parsed, compiled, and SMC-fit by MDA). Note that the language spec names screened/power-law/oscillatory *families* (including the K_1 Yukawa form) as examples, so the proposer is given the physical vocabulary — MDA’s contribution is the inference and VoI design that identify *which* form the data support (see App. C, Fig. 14), not blind form-discovery. Its system message and user template:

```
SYSTEM:
You are a physicist proposing candidate pairwise force laws to explain probe-orbit
↪ data. Reply JSON only.

USER (world context, then the observed data, then the language spec):

A test probe moves in an unknown central force sourced by a fixed body at the origin.
↪ The force MAY be static or MAY vary with time t (e.g. a time-modulated coupling).
↪ Each experiment launches the probe from a position with a velocity, and sets two
↪ knobs p1, p2. <one sentence naming the two experiment knobs p1, p2 for this world
↪ -- e.g. p1 the source coupling, p2 the probe inertia> F_mag is the pairwise force
↪ magnitude between the probe (charge qi) and source (charge qj) at separation r
↪ and time t.

Observed data:
<measurement times and the radius r(t) of each probe run so far>

Propose N distinct plausible force laws (or refinements of those tried).

Propose each force law as the field / Green's-function response of a PHYSICAL FIELD
↪ EQUATION (e.g. 2D Laplacian/Poisson -> 1/r; screened Poisson / Helmholtz -> a
↪ screened form such as exp(-r/lam)/r or K1(r/lam)/lam; fractional Laplacian -> a
↪ power law 1/r^p; 3D inverse-square -> 1/r^2), NOT an arbitrary curve-fit with
↪ softening/offset terms. If the data show the force changing sign or magnitude
↪ over time (not just with r), the coupling itself may be time-dependent (e.g. a
↪ cos(w*t+phi) modulation) -- consider such forms too. State the governing operator
↪ in the rationale. Express each F_mag as a Python expression in the symbols r, qi,
↪ qj, t and your OWN named free parameters ONLY. The source coupling is carried by
↪ qj (the probe is qi); do NOT reference p1 or p2 in the expression -- introduce
↪ named parameters (e.g. k, G, lam, s) for coupling constants and length scales.
↪ Allowed functions: exp, log, sqrt, sin, cos, tanh, k0, k1, gamma, pi, np. Return
↪ JSON {"hypotheses": [{"name": str, "fmag": str, "operator": str, "params":
↪ [{"name": str, "low": float, "high": float}], "rationale": str}]}
```

For the extension worlds (App. C) only the proposer *context* changes — the declared background field (ether/Hubble), the self-interacting cloud (circle), or the known-law hidden sources (dark matter); the language spec is unchanged:

```
ETHER / HUBBLE (central force + declared background):

Here the probe is a neutral test particle (qi=1) orbiting a fixed central anchor that
↪ sources the field (coupling carried by a named parameter); its inertia is 1.
```

Test probes orbit an unknown CENTRAL force sourced by a fixed anchor at the origin (a
 ↳ 2D field-equation response, e.g. a Laplacian giving $F \sim 1/r$). <probe roles>
 ↳ LAYERED ON TOP there is a uniform, mass-independent background acceleration of
 ↳ magnitude α in the +y direction (a constant 'ether' drift), on top of the
 ↳ central force. That background is handled separately by the fitter -- you only
 ↳ need to propose the CENTRAL pairwise force magnitude $F_{\text{mag}}(r, q_i, q_j, t)$ sourced
 ↳ by the anchor. F_{mag} is the magnitude of the attractive central force on the
 ↳ probe at separation r from the anchor.

 CIRCLE (self-interacting N-body):

Eleven identical particles -- one at the centre and ten equally spaced on a ring --
 ↳ ALL interact with each other through the SAME pairwise central force (uniform
 ↳ coupling): every particle both sources the field and feels it. Each experiment
 ↳ sets the ring radius and a tangential launch velocity. Propose the pairwise force
 ↳ magnitude $F_{\text{mag}}(r, q_i, q_j, t)$ between any two particles at separation r (with
 ↳ $q_i=q_j=1$, the uniform coupling); it is attractive and depends only on r for a
 ↳ static field. The many-body motion is the sum of these pairwise forces.

 DARK MATTER (known law, latent hidden sources):

Test probes move in a KNOWN static 2D-Laplacian field (each source contributes $F =$
 ↳ $q/(2\pi r)$, attractive), sourced by 20 VISIBLE particles of coupling 1 whose
 ↳ positions are known, PLUS an unknown number of HIDDEN sources that reveal
 ↳ themselves only through the probes' deflection toward seemingly empty regions.
 ↳ The task is to infer how many hidden sources exist and their positions and
 ↳ couplings.

(three species uses no LLM proposer -- the couplings are inferred by a linear solve;
 ↳ the LLM only verbalizes the recovered species, via the verbaliser above.)

G.1.2 VERBALISER AND JUDGE (BENCHMARK EXPLANATION SCORING; NOT USED FOR DISCOVERY)

These two roles exist only to compute the benchmark's *explanation* metric: the verbaliser turns MDA's *already-selected* law into a short prose explanation, which the benchmark's own judge scores against the world's optimal explanation and rubric (temperature 0, integer 0–10). MDA's posterior, VoI design, and held-out forecasts never call either.

SYSTEM:
 You are a physicist. Reply with a 2-4 sentence explanation only.

USER:

<world context>

A Bayesian model-discovery method fit the data and selected the force law $F_{\text{mag}} =$
 ↳ <the fitted F_{mag} expression> with FITTED parameters {'<parameter names>':
 ↳ $\text{np.float64}(0.0)$ }. Explain the physics of THIS law, and be specific and complete:
 ↳ (1) name the governing field equation / operator; (2) state its temporal
 ↳ character (static vs time-evolving); (3) give the NUMERIC value of any length- or
 ↳ scale-parameter you fitted (e.g. a screening length) and say how the force
 ↳ behaves at short vs long range; (4) explicitly state the physical roles of the
 ↳ knobs p_1 and p_2 as described above.

SYSTEM:
 You are an expert physicist grading how well a student's prose description of a
 ↳ simulated physical system matches the ground-truth description. You are precise,
 ↳ fair, and reward semantic correctness over surface phrasing -- paraphrases and
 ↳ equivalent formulations (e.g. 'inverse-square-like' $\sim$ 'grad²phi' in 2D) should
 ↳ receive credit, but missing or wrong physical content should not.

USER TEMPLATE:

Compare the student's description against the ground-truth description of the
 ↳ physical system.

<ground_truth>
 {ground_truth}
 </ground_truth>

<student>
 {student}
 </student>

Score the student description on a 0-10 integer scale based on how well it captures:

```

1. The correct field equation / governing operator (e.g. Laplacian, fractional
↪ Laplacian, Helmholtz, diffusion, wave).
2. The temporal character (static vs. time-evolving; instantaneous vs. retarded).
3. The force law / coupling structure (how particles couple to the field, including
↪ p1/p2 roles).
4. Any structural features unique to this world: hidden species and their relative
↪ coupling strengths and signs, neutral probes, hidden/dark sources, screening
↪ lengths, etc.

Use the world-specific rubric below to calibrate the bands. A 10/10 represents the
↪ best explanation achievable given the experimental capabilities -- reward
↪ semantically-equivalent phrasings and numeric estimates within the tolerance
↪ specified by the rubric.

<scoring_rubric>
{rubric}
</scoring_rubric>

Respond with 1-3 sentences of justification, then your final integer score inside
↪ <score>...</score> tags. Example: "<score>7</score>".

```

G.1.3 BASELINE FORECASTERS

(The $N_a=0$ and LLM-forecast arms of Fig. 2.) The *zero-shot* baseline reuses the proposer’s system prompt and world context but appends, in place of any data, the instruction: “*No experimental budget is available: you cannot run any experiments or fits. Based ONLY on physical reasoning about the setup described above, submit your single best-guess law now*” (a `<final_law>` discovered_law(...) plus an `<explanation>`). The *LLM-forecast* baseline is the same, but with the collected experiments’ launch configurations and observed radii $r(t)$ listed before the submission instruction — so the LLM *authors a law* from the data in context (rather than MDA templating one from its posterior), scored by the benchmark’s own executor exactly as MDA’s Bayes-forecast is.

G.2 CHEMBENCH (ENZYME RATE LAWS, §4.2)

As in physics, the only LLM call inside MDA’s discovery loop is the *proposer*. It sees an enzyme-kinetics mechanism *grammar*, the experiments collected so far (design inputs $\rightarrow$ observed initial rate r_0), and — when refining an existing pool — the forms already tried together with their residuals (so it does not re-propose dead ends, and knows *which* input a residual correlates with, e.g. a residual growing with temperature suggests an Arrhenius factor) plus the remaining budget and phase. It returns candidate rate laws as JSON, which MDA compiles and SMC-fits; inference, VoI design, and fitting are exact Bayesian computations. The grammar names the standard families (Michaelis–Menten, Hill, competitive/uncompetitive/noncompetitive inhibition, product inhibition, Arrhenius temperature, ping-pong) as the physical vocabulary, so — exactly as in physics — MDA’s contribution is identifying *which* multiplicative composition the data support, not blind symbol search. The residual-directed context engineering (negative evidence + budget/phase) follows LLM-AutoSciLab (Kabra et al., 2026). System message and user template:

```

SYSTEM:
You are an enzyme kineticist proposing candidate rate laws to explain assay data.
↪ Reply JSON only.

USER (world context + mechanism grammar, then the observed data, then -- only when
↪ refining -- residual-directed negative evidence and the remaining budget/phase,
↪ then the language spec):

An enzyme catalyses a reaction with initial rate r0 [mM/min]. Controllable inputs:
↪ C_A [substrate, mM], C_I [inhibitor, mM], C_B [2nd substrate, mM], C_P [product,
↪ mM], Enz [enzyme, mg/mL], T [K], pH. Discover r0 = f(C_A, C_I, C_B, C_P, Enz, T, pH;
↪ theta).

Mechanism families to consider (identify which are active FROM THE DATA):
Substrate C_A: linear | Michaelis-Menten C_A/(Km+C_A) | Hill C_A**n/(Kh**n+C_A**n) |
↪ substrate inhibition C_A/(Km+C_A+C_A**2/Ki)
Inhibitor C_I: none | competitive (raises apparent Km) | uncompetitive (lowers
↪ Vmax) | noncompetitive (lowers Vmax at all C_A)
Product C_P: none | product inhibition (like competitive but in C_P)
Temperature T: none | Arrhenius exp(-Ea/8.314*(1/T-1/310))
Second substrate C_B: none | ping-pong C_A*C_B/(Km*A*C_B+Km*B*C_A+C_A*C_B)
Enzyme Enz: rate is proportional to Enz (Vmax = kcat*Enz).
Compose factors multiplicatively when several mechanisms act together.

```

```

Experiments (inputs -> r0):
  <one line per collected experiment: C_A=.., C_I=.., C_B=.., C_P=.., Enz=.., T=..,
  ↪ pH=.. -> r0=..>

[when refining an existing pool -- residual-directed negative evidence:]
Forms ALREADY TRIED (in the pool) and their residuals -- do NOT re-propose any of
↪ these; propose forms STRUCTURALLY DIFFERENT from all of them:
  <per-form median relative residual; and, for the current best form, which input its
  ↪ residual correlates with -- e.g. residual grows with T -> add Arrhenius; with
  ↪ C_I -> add an inhibitor term>
Refine by proposing a DIFFERENT mechanism combination (add/remove an inhibition,
↪ Hill, Arrhenius, or ping-pong factor) -- do NOT patch with ad-hoc
↪ offset/softening terms.

[budget/phase status, when set:]
Experiment budget remaining: <B>. Current phase: <explore|refine> (explore =
↪ restructure the mechanism; refine = tune an adequate form).

Propose N distinct plausible rate laws (or refinements of those tried).
Each 'expr' is a Python expression in the 7 input names + your declared params, using
↪ only + - * / ** and exp, log, sqrt. Give physically plausible positive param
↪ bounds. JSON schema: {"hypotheses":[{"name":str,"expr":str,"params":[{"name":str}
↪ , "low":float,"high":float}]}}

```

The head-to-head baseline (Section D) is the LLM-AUTOscILAB agent’s own LLM+active-learning+symbolic-regression loop (Kabra et al., 2026), run under matched settings (the *same* LLM, budget, noise, and universal grammar); its prompts are those of that system and are not reproduced here.

G.3 ELECTROPHYSIOLOGY (ION CHANNELS, §4.3)

We group the NEURONBENCH prompts by their role. The **model-proposal** prompt is the *only* LLM call inside MDA’s discovery loop: MDA uses the LLM to propose candidate mechanisms, then fits and selects them by exact SMC and designs experiments by numerical Vol — so this is the prompt MDA’s results depend on. The **baseline** prompts (experiment design and forecasting) are used *only* by the LLM baseline arms MDA is compared against, never by MDA itself. All are *zero-reference*: the LLM is told only to model the cell as a conductance-based neuron and infer its channels from the data — never the candidate models, the posterior, or the true mechanism.

What the LLM sees as “data”. An experiment yields a membrane-voltage trace $V(t)$, but every LLM prompt is shown only its *reduction* to the test-window spike count — one line per protocol run, formatted verbatim as:

```

Experiments run so far and observed spike counts:
- long step (10 uA, 300 ms)                -> 21 spikes
- paired long pulses (12/300, 60 gap, 12/300) -> 16 spikes
- hyperpol step then release (-30/250 -> rebound) -> 4 spikes

```

The raw trace $V(t)$ is never sent to any LLM. It is available only to MDA’s numerical observation model (Section F.3), which chooses whether to reduce it to a spike count or score it directly with a particle filter — the reduction-vs-model choice is the solver’s, not the benchmark’s.

G.3.1 MDA MODEL PROPOSER (THE ONLY LLM CALL IN MDA’S LOOP)

In the released open-world benchmark the proposer returns its *own* parameterised channel hypotheses (reversal potential, activation direction, inactivation, and conductance / half-activation / time-constant bounds), so the hypothesis space is genuinely open:

```

System: You are an electrophysiologist proposing candidate ion-channel mechanisms to explain
current-clamp spike-count data. Reply with a JSON object only.
User: A neuron is recorded in current clamp and fires action potentials under injected current.
Model it as a single-compartment conductance-based (Hodgkin-Huxley) neuron with voltage-gated
channels. From the spike-count data, propose candidate membrane currents (beyond the standard
Na+/K+ spiking currents) that could explain its responses. Each is described by:
  reversal_mV : reversal potential (~+50 Na-like, ~+120 Ca-like, ~-80 K-like, ~-30 mixed)
  opens_on    : 'depol' or 'hyperpol'
  inactivates : true if transient / de-inactivated by a hyperpolarising pre-pulse
  bounds for conductance g, half-activation voltage (mV), and activation time constant (ms).
  <collected protocol -> spike-count data>
Propose N distinct plausible mechanisms (or an empty list if the data look like a plain spiker).

```

```
JSON schema: {"hypotheses":[{"name","reversal_mV","opens_on","inactivates",
"g_bounds","half_mV_bounds","tau_ms_bounds"]}]}
```

The H-SAG deep-dive example (Fig. 23) instead asks for candidate channel *compositions* from a channel menu, given a one-line phenotype description:

```
A neuron recorded in current clamp rests near -65 mV and fires overshooting action
potentials to a supra-threshold current step. Model it as a single-compartment
conductance model with voltage-gated channels from {Na (fast, TTX-sensitive), K
(delayed rectifier), Ca (high-threshold), leak}. Propose 4-5 DISTINCT candidate
channel COMPOSITIONS that could underlie the spiking (plus at least one non-spiking
null), each a list from {Na,K,Ca,L}. Return exactly {"compositions": [{"name": "...",
"channels": ["Na","K","L"]}, ...]}.
```

G.3.2 LLM BASELINES: EXPERIMENT DESIGN AND FORECASTING (NOT USED BY MDA)

As an *experiment proposer* (the LLM-acquisition baseline) the LLM chooses the next protocol from the collected data alone:

```
System: You are an electrophysiologist choosing the next experiment. Reply with ONLY a JSON object.
User: A neuron is recorded in current clamp; we count its action potentials per protocol. Model it
as a single-compartment conductance-based (Hodgkin-Huxley) neuron with voltage-gated channels
(Na, K, Ca, leak, and possibly others), and infer its mechanism from the data.
Experiments run so far and observed spike counts:
- <protocol> -> <count> spikes ...
Available experiments (choose one; each may be run once):
- <protocol label> ...
Which ONE experiment best reveals the neuron's mechanism next?
Return exactly {"experiment": "<one label copied verbatim>"}
```

As the *ICL-forecaster* (the in-context-learning baseline) it predicts the held-out spike counts:

```
System: You are an electrophysiologist forecasting a neuron's response. Reply with ONLY a JSON object.
User: A neuron is recorded in current clamp; we count its action potentials (spikes) per protocol.
Model it as a single-compartment conductance-based (Hodgkin-Huxley) neuron with voltage-gated
channels (Na, K, Ca, leak, and possibly others).
Experiments run and observed spike counts:
- <protocol> -> <count> spikes ...
Predict the spike count for each held-out protocol:
1. <protocol> ...
Return exactly {"counts": [n1, ...]} with one integer per protocol, in order.
```

The $N_a=0$ (zero-shot) neuron point is this forecaster with an empty “experiments run” list. These prompt templates mirror the code (scripts/ephys/hh.worlds_run.py and the released neuronbench), so the documented prompt is the one that is run.

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